Anytime-valid detection of volatility changes in high-frequency returns with robust truncation

Shreyan Thanki
Draft, November 2026

Abstract

We detect volatility changes online in one-minute returns with jumps and fat tails by two Shiryaev-Roberts betting e-detectors on the squared return clipped at a multiple of a robust scale. Under the null, with a known baseline, the average run length to a false alarm is at least A and a false alarm within m steps has probability at most m/A; a daily restart extends it to every day. No nontrivial e-process exists for the null that the variance has not fallen, so a decrease detector needs a bounded statistic, whose clipping level limits the speed of every deterministic valid rule. A confidence-sequence baseline gives $\delta + m/A$ under a tolerance on the conditional truncated variance; we also bound the delay. Proved monitors keep their bounds on permuted calm days; most other real-data analyses use a plug-in baseline without a finite-sample proof. With a confidence-sequence baseline restarted daily, up alarms followed 8 of 16 FOMC statements of 2024 and 2025 on BTC within six hours, against 11% of matched controls, but a pre-registered test on 16 FOMC and CPI releases of 2026 failed (2 of 16 against 6%, $p = 0.25$), so this contrast is exploratory. That monitor alarmed on about half of its days: real days often break its tolerance.

Keywords. Sequential change detection; e-processes; testing by betting; high-frequency volatility; truncation; jumps; anytime validity.

1 Introduction

Volatility in high-frequency markets moves between calm and stressed regimes, and risk limits, position sizes and model refits should react when it moves. We ask for an online monitor that reads one-minute returns, raises an alarm when volatility shifts up or down, and has a proved limit on false alarms however often it is checked. Two obstacles make this hard.

Jumps and fat tails. A few extreme returns dominate sums of squared returns, and a detector built on a Gaussian likelihood reads a single jump as strong evidence of higher volatility. Jump-robust estimators address this for a fixed sample: bipower variation (Barndorff-Nielsen and Shephard 2004), threshold estimators (Mancini 2009), nearest-neighbour truncation (Andersen, Dobrev and Schaumburg 2012), and the kernel quarticity estimator of Bapat (2026), which clips every return at c times its median absolute deviation (MAD).

Repeated looks. A fixed-sample test applied at every minute eventually rejects by chance: in our simulations a 60-step z-test of the variance, repeated at every step, raised a false alarm within one 390-step day on 71% of Gaussian paths. Classical change detection (Page 1954; Shiryaev 1963; Roberts 1966; Lorden 1971; Pollak 1985) controls the run length to a false alarm but needs the pre-change law. E-detectors (Shin, Ramdas and Rinaldo 2024) give the same control for composite, nonparametric nulls, using e-

processes built by betting (Shafer 2021; Ramdas, Grünwald, Vovk and Shafer 2023; Waudby-Smith and Ramdas 2024).

Robust estimators solve the first problem and betting e-detectors the second, but these need a statistic whose conditional mean has a meaning under the null and, to detect a fall in volatility, a bounded one. Clipping supplies it. The tools we combine are known, and we say where each comes from; our contributions are three.

1. **A robust anytime-valid monitor and its limits.** With a known baseline, we monitor the clipped squared return under nulls that restrict only its conditional mean, prove run-length and finite-horizon guarantees, read them over many days, restart the monitor to bound every day and bound the effect of a small drift. No nontrivial e-process exists for "the variance has not fallen", so a decrease detector needs a bounded statistic, whose clipping level limits the speed of every deterministic valid rule (a rule T is *valid* if $\mathbb{P}(T \le n) \le n/A$ for every n and every law in the null, with a known baseline; randomised rules obey a slightly weaker bound).
2. **An estimated baseline with a clean null.** The plug-in baseline keeps its level asymptotically for horizons $m = o(\sqrt{n})$, with n the calibration length (far above 152,100 returns for a 390-step day), and can lose it by an unbounded factor otherwise; a confidence-sequence baseline gives $\delta + m/A$ under a tolerance on the conditional truncated variance that does not depend on the algorithm.
3. **Evidence, including negative results.** A delay theory that splits the cost of robustness into three losses, with a delay floor for every valid rule; simulations against parametric, robust and calibrated classical competitors; null streams built from real returns, on which the proved monitors keep their bounds; and real series, on which the monitor with a plug-in baseline is about as precise as simpler rules matched to its alarm counts and adds little to a forecast that sees the last hour.

Related work. Retrospective tests for volatility breaks include the cumulative sums of squares of Inclán and Tiao (1994) and the high-frequency test of Bibinger, Jirak and Vetter (2017).[1] Monitoring with a training sample and asymptotic control of any false alarm goes back to Chu, Stinchcombe and White (1996) and covers GARCH parameters (Berkes, Gombay, Horváth and Kokoszka 2004; Horváth, Trapani and Wang 2025), the unconditional variance (Horváth, Kokoszka and Zhang 2006) and general parameters (Dette and Gösmann 2020); our guarantee is non-asymptotic and needs no model for the returns, at the price of a null on a conditional mean. Robust change detection censors likelihood ratios (Huber 1965), uses least favourable laws (Unnikrishnan, Veeravalli and Meyn 2009), bounded losses (Fearnhead and Rigaill 2019) or robust divergences in Bayesian online change-point detection (Adams and MacKay 2007; Knoblauch, Jewson and Damoulas 2018); clipping is the betting analogue of censoring. E-value methods for change detection include sequential estimation (Shekhar and Ramdas 2024), conformal detectors (Vovk et al. 2021) and many streams (Dandapanthula and Ramdas 2025); Wang and Ramdas (2023) and Mineiro (2022) give confidence sequences under heavy tails, and e-values have been applied to risk backtesting (Wang, Wang and Ziegel 2026) and high-volatility days (Choe and Ramdas 2024). Tartakovsky, Nikiforov and Basseville (2014) is the standard reference on change detection. Andersen, Todorov and Zhou (2025) monitor high-frequency prices in real time for local no-arbitrage violations. To our knowledge, and subject to the caveat in footnote 1, no published method combines jump-robust high-frequency volatility measures with anytime-valid change detection (our search, of November 2026, was not exhaustive).

Organisation. Sections 2 to 5 give the statistic, the guarantees, estimated baselines and the delay; Sections 6 and 7 report simulations and an application; Section 8 discusses limitations. The supplement holds proofs, further results, settings and sources; a companion note (`paper/extensions.md`) treats a running truncation level, restarts, panels of assets, several frequencies and an adversarial search.

2 Model, statistic and nulls

Returns and information. Let r_t be the one-minute log return ending at minute t. Monitoring starts after a calibration window $r_{-n+1}, \ldots, r_0$, and $\mathcal{F}_t$ is the information up to time t: the returns and, when returns can be missing, the indicators of which were observed. A quantity is *predictable* if its value used at time t is $\mathcal{F}_{t-1}$-measurable. Everything the monitor uses (truncation level, baseline, bets, intraday profile) must be predictable.

The statistic. Bapat (2026) clips returns at $c = 3$ times the MAD of the whole sample before forming a kernel estimate of local quarticity. We keep the clipping and change three things. We use the scaled MAD about zero, $\hat{s} = 1.4826\,\mathrm{median}_i|r_i|$ (the median of one-minute returns is 0 in all our data sets), so that with Gaussian returns the cut-off $\tau = c\,\hat{s}$ sits at c standard deviations rather than at $0.6745\,c$; $\hat{s}$ and τ come from the calibration window, so τ is $\mathcal{F}_0$-measurable; and we monitor

$$X_t = \min(r_t^2, \tau^2) \in [0, \tau^2],$$

the summand of truncated realised variance, rather than a fourth power. Supplement S1.8 compares fourth powers and gives the predictable rule used when most returns of a calibration window are zero.

Truncated moments. For $Z \sim N(0,1)$ with density φ and distribution function Φ, and a cut-off $a > 0$, let

$$\kappa_2(a) = \mathbb{E}[\min(Z^2, a^2)] = 2\Phi(a) - 1 - 2a\varphi(a) + 2a^2(1 - \Phi(a)).$$

For $c = 3$ and the scaled MAD ($a = 3$), 0.27% of Gaussian returns are clipped and $\kappa_2 = 0.995$; with the raw MAD ($a = 2.023$), 4.3% are clipped and $\kappa_2 = 0.925$. For Gaussian returns with standard deviation σ_0 the mean of X_t is $\kappa_2(c)\sigma_0^2$, and τ^2 divided by it is $c^2/\kappa_2(c)$, which equals 9.045 for $c = 3$.

Intraday seasonality and missing returns. Each return is divided by a predictable intraday profile (for crypto, the median of $|r|$ at the same minute over the previous 20 days, smoothed across neighbouring minutes; Andersen and Bollerslev 1997; Boudt, Croux and Laurent 2011), treated as known; errors in it violate the null. Missing returns, and the first return after a gap, are skipped, not set to zero; validity then needs that, given the past, whether r_t is observed carries no information about its size. Horizons m count monitored steps.

Nulls. Let $\mu_t = \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]$ be the conditional truncated variance and $m_0 > 0$ a baseline. For a band $b \in [0, 1)$ fixed in advance, the nulls are

$$H_0^{\uparrow}\colon \mu_t \le (1+b)\,m_0 \ \text{ for all } t \ge 1, \qquad H_0^{\downarrow}\colon \mu_t \ge (1-b)\,m_0 \ \text{ for all } t \ge 1.$$

They restrict one conditional moment and allow any return law, any jumps and any dependence that keeps that moment on the null side at every step. Persistent volatility clustering and price-level microstructure noise both move it, even when the unconditional variance is constant (Section 6.4).

Lemma 1 (volatility meaning). Suppose $r_t = \sigma_t \varepsilon_t$, where $\sigma_t > 0$ is $\mathcal{F}_{t-1}$-measurable and ε_t is independent of $\mathcal{F}_{t-1}$ with a law G that does not depend on t. Let τ be $\mathcal{F}_0$-measurable and $g_\tau(s) = \mathbb{E}_G[\min(s^2\varepsilon^2, \tau^2)]$. Then $\mu_t = g_\tau(\sigma_t)$, and g_τ is nondecreasing on $(0, \infty)$; it is strictly increasing on every interval of values s with $G(0 < |\varepsilon| < \tau/s) > 0$. Hence, with $m_0 = g_\tau(\sigma_0)$ and $b = 0$, $H_0^{\uparrow}$ holds whenever $\sigma_t \le \sigma_0$ for all t, and $H_0^{\downarrow}$ whenever $\sigma_t \ge \sigma_0$ for all t.

The proof is in Supplement S1.5; only the size of the change in μ_t for a given change in σ_t depends on G. Three caveats follow. A change in tail shape at fixed σ_t also moves μ_t: with $c = 3$, Student t innovations at the same variance lower it, which reads as a decrease. With a contemporaneous leverage effect μ_t need not be monotone in σ_t. And returns of prices rounded to a tick are not of the form $\sigma_t \varepsilon_t$ with a fixed G, so for them the null is a statement about μ_t only.

3 Validity

3.1 Betting e-detectors and the false alarm guarantees

Write $m^{\uparrow} = (1 + b)m_0$, $m^{\downarrow} = (1 - b)m_0$ and $B = \tau^2/m^{\downarrow} > 1$, and let $Y_t = X_t/m^{\uparrow}$ on the increase side and $Y_t = X_t/m^{\downarrow} \in [0, B]$ on the decrease side. With a predictable bet λ_t define the betting factors

$$f_t^{\uparrow} = 1 + \lambda_t(Y_t - 1), \ \ \lambda_t \in [0, 1), \qquad f_t^{\downarrow} = 1 + \lambda_t(1 - Y_t), \ \ \lambda_t \in [0, 1/(B - 1)).$$

Both are positive, since $Y_t \ge 0$ and, on the decrease side, $Y_t \le B$. Under the corresponding null, $\mathbb{E}[f_t \mid \mathcal{F}_{t-1}] \le 1$, so for every start time j the wealth $L_t^{(j)} = \prod_{s=j}^{t} f_s$ (with $L_{j-1}^{(j)} = 1$) is a nonnegative supermartingale with initial value 1, a *test supermartingale*.

Definition (Shiryaev-Roberts e-detector). Fix bets $\lambda_1, \ldots, \lambda_K$ in the admissible range and weights $w_k > 0$ with $\sum_k w_k = 1$. For each k let $M_t^{(k)} = f_t^{(k)}(M_{t-1}^{(k)} + 1)$ with $M_0^{(k)} = 0$, so that $M_t^{(k)} = \sum_{j=1}^{t} \prod_{s=j}^{t} f_s^{(k)}$ is the total wealth of bettors who start at every time $j \le t$. Let $M_t = \sum_k w_k M_t^{(k)}$ and raise an alarm at $T_A = \inf\{t \ge 1 : M_t \ge A\}$.

The default grids, used throughout, are $\lambda \in \{0.02, 0.05, 0.1, 0.2, 0.35, 0.5\}$ for increases and $\lambda/\lambda_{\max} \in \{0.1, 0.25, 0.5, 0.75, 0.95\}$ with $\lambda_{\max} = 1/(B - 1)$ for decreases, with equal weights. An update costs $O(K)$ operations.

Lemma 2. Under the null, $\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] \le M_{t-1} + 1$ for every $t \ge 1$, so $M_t - t$ is a supermartingale. The same holds for any predictable bets and for any average of such statistics with fixed weights.

Proof. $\mathbb{E}[M_t^{(k)} \mid \mathcal{F}_{t-1}] = \mathbb{E}[f_t^{(k)} \mid \mathcal{F}_{t-1}](M_{t-1}^{(k)} + 1) \le M_{t-1}^{(k)} + 1$; average over k. ▫

Theorem 3 (average run length). Under the null, $\mathbb{E}[T_A] \ge A$ for every $A > 0$.

Theorem 3 is the run-length guarantee of e-detectors (Shin, Ramdas and Rinaldo 2024); Supplement S1.5 gives the proof with its conditions explicit. A practitioner usually needs a statement for a finite horizon.

Theorem 4 (false alarm within a horizon). Under the null, $\mathbb{P}(T_A \le m) \le m/A$ for every integer $m \ge 1$.

Proof. For $0 \le t \le m$ let $S_t = M_t + (m - t)$. By Lemma 2, S is a nonnegative supermartingale on $\{0, \ldots, m\}$ with $S_0 = m$. Ville's inequality gives $\mathbb{P}(\max_{t \le m} S_t \ge A) \le m/A$, and $M_t \ge A$ implies

$S_t \geq A$. ▫

The argument is elementary and not new, but its reading over many days needs care. With $A = 7800$ for a 390-step trading day, or $A = 28,800$ for a 1440-minute crypto day, Theorem 4 bounds by 5% the probability of a false alarm within the first day after the statistics start at 0 (at the start of monitoring or after a recalibration), not on a later day (Proposition 7). The CUSUM-type statistic $C_t = f_t \max(C_{t-1}, 1)$ satisfies $C_t \leq M_t$ and inherits Theorems 3 and 4.

Proposition 5 (two-sided monitoring). Under the two-sided null $(1-b)m_0 \leq \mu_t \leq (1+b)m_0$ for all t, the average $\bar{M}_t = (M_t^{\uparrow} + M_t^{\downarrow})/2$ of the two side statistics satisfies Lemma 2. Hence the rule that alarms at the first t with $\bar{M}_t \geq A$, and the rule that alarms at the first t with $\max(M_t^{\uparrow}, M_t^{\downarrow}) \geq 2A$, both satisfy Theorems 3 and 4.

Proposition 5 follows from Lemma 2 (Supplement S1.5). All our experiments use the maximum rule, which names the direction of an alarm by the side that crossed; the average rule is never later, but the gain is small.

Corollary 6 (restarts after alarms). Suppose the monitor recalibrates after every alarm, with a truncation level τ_q and baseline $m_0^{(q)}$ for segment q that are known at the start of the segment, and that the two-sided null holds in every segment relative to its own baseline. Let N_m be the number of alarms within the first m monitored steps (calibration steps are not counted). Then $\mathbb{E}[N_m] \leq m/A$.

The proof and a simulation check are in Supplement S1.1. With $b = 0$ the hypothesis asks that each segment's baseline equal the true conditional mean of its statistic, which no estimated baseline achieves; Corollary 18 covers a confidence-sequence baseline.

Later days. Let day d be the steps $(d-1)m+1, \ldots, dm$, and let $\bar{M}_t$ be the statistic of the alarm rule: M_t for one side at level A, or the average of Proposition 5 for the two-sided rule.

Proposition 7 (after the first day). Under the null, with a known baseline and $b = 0$, for every $s \geq 0$ and $m \geq 1$, $\mathbb{P}(\text{an alarm at one of } s+1, \ldots, s+m \mid \mathcal{F}_s) \leq \min\{1, (\bar{M}_s + m)/A\}$, so a false alarm on day d has probability at most $\min(1, dm/A)$. No bound below 1 holds for all days, even over i.i.d. laws: the law with $|r_t| = \sqrt{m_0}$, under which $Y_t = 1$, is in the two-sided null and makes every Shiryaev-Roberts statistic equal to t, so the two-sided rule alarms with certainty at step $\lceil 2A \rceil$, on day 40 for $(m, A) = (390, 7800)$ and for $(1440, 28,800)$.

By Lemma 2 the statistic enters day d with up to $(d-1)m$ in expectation. With 390-step days and i.i.d. Gaussian returns the probability of a first false alarm on day d, given none before, rose from 0.98% on day 1 to 4.7% on average over days 11 to 20, and after long stretches with $Y_t = 1$ it was far higher for a Gaussian day (Supplement S1.7). Over many days the plain detector's guarantee is thus an expected count (Corollary 6); it meets the window-limited criterion of Lai (1998) only for the first window, and Mei (2008) shows why the run length alone can mislead (footnote 1).

A bound for every day. Let $0 = b_0 < b_1 < \ldots$ be stopping times with $b_{i+1} - b_i \leq m$ (in practice the day starts), and at each b_i scale each side's statistics down by a common factor so that the side's average is at most $\iota \geq 0$; $\iota = 0$ is a restart at 0.

Theorem 8 (restart at every day boundary). With a known baseline and under the null of Theorem 4, or of Proposition 5 for the two-sided rule, $\mathbb{P}(\text{an alarm in } (b_i, b_{i+1}] \mid \mathcal{F}_{b_i}) \le \min\{1, (m+\iota)/A\}$ almost surely for every i. If the statistics also restart after every alarm, the expected number of alarms in $(b_i, b_{i+1}]$ given $\mathcal{F}_{b_i}$ is at most $(m+\iota)/A$.

With $\iota = 0$ and $A = m/\alpha$, the false alarm probability, the expected number of false alarms and the hazard of every day are at most α given the past (since $\{T_A > b_i\} \in \mathcal{F}_{b_i}$), and a window of m steps that starts in day i, whatever boundaries it crosses, has an alarm with probability at most $(2m+\iota-1)/A$ given $\mathcal{F}_{b_i}$, though not given the past at its start (Supplement S1.7). With the confidence-sequence baseline the restart gives $\delta + m/A$ per day only marginally, while the tolerance holds during that day (Corollary S14); with the plug-in nothing per day is proved. The price is delay for a change late in a day (Proposition S6). A CUSUM-type statistic bounds the unconditional probability of an alarm in every window, but only for exchangeable returns, fixed bets and a known baseline, and was slower (Theorem S7, Proposition S8; Supplement S1.7).

3.2 A small drift of the conditional mean

The guarantees need the null at every step. The next result says what is left when it holds only up to a drift. Write $\lambda_k^{\uparrow}$ $(w_k^{\uparrow})$ for the increase bets (weights) and $\lambda_l^{\downarrow}$ $(w_l^{\downarrow})$ for the decrease bets, and for $x \ge 0$ and an integer $m \ge 1$ let

$$Y_m(x) = \frac{(1+x)((1+x)^m - 1)}{mx}, \qquad Y_m(0) = 1,$$

the average of $(1+x)^j$ over $j = 1, \dots, m$.

Theorem 9 (drift of the conditional mean). Suppose that $\mu_t \le (1+\xi^{\uparrow})\, m^{\uparrow}$ and $\mu_t \ge (1-\xi^{\downarrow})\, m^{\downarrow}$ for all t, with constants $\xi^{\uparrow}, \xi^{\downarrow} \ge 0$ known at time 0. Then for the two-sided rule of Proposition 5 with the maximum and every $m \ge 1$,

$$\mathbb{P}(T_A \le m) \le \min\{1, \frac{m}{A}\Gamma_m(\xi^{\uparrow}, \xi^{\downarrow})\}, \qquad \Gamma_m(\xi^{\uparrow}, \xi^{\downarrow}) = \frac{1}{2}[\sum_k w_k^{\uparrow} Y_m(\lambda_k^{\uparrow}\xi^{\uparrow}) + \sum_l w_l^{\downarrow} Y_m(\lambda_l^{\downarrow}\xi^{\downarrow})].$$

With $\xi^{\uparrow} = \xi^{\downarrow} = 0$ this is Theorem 4.

The proof is in Supplement S1.2. For the default detector ($c = 3$, $b = 0$, $A = 7800$), Γ_m is close to 1 while the drift times the horizon is small (5.80% instead of 5% at one day for a drift of 0.5% on both sides) and then compounds: the one-day bound doubles at a drift of 1.63%, and a drift of 0.5% exhausts the five-day bound (Table S1). A drift that alternates costs far less, but no bound in the mean drift alone exists (Supplement S1.3). So the null holds approximately when the conditional mean stays within a few tenths of a percent of the baseline over a day, or when a band b absorbs the drift.

3.3 Why a decrease detector needs a bounded statistic

Proposition 10. Let $\mathcal{N}_{\infty}$ be the set of all probability laws on $[0, \infty)$ with mean at least 1. If $E: [0,\infty) \to [0,\infty]$ is measurable and $\mathbb{E}_P[E(Y)] \le 1$ for every $P \in \mathcal{N}_{\infty}$, then $E(y) \le 1$ for every $y \ge 0$.

Corollary 11 (no e-process for "the variance has not fallen"). Let $\mathcal{P}^{\downarrow}$ be the set of laws of real sequences $(r_t)_{t \ge 1}$ with $\mathbb{E}[r_t^2 \mid r_1, \dots, r_{t-1}] \ge m_0$ for every t. If $W_t = W_t(r_1, \dots, r_t) \ge 0$ satisfies $\mathbb{E}_P[W_t] \le 1$ for every $P \in \mathcal{P}^{\downarrow}$ and every fixed t, then $W_t \le 1$ for every t and every path. Consequently a Shiryaev-Roberts statistic built from such processes started at each time satisfies $M_t \le t$ on every path, and cannot alarm before time A, whatever the data.

The proof (Supplement S1.4) builds, for any path, a law in $\mathscr{P}^{\downarrow}$ that follows the path with probability close to 1, a one-sided relative of the result of Bahadur and Savage (1956). The increase null admits nontrivial e-variables such as $E(y) = y$, so an increase detector works on raw squared returns, but no nonparametric e-test detects a fall in the variance itself. For a statistic in $[0, B]$ the null "mean at least 1" admits the bets $1 + \lambda(1 - y)$, $\lambda \in [0, 1/(B - 1)]$, and up to domination only these (Lemma 20). Truncation is one bounded functional with a volatility meaning (Lemma 1); ranks are another (Section 6.5). Its price is a speed limit: one step multiplies a decrease bet's wealth by at most $B/(B - 1)$ (Corollary 22); for increases, clipping bounds the effect of one jump.

Every valid rule. The next result bounds every valid rule (Section 1), with a known baseline and $b = 0$. Let $\pi(y)$ be the largest probability that a law on $[0, B]$ in the null can put on the value y: under the decrease null (conditional mean at least 1), $\pi(y) = (B - 1)/(B - y)$ for $y < 1$ and 1 otherwise; under the increase null (at most 1), $\pi(y) = 1/y$ for $y > 1$ and 1 otherwise; under the two-sided null (equal to 1), the product of the two. For a path $y = (y_1, y_2, \ldots)$ of $Y_t = X_t/m_0$ let $\Pi_n(y) = \prod_{s \le n} 1/\pi(y_s)$.

Theorem 12 (pathwise floor). Let T be a stopping rule with $\mathbb{P}(T \le n) \le n/A$ for every n and every law of (Y_t) on $[0, B]$ in one of the three nulls, with a known baseline and $b = 0$. (a) If T is deterministic, it stops at step n on the path y only if $\Pi_n(y) \ge A/n$. (b) If T is randomised independently of the data, $\mathbb{P}(T \le n \mid y) \le (n/A)\Pi_n(y)$. (c) For every path y and every n with $\Pi_n(y) \ge A/n$, some deterministic rule with the same guarantee stops at n on y.

Corollary 13. (a) For the decrease null on $[0, \infty)$, that is without truncation, no deterministic rule valid at levels n/A alarms before step A, on any path. (b) On a complete collapse, $y_s = 0$ for every s, every deterministic rule valid at levels n/A for the decrease or the two-sided null needs at least $\min\{n : n \log(B/(B - 1)) \ge \log(A/n)\}$ steps: 45 at $c = 3$ and $A = 7800$. A randomised valid rule alarms by step n on that path with probability at most $\min(1, n\Pi_n/A)$, which is 0.98 at step 44; this cap alone gives a mean delay there of at least 37.8, and a linear programme that imposes validity at every n gives 43.4 (Supplement S1.6).

The proof (Supplement S1.6) places at each step the two-point null law that gives y_s its largest probability; under this product law the path has probability $1/\Pi_n(y)$, so a rule that stops there spends at least that much of its budget n/A. The laws can be placed on the returns, so the floor binds rules that see r_t, and part (a) extends Corollary 11 to every deterministic procedure. We claim no priority for this elementary argument; Clerico (2025) shows a related optimality of betting for a bounded mean (footnote 1).

4 Estimated baseline

The guarantees above assume that the baseline m_0 and the level τ are fixed in advance, or at least that the null holds with whatever values the monitor uses. In practice both are estimated from a calibration window.

4.1 The plug-in baseline

The *plug-in* baseline $\hat{m}$ is the mean of X over the window, with τ from the same window. With 390 calibration steps its one-day false alarm rate on the increase side was 4.58% to 5.10% across Gaussian, Student t_3 and jump returns, against 1.62% to 1.75% with the true baseline, and over five days 41% to 44% against a guaranteed 25%: a baseline

slightly too low gives the log wealth a positive drift that the bets eventually find (Supplement S2).

Theorem 9 explains this. Given the window W, the plug-in's error is a fixed drift: if the conditional mean of X_t after calibration stays at a level $\mu^*(W)$ fixed by the window, for example $\mathbb{E}_F[\min(r^2, \hat{\tau}^2)]$ for i.i.d. returns with law F, the increase side sees the drift $\xi^{\uparrow}(W) = (\mu^*(W)/\hat{m} - 1)^+$ and the decrease side $\xi^{\downarrow}(W) = (1 - \mu^*(W)/\hat{m})^+$, both known at time 0.

Corollary 14 (plug-in baseline). Under this condition, $\mathbb{P}(T_A \le m) \le \mathbb{E}_W[\min\{1, (m/A)\Gamma_m(\xi^{\uparrow}(W), \xi^{\downarrow}(W))\}]$, where Γ_m uses the decrease bets of the window.

The proof applies Theorem 9 given the window (Supplement S1.2). The proved one-day bound, 42% with 390 calibration steps, is within a factor of 1.3 of the worst stream found by the adversarial search, and falls to 7.3% at 11,520 calibration steps and 6.0% at 23,040: the plug-in has no guarantee of the form Cm/A at practical calibration lengths, but it does have a computable one.

When n, m and A grow together. Assume that the calibration returns are i.i.d. with a law F that has no atom at 0 and under which $|r|$ has a unique median, and let $\mathscr{C}(W)$ be the class of monitored streams whose conditional truncated mean at the window's level, $\mathbb{E}[\min(r_t^2, \hat{\tau}^2) \mid \mathcal{F}_{t-1}]$, equals its value under F; i.i.d. returns from F are one such stream. This null is defined given the window W, after calibration, unlike the null of Theorem 16. Write T for the first alarm of the two-sided rule with the plug-in baseline, $\bar{\Phi}$ for the standard normal upper tail and $\hat{\mathrm{se}}$ for the standard error of $\hat{m}$.

Theorem 15 (the plug-in as n, m and A grow). Let $n \to \infty$, with $m = m_n$ and $A = A_n \ge m$. (a) If $m/\sqrt{n} \to 0$, then for every stream in $\mathscr{C}(W)$, $\mathbb{P}(T \le m) \le (m/A)(1 + C(m+1)/\sqrt{n}(1 + o(1)))$ uniformly in A, with $C = 0.037$ for Gaussian returns and 0.047 for Student t_3 returns at $c = 3$, and $C = 1.79$ for every F. (b) If $m/\sqrt{n} \to \infty$, $m/\log A \to \infty$ and $m/A \to 0$, some stream in $\mathscr{C}(W)$ makes $\mathbb{P}(T \le m)/(m/A) \to \infty$. (c) Baselines moved apart to $\hat{m} \pm z\hat{\mathrm{se}}$ give $\mathbb{P}(T \le m) \le m/A + 2\bar{\Phi}(z) + o(1)$ uniformly in m and A, and some stream attains the excess $2\bar{\Phi}(z)$ when $m/(\sqrt{n}\log A) \to \infty$; so $z = \Phi^{-1}(1 - \delta/2)$ restores $m/A + \delta$.

The proof (Supplement S2) bounds the drift of the plug-in by $\hat{B}$ times the Kolmogorov distance between the window and F; the stream of (b) holds the conditional mean fixed and then bets boldly on 0 and $\hat{B}$. After a 390-step Gaussian window the worst one-day false alarm probability over $\mathscr{C}(W)$ is about 33% to 42%, while for i.i.d. returns the largest simulated rate was 5.14% (4.72% to 5.59%; Supplement S2). The confidence-sequence baseline achieves (c) without asymptotics, at an effective z of 3.8 to 5.7 against 2.576 for $\delta = 0.01$.

4.2 A baseline from a confidence sequence

A confidence sequence restores a distribution-free proof. It needs a sample split, because a τ computed from the whole window makes each X_i in the window depend on later returns.

Construction. The first n_τ returns of the calibration window fix $\tau = c\hat{s}$ (25% of the window by default). On the later returns, $\chi_i = \min(r_i^2, \tau^2)/\tau^2 \in [0, 1]$ has conditional mean $\eta_i = \mu_i/\tau^2$, and predictable empirical Bernstein bounds in the style of Waudby-Smith and Ramdas (2024), with one-sided ranges about a predictable centre, give an upper bound U_t

and a lower bound L_t with $\mathbb{P}(\exists t: \bar{\eta}_t^{\uparrow} \geq U_t) \leq \delta_u$ and $\mathbb{P}(\exists t: \bar{\eta}_t^{\downarrow} \leq L_t) \leq \delta_l$. Here $\bar{\eta}_t^{\uparrow}$ and $\bar{\eta}_t^{\downarrow}$ are averages of $\eta_1, \ldots, \eta_t$ weighted by the predictable bets of the two bounds, and nothing is assumed about dependence or about how η_i varies (Supplement S2 gives the formulas, and Lemma S10 the proof).

Monitoring rule. Calibration ends at a stopping time $T_0 > n_\tau$, a fixed length or a rule that stops when the relative width of the bounds falls below a target. Set $m^{\uparrow} = (1+b)\tau^2 \min(U_{T_0}, 1)$ and $m^{\downarrow} = (1-b)\tau^2 \max(L_{T_0}, 0)$, run the two statistics of Section 3.1 from T_0 with these baselines and with bets fixed at T_0 (the decrease side is switched off when $m^{\downarrow} = 0$ or $\tau^2/m^{\downarrow} \leq 1$), and alarm at the first $t > T_0$ with $\max(M_t^{\uparrow}, M_t^{\downarrow}) \geq 2A$.

The null as a tolerance. Let $h \geq 1$ be a horizon, possibly infinite. The tolerance condition with horizon h is

$$\sup_{T_0 < t \leq T_0 + h} \mu_t \leq (1+b) \min_{n_\tau < i \leq T_0} \mu_i \quad \text{and} \quad \inf_{T_0 < t \leq T_0 + h} \mu_t \geq (1-b) \max_{n_\tau < i \leq T_0} \mu_i$$

almost surely, with nothing assumed about the calibration window beyond $0 \leq X \leq \tau^2$: during the first h monitored steps the conditional truncated variance never rises above its smallest calibration value by more than the factor $1+b$, and never falls below its largest calibration value by more than the factor $1-b$. Let $\mathcal{O}_m$ be the event that one of the two inequalities fails at some $t \in (T_0, T_0 + m]$.

Theorem 16 (guarantee with an estimated baseline). Let $\delta = \delta_u + \delta_l$.

1. Under the tolerance condition with $h = m$, for every integer $m \geq 1$, $\mathbb{P}(\text{an alarm at one of } T_0 + 1, \ldots, T_0 + m) \leq \delta + m/A$, and $\leq \delta + (1-\delta)m/A$ when $m < A$.
2. Under the condition with $h = \infty$, if $T_0 < \infty$ almost surely, the expected number of monitored steps before the first alarm is at least $(1-\delta)A$.
3. One side alone at level A satisfies part 1 with δ_u (increase) or δ_l (decrease) in place of δ, under its half of the condition with $h = m$.
4. With no assumption on the law of the returns, $\mathbb{P}(\text{an alarm at one of } T_0 + 1, \ldots, T_0 + m) \leq \delta + \mathbb{P}(\mathcal{O}_m) + m/A$.

The condition does not involve the bets, so with a fixed calibration length it concerns the data-generating process alone. The proof (Supplement S2) uses two facts: the time-uniform bounds hold at the random time T_0 with probability at least $1-\delta$, and on that event the condition puts every μ_t of the horizon on the null side of the baselines. Part 4 charges the probability that the conditional truncated variance leaves the band; Theorem 9 charges the size of a drift instead.

What the tolerance excludes. With $b = 0$ the condition rules out any volatility clustering: the confidence sequence protects against estimation error, not against a moving conditional mean. Under GARCH(1,1) with 390 calibration steps and a 390-step day, the smallest band that held on a path had a median of 0.90 to 2.58 across persistences 0.95 to 0.9999 (2.05 to 5.01 with 1440 steps), and no path met $b = 0.25$, so parts 1 to 3, which need the tolerance almost surely, never apply to these models, and part 4 charges a failure of probability 1.

Corollary 17 (a constant conditional law). If from step $n_\tau + 1$ until the change the conditional law of r_t given $\mathcal{F}_{t-1}$ is one fixed law (for example i.i.d. returns with jumps and fat tails), then μ_t is constant, the tolerance condition holds with $b = 0$, and with no change $\mathbb{P}(\text{false alarm within } m \text{ monitored steps}) \leq \delta + m/A$ with the baseline estimated from the data.

Corollary 18 (recalibration after every alarm). If every segment has its own split, bounds and error budget δ, and the tolerance condition with $h = m$ holds in every segment relative to its own calibration, then the number N_m of alarms within the first m monitored steps satisfies $\mathbb{E}[N_m] \le (\delta + m/A)/(1 - \delta)$. With $\delta = 0$ this is Corollary 6.

Choosing A and δ. For a target α for the first day of m steps after each calibration, take $\delta < \alpha$ and $A = m/(\alpha - \delta)$: with $\delta = 0.01$ and $\alpha = 5\%$, $A = 9750$ for a 390-step day and $36,000$ for a 1440-minute day. The larger A costs little delay ($\log 9750/\log 7800 \approx 1.025$); the real cost is the width of the bounds (Section 6.3). Shekhar and Ramdas (2024) combine a confidence sequence for the pre-change parameter with a detector in a similar way; to our knowledge, and subject to the caveat in footnote 1, the one-sided ranges, the sample split and the explicit tolerance are new.

5 Detection delay

This section bounds the delay after a change, splits the cost of robustness and gives speed limits for decreases. Proofs are in Supplement S3.

5.1 Growth rate, information and speed limits

Post-change model. From a change time ν on, $Y_\nu, Y_{\nu+1}, \dots$ are i.i.d. with law Q and, given τ and the baseline (through which Y is defined), independent of $\mathcal{F}_{\nu-1}$. Write $V = Y - 1$ on the increase side and $V = 1 - Y$ on the decrease side; V takes values in $[v_{\mathrm{lo}}, v_{\mathrm{hi}}]$, equal to $[-1, B-1]$ or $[1-B, 1]$, and the admissible bets are $\lambda \in [0, \lambda_{\max})$ with $\lambda_{\max} = 1/|v_{\mathrm{lo}}|$. A change means $\mathbb{E}_Q[V] > 0$. The worst-case delay of a stopping rule T is Lorden's (1971)

$$\mathrm{WADD}(T) = \sup_{\nu \ge 1} \operatorname{ess\,sup} \mathbb{E}_\nu[(T - \nu + 1)^+ \mid \mathcal{F}_{\nu-1}].$$

The *growth rate* of a bet is $g(\lambda) = \mathbb{E}_Q[\log(1 + \lambda V)]$. It is strictly concave with $g(0) = 0$ and $g'(0) = \mathbb{E}_Q[V] > 0$, so it has a unique maximiser $\lambda^*(Q)$ in $[0, \lambda_{\max}]$. Let $\mathcal{N}_B = \{P \text{ on } [0,B] : \mathbb{E}_P[V] \le 0\}$ be the nonparametric null class of one step, and $\mathrm{KL}_{\inf}(Q) = \inf_{P \in \mathcal{N}_B} \mathrm{KL}(Q \| P)$.

Lemma 19 (growth is bounded by information). If $E \ge 0$ and $\mathbb{E}_P[E(Y)] \le 1$ for a law P, then $\mathbb{E}_Q[\log E(Y)] \le \mathrm{KL}(Q \| P)$.

Lemma 20 (the best nonparametric growth). Let $B < \infty$. (a) Every e-variable for the whole class $\mathcal{N}_B$ is dominated by a bet: there is $\lambda \in [0, \lambda_{\max}]$ with $E(y) \le 1 + \lambda v(y)$ for all y. (b) If $0 < \lambda^* < \lambda_{\max}$, then $\max_\lambda g(\lambda) = g(\lambda^*) = \mathrm{KL}_{\inf}(Q)$, and the infimum is attained by $P^*(dy) = Q(dy)/(1 + \lambda^* v(y))$.

Part (b) is the dual form, for a bounded statistic, of the information number of Honda and Takemura (2010), and P^* is the reverse information projection of Q onto the null (Grünwald, de Heide and Koolen 2024; Larsson, Ramdas and Ruf 2024). For decreases under light tails the maximiser is only barely interior (Supplement S3), so the relevant decrease bets lie just below $\lambda_{\max}$, and the constants of the asymptotic theory below are large.

Since every bet is an e-variable for the true pre-change law P_0 when $P_0 \in \mathcal{N}_B$, and since X and Y are functions of r, Lemmas 19 and 20 and the data processing inequality give:

Corollary 21 (information ladder). With $g_{\mathrm{grid}} = \max_k g(\lambda_k)$,

$$\mathrm{KL}(Q_r \,\|\, P_{0,r}) \;\geq\; \mathrm{KL}(Q_X \,\|\, P_{0,X}) \;\geq\; \mathrm{KL}_{\inf}(Q) \;\geq\; g_{\mathrm{grid}}.$$

The three ratios between neighbouring terms are the *truncation loss*, the *nonparametric loss* (the price of a null that restricts only the mean of Y) and the *grid loss*. Their product is the first-order delay ratio of the robust detector against the likelihood-ratio detector that knows both laws.

Corollary 22 (speed limits). Every bet satisfies $f_\lambda \leq 1 + \lambda_{\max} v_{\mathrm{hi}}$, so every e-variable for $\mathcal{N}_B$ satisfies $\log E \leq \log B$ on the increase side and $\log E \leq \log (B/(B-1))$ on the decrease side, and $\mathrm{KL}_{\inf}(Q)$ obeys the same bounds however large the change. For the detector the statement is deterministic: $M_n \leq \sum_k w_k \sum_{j=1}^{n} F_k^{n-j+1}$ with $F_k = 1 + \lambda_k v_{\mathrm{hi}}$, so T_A is never smaller than the first n at which this sum reaches A.

With $c = 3$ and Gaussian returns, $\log (B/(B-1)) = 0.117$ nats per step. On a complete collapse (all returns zero) no Shiryaev-Roberts e-detector with bets valid over the whole null can alarm before 57.7 steps at $A = 7800$, and the default grid needs 73, while every deterministic valid rule needs 45 (Corollary 13) and a randomised one 43.4 on average. The gap between 45 and 58 is the factor n/B in the level: a valid rule may spend its whole budget n/A on one path, while an e-detector must spread $\mathbb{E}M_n \leq n$ over all paths (Supplement S1.6).

A delay floor for every valid rule. For a change at the first step with post-change law Q, let $N_A = \min \{n : \Pi_n \geq A/n\}$, with Π_n of Theorem 12 evaluated on the data.

Theorem 23 (delay floor). Let T be valid at levels n/A for one of the three nulls. (a) If T is deterministic, $T \geq N_A$ almost surely under Q, so $\mathrm{WADD}(T) \geq \mathbb{E}_Q[N_A]$. (b) For every null law P° with respect to which Q, restricted to the first n steps, is absolutely continuous, $Q(T \leq n) \leq \beta_n$, the largest Q-probability of rejection of a test of P° at level n/A that rejects only where $N_A \leq n$ if T is deterministic, or with probability at most $\min \{1, (n/A)\Pi_n\}$ if T is randomised; hence $\mathrm{WADD}(T) \geq 1 + \sum_{n \geq 1} (1 - \beta_n)$.

Both parts follow from Theorem 12 and the Neyman-Pearson lemma (Supplement S3). With $P^\circ = P^*$ of Lemma 20 and β_n estimated by importance sampling, over 18 post-change laws (Gaussian, Student t_3 and jumps, standard deviation ratios 0.5 to 2, $A = 7800$) the default grid took 1.69 to 2.16 times the bound for deterministic rules and the best single bet 1.34 to 1.76 times: at least 46% of the grid's delay, about half, is unavoidable for deterministic procedures (45% for randomised ones). The floor grows like $\log A/\gamma_Q$ with $\gamma_Q = \mathbb{E}_Q \log (1/\pi(Y)) \geq \mathrm{KL}_{\inf}(Q)$ (Proposition S11).

5.2 Delay bounds and first-order optimality

Theorem 24 (non-asymptotic delay bound). Under the post-change model, for one side of the Shiryaev-Roberts mixture, or of its CUSUM-type version, at level A:

a. for every change time v and every pre-change history,

$$\mathbb{E}[(T_A - v + 1)^+ \mid \mathcal{F}_{v-1}] \;\leq\; \min_{k:\, g_k > 0} \frac{\log (A/w_k) + o_k}{g_k},$$

where $g_k = g(\lambda_k)$, $\Lambda_k = \log (1 + \lambda_k V)$ with V drawn from Q, and $o_k = \min (\sup \Lambda_k, \mathbb{E}[(\Lambda_k^+)^2]/\mathbb{E}[\Lambda_k])$ bounds the mean overshoot; for the two-sided monitor, whose sides alarm at $2A$, replace $\log (A/w_k)$ by $\log (2A/w_k)$;

b. $\mathrm{WADD}(T_A) = \mathbb{E}_Q[T_A^0]$, where T_A^0 is the alarm time of the detector started from $M_0 = 0$ on i.i.d. data from Q: the worst case is a change at the first monitored step.

Part (a) combines Wald's identity with Lorden's (1970) bound on the overshoot for one bet; Shin, Ramdas and Rinaldo (2024) give such bounds for mixtures of exponential e-detectors. It pays the full $\log(1/w_k)$ and ignores the other bets and start times; in simulations of one side at level A it is 1.3 to 2.7 times the mean delay. By part (b), simulations with the change at the first monitored step measure the worst-case delay.

Theorem 25 (first-order asymptotics). Fix Q with $\mathbb{E}_Q[V] > 0$ and an interior maximiser $0 < \lambda^* < \lambda_{\max}$, and treat τ and the baseline as fixed (condition on $\mathcal{F}_0$). As $A \to \infty$: (a) every stopping rule whose average run length is at least A under every i.i.d. return law in the null has $\mathrm{WADD}_Q(T) \geq (\log A/\mathrm{KL}_{\inf}(Q))(1 + o(1))$, where the worst-case delay is evaluated with the pre-change law P^* of Lemma 20, not with the true pre-change law; (b) a fixed grid with $g_{\mathrm{grid}} > 0$ has $\mathrm{WADD}(T_A)/(\log A/\mathrm{KL}_{\inf}) \to \mathrm{KL}_{\inf}/g_{\mathrm{grid}} \geq 1$; (c) $K_A = 1 + \lceil (b' - a)\sqrt{\log A} \rceil$ equally spaced bets on an interval $[a, b'] \ni \lambda^*$ with $b' < \lambda_{\max}$ give $\mathrm{WADD}(T_A) \leq (\log A + \frac{1}{2}\log\log A + O(1))/\mathrm{KL}_{\inf}$; (d) a mixture over bets with a density bounded below near λ^* gives $\mathrm{WADD}(T_A) \leq (\log A + \log\log A + O(1))/\mathrm{KL}_{\inf}$.

Ram and Ramdas (2026) prove a universal lower bound of the form (a) for composite pre- and post-change classes and show that it is attained for bounded means (footnote 1); Theorem 25 is an instance with explicit constants for linear bets, and we claim no priority. A refined grid is first-order optimal over the nonparametric null; the default grid is not, but its ratio $\mathrm{KL}_{\inf}/g_{\mathrm{grid}}$ is 1.00 to 1.08 in our cases. For decreases the constant behind the $O(1)$ in (c) grows without bound as λ^* approaches $\lambda_{\max}$.

5.3 Practical alarm levels and the price of robustness

At $A = 7800$ the first-order formula $\log A/g$ overstates the delay of small changes by up to 112%; a second-order expansion from nonlinear renewal theory (Proposition S20), with a heuristic value of its constant, was within 3% of the simulated mean delay of a single bet in all 20 cases of Table S2 (Supplement S4), and the default grid costs 9% to 26% against the best single bet.

Table 1 splits the first-order delay ratio against the Gaussian likelihood-ratio detector into the three factors of Corollary 21, for clean Gaussian data and $c = 3$.

sd ratio	1.25	1.5	2	0.8	0.67	0.5
truncation loss	1.03	1.08	1.27	1.00	1.00	1.00
nonparametric loss	1.10	1.23	1.48	1.25	2.00	3.61
grid loss	1.02	1.02	1.04	1.02	1.05	1.05
predicted penalty $\mathrm{KL}/g_{\mathrm{grid}}$	1.16	1.37	1.95	1.27	2.09	3.79
simulated penalty against the Gaussian LR mixture	1.14	1.35	1.73	1.48	2.19	3.45

Table 1. The price of robustness on clean Gaussian data, $c = 3$; simulated data. Penalties are ratios of mean delays with the change at the first step, 3000 paths with common random numbers.

The predicted penalty agrees to within about 15% with the 10% to 70% extra delay measured in Section 6.2. The largest factor is the nonparametric loss, not truncation; for decreases it is the speed limit of Corollary 22. The robust detector keeps 53% to 88% of the asymptotic speed of the Gaussian likelihood-ratio detector for increases and 28% to 80% for decreases, and under Student t_3 returns 94% to 98% of the speed of the correctly specified detector for increases, while its guarantee covers every return law.

The truncation constant. For increases on clean data $c = 3$ costs 3% to 12% of delay against no truncation and keeps a burst of jumps from looking like a volatility increase; a separate decrease clip $c_{\downarrow} = 2$ cut median decrease delays by 13% to 51%, but was slower at a coarse price tick (Supplement S5).

6 Simulations

All results in this section are from simulated data, with settings in Supplement S8. Unless stated otherwise the monitor uses $c = 3$, the default grids and $A = 7800$, so that with a known baseline Theorem 4 bounds by 5% the probability of a false alarm within the first 390-step day. “Jumps” adds to Gaussian returns, with probability 0.005 per step, a normal jump with 8 times the diffusive standard deviation (10 times with the true baseline); Student t_ν returns have unit variance; “price ticks” round Gaussian returns to multiples of half a standard deviation (about 20% zero returns).

6.1 Validity

Table 2 compares the robust monitor with an e-detector on raw squared returns, a Shiryaev-Roberts mixture of Gaussian likelihood ratios (standard deviation ratios 1.25, 1.5 and 2), Page’s (1954) CUSUM on r_t^2 calibrated by simulation to a run length of about 7800 on Gaussian data, and a z-test of the variance over the last 60 steps at level 0.05, repeated at every step. Every baseline is estimated from 2000 calibration steps of the same law, so jumps and fat tails are part of “normal”.

Detector	Gaussian	jumps	Student t_5	price noise	slow SV
Robust e-detector, increase	1.7%	2.3%	0.7%	3.7%	26.3%
Robust e-detector, decrease	0.7%	0.7%	0.3%	0.3%	27.0%
E-detector on raw r_t^2, increase	1.7%	2.3%	1.3%	3.0%	26.3%
Gaussian likelihood-ratio e-detector, increase	2.7%	**70.7%**	**96.3%**	6.3%	29.0%
Page CUSUM, calibrated on Gaussian data	5.0%	**70.3%**	**97.3%**	9.7%	32.3%

Detector	Gaussian	jumps	Student t_5	price noise	slow SV
Repeated 60-step z-test	**70.7%**	**90.0%**	**100%**	**78.0%**	**95.0%**

Table 2. Simulated data. Share of paths with a false alarm within one 390-step day, no change, plug-in baselines from 2000 steps, 300 paths per cell (Monte Carlo standard error about 0.8 points at 2% and 1.3 points at 5%). "Price noise" adds i.i.d. noise with standard deviation 0.5 per-step return standard deviations to log prices; "slow SV" is a log-normal stochastic volatility with log-sd 0.15 and half-life 390 steps.

Where the null holds (first three columns) all e-detectors stayed below 5%, including the increase detector on raw squared returns, as Section 3.3 predicts. Calibrated on Gaussian data, the Gaussian likelihood-ratio and CUSUM detectors failed under jumps or fat tails, and the repeated z-test everywhere. In the last two columns the null is false: price noise makes the conditional mean of r_t^2 depend on the previous return, and slow stochastic volatility moves it.

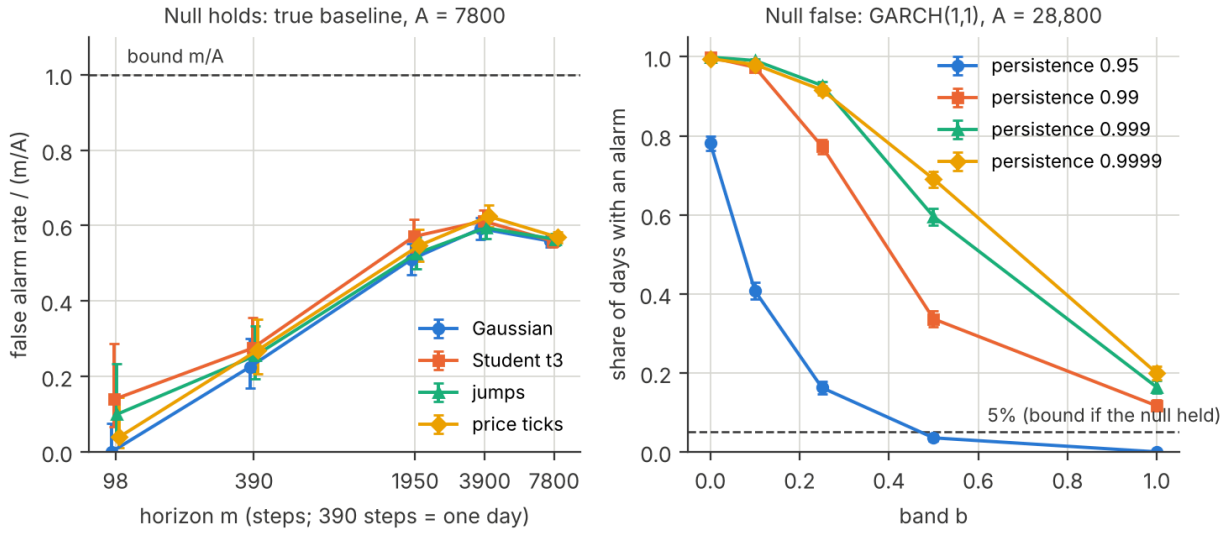


Figure 1. Simulated data. Left: false alarm rate within m steps divided by the bound m/A, for four return laws with the true baseline, $A = 7800$, with 95% intervals. Right: share of 1440-step days with an alarm under GARCH(1,1) clustering, where the null is false, as a function of the band b, $A = 28,800$, with 95% intervals; the dashed line is the 5% that Theorem 4 would give if the null held.

With the true baseline, to which the theorems apply, the bound held for every law, with almost the same rates for all four (Figure 1, left; Table S4): 1.1% to 1.4% at one day against 5%, and 55.9% to 56.9% at $m = A$. The bound is conservative by a factor of about 4 at one day and 2 at $m = A$, mostly because of expected wealth stranded on paths that never alarm, but for a single bet of 0.5 a law found by dynamic programming over the whole nonparametric null reaches 95% of it (Supplement S5).

An adversarial search. The worst null stream found, which watches the detector's own state, alarmed within one day on 2.62% of paths (95% interval 2.50% to 2.75%) against the bound of 5%, and within five days on 18.1% against 25%; outside the class each failure matched the broken condition (companion note).

6.2 Delay against parametric and robust competitors

With the designs of Table 2 and a change 500 steps into monitoring (Table S5), the Gaussian likelihood-ratio e-detector detected increases of the standard deviation by 25%, 50% and 100% in a median of 78, 26 and 10 steps on clean Gaussian data, against 87, 35 and 17 for the robust detector, an extra delay of 10% to 70%, close to Table 1, but under jumps or fat tails the parametric detectors alarmed before the change on 74% to 99% of paths. Truncation matters under jumps (Table S5). For decreases the Gaussian detector was faster, at 3% to 9% false alarms before the change, but this rests on normality: real one-minute returns contain many exact zeros (5.7% of Bitstamp BTC/USD minutes; 5.4% when the provider's filled minutes are treated as missing), which a Gaussian likelihood reads as low volatility.

BOCPD and a two-window scan. Bayesian online change-point detection (BOCPD; Adams and MacKay 2007) with a Gaussian model, on raw or clipped returns, and a two-window variance-ratio scan inspired by Bibinger, Jirak and Vetter (2017) but of our own construction (footnote 1; Supplement S5) were matched to the e-detector's first-day false alarm rate on clean Gaussian paths. BOCPD then alarmed within the first day on 73.2% of paths with jumps and 53.2% with Student t_5 returns (e-detector 1.3% and 1.2%); clipping fixed jumps (1.6%) but not fat tails (7.5%). On clean Gaussian data BOCPD was faster for large changes (median 26 against 81 steps for a halving of the standard deviation) but detected fewer small changes.

Competitors calibrated for the data at hand. We also ran classical rules on the e-detector's own statistic, and a Student t BOCPD and CUSUM monitoring rules with a training sample of the type of Chu, Stinchcombe and White (1996) and Horváth, Kokoszka and Zhang (2006), not reproductions (footnote 1); the last two were slower or missed more small changes (Supplement S5). At its first-day Gaussian rate (1.3%), the e-detector's own betting statistic with a simulated threshold in place of $2A$ was slightly slower than the e-detector, and Page's CUSUM on X was as fast for increases and 15% to 40% faster for decreases, where validity caps the e-detector's bets. But calibrated thresholds hold only for their calibration law: Gaussian-calibrated CUSUMs on X had 5.0% to 8.2% first-day false alarms under Student t returns, and per-path bootstrap thresholds were significantly above target in 5 of 24 cells. At a matched rate, Ville's threshold costs this statistic little delay and adds a guarantee that needs no calibration.

6.3 The cost of an estimated baseline

With $\delta = 0.01$ per side and $A = 9750$, the confidence-sequence monitor of Theorem 16 had at most 0.27% false alarms in the first day in nine settings without a change (Supplement S5), and its bound failed on at most 0.08% of paths against an allowed 1%; the plug-in was at 4.5% to 4.9% on the increase side at 390 calibration steps. The price is delay: 1.8 to 7.6 times the plug-in's median delay at 390 calibration steps, 1.4 to 3.0 times at 1440 and 1.2 to 1.7 times at 5000, because the confidence bound is wide when calibration is short (Table S6; Supplement S5).

6.4 When the null is false: clustering, forecasts and price noise

Under GARCH(1,1) returns with constant unconditional variance and the true unconditional baseline, the monitor alarmed on 78% of 1440-step days at persistence 0.95 and on 100% at 0.99 or more; a band $b = 0.5$ absorbs short-lived clustering, but even $b = 1$ leaves 12% to 20% of days with an alarm when the half-life is an hour or more (Figure 1, right). Dividing returns by a frozen GARCH(1,1) forecast (Bollerslev 1986) brought the

one-day rate to 3.0% to 6.2%, but at persistence 0.99 a doubling of the long-run variance was detected within a day on only 34% to 43% of paths: with GARCH parameters (ω, α, β), no break of ω lifts the ratio of the mean squared return to the mean forecast above $(1-\beta)/\alpha$ (Proposition S13). Price-level noise also breaks the null, but not for a statistic pre-averaged over non-overlapping blocks of prices, to which Theorems 3 and 4 apply (Supplement S6).

6.5 Extensions

A null without a baseline. The *rank detector* bets $f(p_t)$ on the smoothed conformal p-value p_t of Vovk, Nouretdinov and Gammerman (2003), the share of $|r_1|, \ldots, |r_t|$ above $|r_t|$ with ties split at random, where $f \geq 0$ and $\int_0^1 f = 1$. For exchangeable returns the p_t are i.i.d. uniform, so Lemma 2 holds in their filtration and Theorems 3 and 4 hold with no baseline, no τ and no δ; for i.i.d. returns, and for every infinite exchangeable sequence, they hold also when the detector restarts at once after an alarm with an empty reference set (Supplement S1.9). Its null, exchangeability, neither implies nor is implied by the conditional-mean null: it needs no baseline, but it excludes all time dependence.

Other extensions. A running truncation level with a band for all levels, and restarts that learn their band while they monitor, keep a proof for i.i.d. returns (Supplement S1.9). Restarts remove the blind period: where volatility moved about once a day, the rank restart detected 99.6% of Gaussian changes within a day, against 56% to 66% for monitors that recalibrate. The companion note also treats many assets, with e-BH (Wang and Ramdas 2022), and several sampling frequencies.

7 Application

7.1 Data, design and what real data can test

Data. Binance spot one-minute bars for BTC, ETH, XRP, SOL and DOGE against USDT, with no missing minutes: 1 July to 31 December 2025 (264,959 returns per coin) and, out of sample, 21 January 2024 to 30 June 2025 (527 days), deseasonalised as in Section 2. The monitor uses $c = 3$ and the band $b = 0.25$, each side alarming at $2A$, and recalibrates on one day of data after every alarm, with $A = 28,800$ for the plug-in baseline and $\delta = 0.01$ and $A = 36,000$ for the confidence-sequence baseline (5% for the first 1440-minute day). The band tolerates slow drift of the conditional variance: under slow log-normal stochastic volatility it cut the first-day alarm rates from 24.7% and 26.7% to 8.3% (still above 5%) and 5.0%, and raised the median delay to a doubling or a halving of volatility from 19 and 74 steps to 28 and 111 (Supplement S6). The settings were fixed after one run on Bitstamp BTC/USD for July to December 2025, so only January to June 2025 is fully held out.

What the real data test. Four pre-registered analyses use proved monitors, each specified in a document committed before any of its real-data runs (later decisions are called post hoc): the null streams below, on which the null holds by construction, the confidence-sequence configuration of Section 7.2, the rerun of the event test with a daily restart, and the same test on the releases of 2026, the only one on data not seen before (Section 7.3). All other real-data analyses use the plug-in baseline, whose level is measured (Sections 4.1 and 6.1), not proved, and show how the monitor behaves as a tool. Of these, the FOMC and CPI test on BTC (Section 7.3) is primary, fixed before the

first run in analysis code first saved with its results, and was not confirmed on the releases of 2026; the Nifty 50 and Budget analyses, the forecast regressions and the comparisons with classical rules are secondary (described in advance); the rest is exploratory. Secondary results carry 95% intervals from a block bootstrap over calendar days and p-values adjusted by Holm's (1979) method within families (Supplement S7.4).

Null streams built from real returns. Real data cannot show where the null holds, but streams built from real returns can make it hold; they check the bounds and their slack under the real marginal law of returns, which no simulation from a parametric law does. The primary endpoint uses 20,000 streams that permute the returns of the calm days of July to December 2025, pooled over the five coins (days whose robust volatility is near its median over the previous 20 days; Supplement S7.6); they keep the real zeros, ticks and heavy tails and remove all time dependence, so the null holds exactly for the rank detector and, for the other monitors, up to the small drift of a finite pool. At band 0 the first-day false alarm probability was 2.26% (95% interval 2.06% to 2.48%) with the known baseline, 0.01% (0.00% to 0.04%) with the confidence-sequence baseline and 3.38% (3.14% to 3.64%) for the rank detector, against 5%, and no interval of these three lay above its bound in any of 252 cells. The plug-in gave 5.50% (5.19% to 5.82%), and 41% within five days against 25%. Streams that bootstrap blocks of calm days, keeping their real order, made every monitor alarm within a day on 86% to 100% of streams: real minute-level data rarely satisfy the null, so real alarm rates are not false alarm rates.

7.2 Five cryptocurrencies

The confidence-sequence baseline (pre-registered). In the primary configuration (July to December 2025, band 0.25), one of the real-data analyses with a proved monitor (Section 7.1), under the tolerance of Theorem 16, it raised 0.51 to 0.53 alarms per day per coin, against 0.66 to 0.70 for the plug-in. These data had already been analysed with the plug-in, so the pre-registration fixed the configuration, not the novelty of the data. Of its up alarms 78.2% were followed by higher volatility, against 79.6% for the plug-in (difference −1.3 points, day-block bootstrap interval −5.8 to 3.2), and of its down alarms 96.3% by lower volatility, against 88.3% (+8.0, 2.7 to 13.7); out of sample, 80.8% and 90.9% against 77.9% and 84.5%. Its alarms came later, it monitored more of the time, and in 5 of 7031 segments, all on BTC, it could not raise an increase alarm, for up to 136 days (post hoc; Supplement S7.6).

The plug-in baseline (applied). With the plug-in baseline the monitor raised 0.66 to 0.71 alarms per day for every coin in both periods. Of its increase alarms, 75% to 84% were followed by a day whose robust volatility over the next 1440 minutes was above the calibration baseline, and of its decrease alarms 83% to 93% by a day below it, against about 50% at random times with the same delay (pooled differences 27 and 38 points in July to December 2025, 95% intervals 22 to 33 and 34 to 44); out of sample the shares fell by 1 to 4 points.

Forecasts. The outcome above partly measures the persistence of volatility, so at each alarm we also forecast the robust volatility of the next 1440 minutes (EWMA with a half-life of one day, or HAR; Corsi 2009). In July to December 2025 the next day was on the flagged side of the EWMA forecast after 78.5% of the plug-in monitor's up alarms and 80.5% of its down alarms, against 48.7% to 51.3% at random times (Holm $p \le 0.0016$); ratio rules on the last 60 or 240 minutes, tuned with hindsight to the same alarm counts, reached 73.9% to 82.3%, and only 1 of 48 comparisons with them held after Holm (Table

S8; Supplement S7.3). In hourly regressions the alarms added little to a HAR forecast that includes the last hour's realised variance (Supplement S7.4): such a forecast mostly knows that volatility has just moved.

Competitors at matched alarm counts. Against BOCPD and the two-window scan of Section 6.2, the e-detector's alarms were confirmed more often in 15 of 16 comparisons with windows fixed before the first run (Holm-adjusted $p \le 0.021$), but not significantly more often than the decrease alarms of competitors with day-long windows, added later. Classical rules on the same statistic, matched in the same way, did as well: the calibrated betting rule and both CUSUMs on X were never significantly less precise in 24 comparisons (unadjusted p-values; Supplement S7.3). So the advantage over BOCPD and the scan may come from the clipped statistic and the restart design, and the e-detector is not shown to be more precise than a well-calibrated classical rule.

7.3 Scheduled announcements

FOMC and CPI: the pre-specified primary test. For BTC, ETH and SOL in 2024 and 2025, with 16 FOMC statements, 22 CPI releases and control windows at the same weekday and clock time, the primary family (BTC, the streaming monitor, Holm over FOMC and CPI) found an increase alarm within six hours after 9 of 16 FOMC statements against 14% of matched controls (Holm $p = 0.00025$); the CPI test was not significant on BTC ($p = 0.11$) but was on ETH and SOL, as secondary replications. This test used the plug-in baseline, which has no finite-sample guarantee. Its contrast was not confirmed on the releases of 2026 (below).

The rerun with a proved monitor (pre-registered). Before any of its real-data runs, but with the result above known, we pre-registered the same test on the same releases with the confidence-sequence baseline restarted at every UTC midnight, whose false alarm probability is at most $\delta + m/A = 5\%$ on each day of a segment, marginally, while the tolerance holds during that day (Corollary S14). The primary test passed: an up alarm came within six hours of 8 of 16 FOMC statements on BTC, against 10.9% of matched controls (Holm $p = 0.00033$); CPI on BTC, a secondary hypothesis, passed but is fragile (Supplement S7.2). Yet this monitor raised an alarm on 48.1% to 53.5% of the UTC days on which it monitored at least one minute, against the 5% per day that it allows under the tolerance: real one-minute returns often break the tolerance, so the rerun shows a monitor that carries the guarantee flagging the releases of 2024 and 2025, not the guarantee at work.

The test on fresh releases (pre-registered). Because the rerun was designed on data already seen, we pre-registered one more test before loading any return of 2026: the primary monitor of the rerun on BTC, with one stratified test over the FOMC statements and CPI releases of January to 24 October 2026 together (6 and 10), chosen for its power (0.82 at the rates of the rerun, 0.78 for the 16 releases in the data). It failed: an up alarm came within six hours of 2 of the 16 releases, against 5.9% of matched controls ($p = 0.25$; 95% interval for the release rate 3.5% to 36.0%), and of 1 of the 6 FOMC statements, against 10.3%. No secondary family of this monitor passed after Holm, and no monitor showed a CPI contrast on any coin. On BTC the monitor was recalibrating or unable to raise an increase alarm at 5 of the 16 releases, in one segment for 47 days. The rank detector, which is never blind, flagged 4 of the 6 FOMC statements on each coin (Holm $p = 0.016$, 0.033 and 0.055), a secondary result. The FOMC contrast of 2024 and 2025 is therefore exploratory: the only confirmatory test on releases not seen before did not

confirm it, and with power 0.78 this says that the contrast was smaller in 2026 or absent, not that there is none (Supplement S7.2). A guard that discards a near-blind calibration window and calibrates again, with the error budget split over the windows tried, keeps the bounds of Section 4.2 (Supplement S7.6); on BTC it removed all blind time, but on these data, now seen, it raised the count only to 4 of 16 ($p = 0.056$, post hoc). A forward test of the rank detector on the BTC releases up to 1 June 2027 is pre-registered. So is an anytime-valid version of it (Supplement S7.7): each release, with control windows within four weeks that no other release's windows overlap, contributes a likelihood-ratio e-value, and the running product over releases is a test martingale if each release window is exchangeable with its controls given the earlier releases, so it may be checked after every release and continued without an end date. At the planning rates it reaches 20 by 1 June 2027 with probability 0.43, against power 0.82 for the fixed-date test, and by 1 January 2028 with probability 0.82; on the releases of 2024 and 2025, already seen, it would have reached 20 in June 2024.

7.4 Other data, and what the real data say about the method

On the Nifty 50 index (2,361 sessions from April 2017 to October 2026; secondary, with a configuration pre-registered on 2017 to 2021), 78% of the held-out increase alarms and 76% of the decrease alarms were followed by a session on the flagged side of the baseline, against 47% and 53% at random times (Holm $p = 0.0008$). In a secondary event analysis, Union Budget sessions were followed by alarms far more often than ordinary sessions (9 of 11 against 25%, Holm $p = 0.0006$ over five event types); Reserve Bank of India policy days showed no significant excess (Supplement S7.1).

Two lessons qualify the simulations. With the plug-in the coins raised 2.0 to 2.35 alarms per monitored day (July to December 2025, band 0.25), far above the budget of 0.05 false alarms per monitored day of Corollary 6, because within real calm days the conditional variance moves (Section 7.1). And a monitor that recalibrates for a day after every alarm is blind most of the time when alarms come every 1.4 days (about 70% of the time in the event study); at the four landmark shocks of Supplement S7.5, fixed in advance, it was recalibrating at all fifteen crypto cases, which restarts without a blind period address (Section 6.5).

8 Discussion

What is new and what is not. The probabilistic tools (Ville's inequality, test supermartingales, Shiryaev-Roberts e-detectors, empirical Bernstein confidence sequences, conformal p-values, the Bahadur-Savage phenomenon, Lorden's and Lai's lower bounds) are known, and Theorems 3, 4 and 25 and Proposition 10 are known results or direct instances of them. To our knowledge, and subject to the caveat in footnote 1, what we add is their combination for jump-robust online volatility monitoring, the reading over many days and the daily restart, the drift bound, the plug-in regime, the bounded statistic with a speed limit for every deterministic valid rule, a confidence-sequence baseline with a null that does not depend on the algorithm, the decomposition of the delay cost, and null streams built from real returns. On real data the hypotheses of the proofs fail on many days, and the only confirmatory test on releases not seen before failed (Section 7.3), so the evidence for the guarantees comes from null streams built from real returns, and the real-data analyses describe how the monitor behaves as a tool.

Limitations.

- *The null is a statement about conditional means.* Clustering, price-level noise, estimated baselines, errors in the intraday profile (which no experiment isolates), a change of tail shape at constant volatility and a leverage effect all move it without a structural change, and on real calm days it fails within a day. Theorem 9 bounds a small drift; the remedies (a band, forecast-relative monitoring, non-overlapping blocks, the confidence-sequence baseline) cost speed.
- *Per-period guarantees.* Over many days the plain detector's guarantee is an expected count; a bound for every day needs the restart of Theorem 8, proved for a known baseline and, marginally, for the confidence-sequence baseline (Corollary S14), at a cost in delay.
- *Estimated baselines.* The plug-in keeps its level only asymptotically, for horizons $m = o(\sqrt{n})$; the confidence-sequence baseline needs long calibration to be fast, can be blind to increases after an unusually quiet start of its window (on BTC in 2026, for 47 days, at two releases), and was run only with a quarter of the window fixing τ; a guard proved in Supplement S7.6 removes the blind segments, at the cost of a fresh calibration where a window is discarded.
- *Speed and precision.* Classical rules on the same statistic, calibrated for the law at hand, were as fast for increases, faster for decreases and as precise on real data; BOCPD found large decreases two to three times faster on clean Gaussian data; against a forecast that sees the last hour the alarms added little.
- *Delay theory.* It assumes i.i.d. post-change data and a fixed baseline; the delay floor covers a change at the first step and is a Monte Carlo estimate; the constant of the second-order expansion is heuristic; Theorem 25 is asymptotic.
- *Data and benchmarks.* One-minute bars from public mirrors, no tick data and no ground truth; apart from the null streams, one confidence-sequence configuration and the two pre-registered event tests, the real-data analyses use the plug-in baseline, whose level is measured, not proved; the only confirmatory test on data not seen before, on the FOMC and CPI releases of 2026, failed (Section 7.3); the training-sample rules and the robust BOCPD are of the type of the published procedures, not reproductions (footnote 1).

Open problems. (1) A certified threshold for mixtures sharper than m/A: the worst stream found reaches about half of the bound at one day, but the best proved bound is barely below m/A (companion note, Section 4). (2) A running truncation level and a restart without a blind period that are as fast as the plug-in. (3) A narrower baseline, for example from a betting confidence sequence. (4) Theory under clustering: a bound for the mean of the ratio $\mathbb{E}[r_t^2/\hat{\sigma}_t^2]$ that the monitor sees, and a principled band. (5) Monitoring the truncated variance and the truncated quarticity under one false alarm budget.

Reproducibility

The code, data manifests and result files are in the accompanying repository, and all simulations use fixed seeds. Supplement S8 gives the settings and Supplement S9 the script and result file behind every table and figure. The real data are one-minute bars from public mirrors of exchange files (Binance; Bitstamp for one check) and a public archive of Nifty 50 bars, with event dates checked against published schedules. A script traces every number of the paper to a committed source.

References

Issue numbers and DOIs are given where we could confirm them.

Adams, R. P. and MacKay, D. J. C. (2007). Bayesian online changepoint detection. arXiv:0710.3742.

Andersen, T. G. and Bollerslev, T. (1997). Intraday periodicity and volatility persistence in financial markets. *Journal of Empirical Finance* 4(2-3), 115-158. doi:10.1016/S0927-5398(97)00004-2

Andersen, T. G., Dobrev, D. and Schaumburg, E. (2012). Jump-robust volatility estimation using nearest neighbor truncation. *Journal of Econometrics* 169(1), 75-93. doi:10.1016/j.jeconom.2012.01.011

Andersen, T. G., Todorov, V. and Zhou, B. (2025). Real-time detection of local no-arbitrage violations. *Quantitative Economics* 16(2), 459-495. doi:10.3982/QE2585

Bahadur, R. R. and Savage, L. J. (1956). The nonexistence of certain statistical procedures in nonparametric problems. *Annals of Mathematical Statistics* 27(4), 1115-1122. doi:10.1214/aoms/1177728077

Bapat, S. R. (2026). A jump and noise robust kernel based estimator of quarticity for high frequency data. *International Review of Economics and Finance* 109, 105482. doi:10.1016/j.iref.2026.105482

Barndorff-Nielsen, O. E. and Shephard, N. (2004). Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics* 2(1), 1-37. doi:10.1093/jjfinec/nbh001

Berkes, I., Gombay, E., Horváth, L. and Kokoszka, P. (2004). Sequential change-point detection in GARCH(p,q) models. *Econometric Theory* 20(6), 1140-1167.

Bibinger, M., Jirak, M. and Vetter, M. (2017). Nonparametric change-point analysis of volatility. *Annals of Statistics* 45(4), 1542-1578. doi:10.1214/16-AOS1499

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics* 31(3), 307-327. doi:10.1016/0304-4076(86)90063-1

Boudt, K., Croux, C. and Laurent, S. (2011). Robust estimation of intraweek periodicity in volatility and jump detection. *Journal of Empirical Finance* 18(2), 353-367. doi:10.1016/j.jempfin.2010.11.005

Choe, Y. J. and Ramdas, A. (2024). Combining evidence across filtrations. arXiv:2402.09698.

Chu, C.-S. J., Stinchcombe, M. and White, H. (1996). Monitoring structural change. *Econometrica* 64(5), 1045-1065.

Clerico, E. (2025). On the optimality of coin-betting for mean estimation. *International Journal of Approximate Reasoning* 187, 109550. doi:10.1016/j.ijar.2025.109550

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics* 7(2), 174-196. doi:10.1093/jjfinec/nbp001

Dandapanthula, S. and Ramdas, A. (2025). Multiple testing in multi-stream sequential change detection. arXiv:2501.04130.

Dette, H. and Gösmann, J. (2020). A likelihood ratio approach to sequential change point detection for a general class of parameters. *Journal of the American Statistical Association* 115(531), 1361-1377. doi:10.1080/01621459.2019.1630562

Diebold, F. X. (1986). Modeling the persistence of conditional variances: a comment. *Econometric Reviews* 5(1), 51-56.

Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business and Economic Statistics* 13(3), 253-263.

Driscoll, J. C. and Kraay, A. C. (1998). Consistent covariance matrix estimation with spatially dependent panel data. *Review of Economics and Statistics* 80(4), 549-560. doi:10.1162/003465398557825

Dufresne, D. (1990). The distribution of a perpetuity, with applications to risk theory and pension funding. *Scandinavian Actuarial Journal* 1990(1), 39-79.

Dvoretzky, A., Kiefer, J. and Wolfowitz, J. (1956). Asymptotic minimax character of the sample distribution function and of the classical multinomial estimator. *Annals of Mathematical Statistics* 27(3), 642-669. doi:10.1214/aoms/1177728174

Fan, X., Grama, I. and Liu, Q. (2015). Exponential inequalities for martingales with applications. *Electronic Journal of Probability* 20, paper 1, 1-22. doi:10.1214/EJP.v20-3496

Fearnhead, P. and Rigaill, G. (2019). Changepoint detection in the presence of outliers. *Journal of the American Statistical Association* 114(525), 169-183. doi:10.1080/01621459.2017.1385466

Grünwald, P., de Heide, R. and Koolen, W. (2024). Safe testing. *Journal of the Royal Statistical Society Series B* 86(5), 1091-1128. doi:10.1093/jrsssb/qkae011

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics* 6(2), 65-70.

Honda, J. and Takemura, A. (2010). An asymptotically optimal bandit algorithm for bounded support models. *Proceedings of the 23rd Conference on Learning Theory (COLT 2010)*, 67-79.

Horváth, L., Hušková, M., Kokoszka, P. and Steinebach, J. (2004). Monitoring changes in linear models. *Journal of Statistical Planning and Inference* 126, 225-251.

Horváth, L., Kokoszka, P. and Zhang, A. (2006). Monitoring constancy of variance in conditionally heteroskedastic time series. *Econometric Theory* 22(3), 373-402.

Horváth, L., Trapani, L. and Wang, S. (2025). Sequential monitoring for changes in GARCH(1,1) models without assuming stationarity. *Journal of Time Series Analysis* 46(5), 981-996. doi:10.1111/jtsa.12824

Howard, S. R., Ramdas, A., McAuliffe, J. and Sekhon, J. (2021). Time-uniform, nonparametric, nonasymptotic confidence sequences. *Annals of Statistics* 49(2), 1055-1080. doi:10.1214/20-AOS1991

Huber, P. J. (1965). A robust version of the probability ratio test. *Annals of Mathematical Statistics* 36(6), 1753-1758. doi:10.1214/aoms/1177699803

Inclán, C. and Tiao, G. C. (1994). Use of cumulative sums of squares for retrospective detection of changes of variance. *Journal of the American Statistical Association* 89(427), 913-923. doi:10.1080/01621459.1994.10476824

Knoblauch, J., Jewson, J. and Damoulas, T. (2018). Doubly robust Bayesian inference for non-stationary streaming data with β-divergences. *Advances in Neural Information Processing Systems* 31.

Lai, T. L. (1998). Information bounds and quick detection of parameter changes in stochastic systems. *IEEE Transactions on Information Theory* 44(7), 2917-2929. doi:10.1109/18.737522

Lamoureux, C. G. and Lastrapes, W. D. (1990). Persistence in variance, structural change, and the GARCH model. *Journal of Business and Economic Statistics* 8(2), 225-234.

Larsson, M., Ramdas, A. and Ruf, J. (2024). The numeraire e-variable and reverse information projection. arXiv:2402.18810.

Lorden, G. (1970). On excess over the boundary. *Annals of Mathematical Statistics* 41(2), 520-527. doi:10.1214/aoms/1177697092

Lorden, G. (1971). Procedures for reacting to a change in distribution. *Annals of Mathematical Statistics* 42(6), 1897-1908. doi:10.1214/aoms/1177693055

Mancini, C. (2009). Non-parametric threshold estimation for models with stochastic diffusion coefficient and jumps. *Scandinavian Journal of Statistics* 36, 270-296.

Massart, P. (1990). The tight constant in the Dvoretzky-Kiefer-Wolfowitz inequality. *Annals of Probability* 18(3), 1269-1283. doi:10.1214/aop/1176990746

Mei, Y. (2008). Is average run length to false alarm always an informative criterion? *Sequential Analysis* 27(4).

Mikosch, T. and Starica, C. (2004). Nonstationarities in financial time series, the long-range dependence, and the IGARCH effects. *Review of Economics and Statistics* 86(1), 378-390. doi:10.1162/003465304323023886

Mineiro, P. (2022). A lower confidence sequence for the changing mean of non-negative right heavy-tailed observations with bounded mean. arXiv:2210.11133.

Page, E. S. (1954). Continuous inspection schemes. *Biometrika* 41(1-2), 100-115. doi:10.1093/biomet/41.1-2.100

Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics* 160(1), 246-256. doi:10.1016/j.jeconom.2010.03.034

Pollak, M. (1985). Optimal detection of a change in distribution. *Annals of Statistics* 13(1), 206-227. doi:10.1214/aos/1176346587

Pollak, M. (1987). Average run lengths of an optimal method of detecting a change in distribution. *Annals of Statistics* 15(2), 749-779. doi:10.1214/aos/1176350373

Ram, A. and Ramdas, A. (2026). Asymptotically optimal sequential change detection for bounded means. arXiv:2602.05272.

Ramdas, A., Grünwald, P., Vovk, V. and Shafer, G. (2023). Game-theoretic statistics and safe anytime-valid inference. *Statistical Science* 38(4), 576-601. doi:10.1214/23-STS894

Roberts, S. W. (1966). A comparison of some control chart procedures. *Technometrics* 8(3), 411-430.

Shafer, G. (2021). Testing by betting: a strategy for statistical and scientific communication. *Journal of the Royal Statistical Society Series A* 184(2), 407-431. doi:10.1111/rssa.12647

Shekhar, S. and Ramdas, A. (2024). Reducing sequential change detection to sequential estimation. *Proceedings of the International Conference on Machine Learning (ICML 2024)*, PMLR 235, 44628-44642.

Shin, J., Ramdas, A. and Rinaldo, A. (2024). E-detectors: a nonparametric framework for sequential change detection. *New England Journal of Statistics in Data Science* 2(2), 229-260. doi:10.51387/23-NEJSDS51

Shiryaev, A. N. (1963). On optimum methods in quickest detection problems. *Theory of Probability and Its Applications* 8(1), 22-46. doi:10.1137/1108002

Tartakovsky, A., Nikiforov, I. and Basseville, M. (2014). *Sequential Analysis: Hypothesis Testing and Changepoint Detection*. Chapman and Hall/CRC.

Unnikrishnan, J., Veeravalli, V. V. and Meyn, S. (2009). Minimax robust quickest change detection. arXiv:0911.2551.

Ville, J. (1939). *Étude critique de la notion de collectif*. Thèse, Paris.

Vovk, V., Nouretdinov, I. and Gammerman, A. (2003). Testing exchangeability on-line. *Proceedings of the 20th International Conference on Machine Learning (ICML 2003)*, 768-775.

Vovk, V., Petej, I., Nouretdinov, I., Ahlberg, E., Carlsson, L. and Gammerman, A. (2021). Retrain or not retrain: conformal test martingales for change-point detection. *Proceedings of Machine Learning Research* 152, 191-210.

Wang, H. and Ramdas, A. (2023). Huber-robust confidence sequences. *Proceedings of the International Conference on Artificial Intelligence and Statistics (AISTATS 2023)*, PMLR 206, 9662-9679.

Wang, Q., Wang, R. and Ziegel, J. (2026). E-backtesting. *Management Science* 72(6), 4952-4973. doi:10.1287/mnsc.2023.01659

Wang, R. and Ramdas, A. (2022). False discovery rate control with e-values. *Journal of the Royal Statistical Society Series B* 84(3), 822-852. doi:10.1111/rssb.12489

Waudby-Smith, I. and Ramdas, A. (2024). Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society Series B* 86(1), 1-27. doi:10.1093/jrsssb/qkad009

Woodroofe, M. (1982). *Nonlinear Renewal Theory in Sequential Analysis*. CBMS-NSF Regional Conference Series in Applied Mathematics 39. SIAM, Philadelphia.

Zeileis, A., Leisch, F., Hornik, K. and Kleiber, C. (2002). strucchange: An R package for testing for structural change in linear regression models. *Journal of Statistical Software* 7(2), 1-38. doi:10.18637/jss.v007.i02

Supplementary material

The supplement holds the proofs of the results of the main text, the details of several results that the main text summarises, the settings of every experiment and a map from each table and figure to its script and result file. Section, table and figure numbers refer to the main text unless they start with S. Supplement S1.9 proves the results behind the rank detector, the running truncation level and the restarts of Section 6.5. A companion note, `paper/extensions.md`, reports the experiments on these extensions, treats the others summarised in Section 6.5 and the adversarial search of Section 6.1.

S1. Proofs and details for Sections 2 and 3

S1.1 Corollary 6: proof and numbers

Let W_t be the sum of $\bar{M}$ at all past alarms, plus the current $\bar{M}$, minus the number of monitored steps so far. By Lemma 2 applied in each segment, W is a supermartingale with $W_0 = 0$: at an alarm the statistic moves into the sum unchanged, and calibration steps leave W unchanged. Each alarm adds at least A to the sum. Optional stopping at the bounded time $\vartheta \wedge n$, with ϑ the time of the m-th monitored step, and monotone convergence as $n \to \infty$ give $A\,\mathbb{E}[N_m] - m \le 0$. ▫

Numbers. With the true baseline and $A = 300$ the restarted monitor raised 26.5 alarms per 10,000 monitored steps against the bound of 33.3; with the confidence-sequence baseline and $A = 36,000$ it raised 0.000 (Gaussian) and 0.001 (Student t_3) false alarms per 1440-step day in 30-day streams, against a bound of 0.041. At operating levels with a known baseline, the monitor restarted after every alarm raised 0.037 to 0.048 false alarms per day over 20 days at $A = 7800$ and $28,800$ (Supplement S1.7), and 0.040 per monitored day on null streams built from real returns (Supplement S7.6), against the 0.05 of Corollary 6.

S1.2 Theorem 9: proof, design numbers and sharpness

For one bet write $R_t = f_t(R_{t-1} + 1)$ with $R_0 = 0$, so that $M_t = \sum_k w_k R_t^{(k)}$ on each side.

Lemma S1 (expected wealth under a drift). Let $f_t \ge 0$ be adapted and bounded, and let $h_1, h_2, \ldots$ be numbers at least 1 with $\mathbb{E}[f_t \mid \mathcal{F}_{t-1}] \le h_t$ for every t. Then for every stopping time T (possibly infinite) and every $t \ge 0$, $\mathbb{E}[R_{T \wedge t}] \le s_t := \sum_{j=1}^{t} \prod_{i=j}^{t} h_i$. With a constant $h_i = 1 + x$, $s_t = t\,Y_t(x)$.

Proof. Put $a_t = \mathbb{E}[R_{T \wedge t}]$, which is finite because the factors are bounded. Split $R_{T \wedge t} = R_T 1\{T \le t-1\} + R_t 1\{T > t-1\}$. The event $\{T > t-1\}$ is known at time $t-1$, so $\mathbb{E}[R_t 1\{T > t-1\}] = \mathbb{E}[1\{T > t-1\}\,\mathbb{E}[f_t \mid \mathcal{F}_{t-1}](R_{t-1} + 1)] \le h_t\,\mathbb{E}[1\{T > t-1\}(R_{t-1} + 1)]$. Because $h_t \ge 1$ and $R \ge 0$, also $\mathbb{E}[R_T 1\{T \le t-1\}] \le h_t\,\mathbb{E}[R_T 1\{T \le t-1\}]$. Adding gives $a_t \le h_t(a_{t-1} + \mathbb{P}(T > t-1)) \le h_t(a_{t-1} + 1)$. Since $a_0 = 0 = s_0$ and $s_t = h_t(s_{t-1} + 1)$, induction gives $a_t \le s_t$. ▫

Proof of Theorem 9. On the increase side $\mathbb{E}[f_t^{(k)} \mid \mathcal{F}_{t-1}] = 1 + \lambda_k^{\uparrow}(\mu_t/m^{\uparrow} - 1) \le 1 + \lambda_k^{\uparrow}\xi^{\uparrow}$, and on the decrease side $\mathbb{E}[f_t^{(l)} \mid \mathcal{F}_{t-1}] = 1 + \lambda_l^{\downarrow}(1 - \mu_t/m^{\downarrow}) \le 1 + \lambda_l^{\downarrow}\xi^{\downarrow}$. An alarm of the two-sided rule at $T_A \le m$ implies $\bar{M}_{T_A \wedge m} \ge A$, and $\bar{M} \ge 0$, so $A\,\mathbb{P}(T_A \le m) \le \mathbb{E}[\bar{M}_{T_A \wedge m}]$. Lemma S1 with $h_t = 1 + \lambda_k^{\uparrow}\xi^{\uparrow}$ bounds the expected wealth of increase bet k at $T_A \wedge m$ by $mY_m(\lambda_k^{\uparrow}\xi^{\uparrow})$, and likewise on the decrease side. Averaging with the weights and over the two sides gives $\mathbb{E}[\bar{M}_{T_A \wedge m}] \le m\Gamma_m$. When $\xi^{\uparrow}$ and $\xi^{\downarrow}$ are random but known at time 0, apply Lemma S1 conditionally on $\mathcal{F}_0$. ▫

Proof of Corollary 14. Given the window, $\hat{\tau}$, $\hat{m}$, the bets and the drifts $\xi^{\uparrow}(W)$ and $\xi^{\downarrow}(W)$ are known at time 0, and the conditional means stay within these drifts by assumption. Theorem 9 conditionally on $\mathcal{F}_0$ gives $\mathbb{P}(T_A \le m \mid \mathcal{F}_0) \le \min\{1, (m/A)\Gamma_m(\xi^{\uparrow}(W), \xi^{\downarrow}(W))\}$; take expectations over the window. ▫

Design numbers. Table S1 gives the inflation factor for the default detector with $c = 3$, $b = 0$ and $A = 7800$, for a drift of ξ on both sides; the bound is $\min(1, (m/A)\Gamma_m)$. At one day a drift of 0.1%, 0.5% and 1% on both sides gives 5.14%, 5.80% and 7.06% instead of 5%. To first order $\Gamma_m \approx 1 + (m+1)(\bar{\lambda}^{\uparrow}\xi^{\uparrow} + \bar{\lambda}^{\downarrow}\xi^{\downarrow})/4$, with $\bar{\lambda}$ the weighted mean bet of each side, while the largest bet times the drift times m is small; beyond that it grows like $(1 + \lambda_{\max}\xi)^m$. The one-day bound reaches 10% at a drift of 1.63% and 25% at 2.66% on both sides; with an upward drift only (a conditional mean between m_0 and $m_0(1+\xi)$) at 1.72% and 2.70%; with a downward drift only at 6.36% and 10.51%, because the decrease bets are smaller; and for a 1440-step day with $A = 28,800$, at 0.44% and 0.72%.

drift ξ on both sides	Γ_m, one day	Γ_m, five days	first horizon at which the bound reaches 1
0.10%	1.027	1.160	4,880
0.50%	1.160	4.239	1,919
1.00%	1.412	158	1,140
2.00%	2.638	1.2×10^6	654
3.00%	7.452	1.2×10^{10}	468

Table S1. Inflation factor of Theorem 9 for the default detector, $c = 3$, $b = 0$, $A = 7800$; a day is 390 steps. Closed form, no simulation.

Proposition S2 (where the bound is attained). For one side with one bet λ at level A, the deterministic stream $Y_t = 1 + \xi$ (allowed when $1 + \xi \le B$) satisfies the hypothesis of Theorem 9 with equality and alarms at $m^* = \min\{t : tY_t(\lambda\xi) \ge A\}$. No bound of the form $c\,mY_m(\lambda\xi)/A$ that is valid for this stream can have $c < 1/((1+\lambda\xi)(1+1/A))$ at $m = m^*$.

Proof. Every factor equals $1 + \lambda\xi$, so $R_t = tY_t(\lambda\xi)$ and the alarm time is m^* with probability 1. Since $R_{m^*} = (1+\lambda\xi)(R_{m^*-1} + 1) < (1+\lambda\xi)(A+1)$, a valid bound $cR_{m^*}/A \ge 1$ needs $c > A/((1+\lambda\xi)(A+1))$. ▫

For the default two-sided detector the one-day bound reaches 1 at an upward drift of 3.740%, and the deterministic stream alarms within the day from 3.756%. Below that drift the deterministic stream does not alarm within the day, and the strongest stream we found, which holds Y_t at its mean for a searched number of steps and then plays bold (Section 6.1), reaches 0.33 to 0.63 of the bound in one day; the worst case there is open. Noisy streams are far below the bound: the two-point law $\{0, B\}$ gives 2.0% in one day at a drift of 0.25% and 2.4% at 5%.

S1.3 A drift that varies

Theorem 9 needs a bound on the drift at every step. Under GARCH clustering the conditional mean comes arbitrarily close to τ^2 when σ_t is large, so the only valid constant is $\xi^{\uparrow} = B - 1$ and the bound is 1, although the drift averages out over time. Take $b = 0$, so both sides normalise by m_0, and let $d_t = \mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] - 1$ be the signed drift.

Lemma S3 (a pathwise inequality). Let $u_t^{\uparrow}$ and $u_t^{\downarrow}$ be predictable with $u_t^{\uparrow} \geq d_t$, $u_t^{\downarrow} \geq -d_t$, $u_t^{\uparrow} \geq -1$ and $u_t^{\downarrow} \geq 1 - B$, and put $h_{k,t}^{\uparrow} = 1 + \lambda_k^{\uparrow} u_t^{\uparrow}$ and $h_{l,t}^{\downarrow} = 1 + \lambda_l^{\downarrow} u_t^{\downarrow}$. Build the two-sided average $\tilde{M}_t$ of the Shiryaev-Roberts statistics of the drift-corrected factors $f_t^{(k)}/h_{k,t}^{\uparrow}$ and $f_t^{(l)}/h_{l,t}^{\downarrow}$. Then (1) $\tilde{M}$ satisfies Lemma 2, so $\mathbb{P}(\max_{t \leq m} \tilde{M}_t \geq a) \leq m/a$ for every $a > 0$; and (2) path by path $\bar{M}_t \leq D_t^* \tilde{M}_t$, where D_t^* is the largest window product $\prod_{i=j}^{t} h_{k,i}^{\uparrow}$ or $\prod_{i=j}^{t} h_{l,i}^{\downarrow}$ over all bets and all starts $j \leq t$.

Proof. (1) The corrected factors have conditional mean at most 1, so Lemma 2 and the proof of Theorem 4 apply. (2) The wealth of a bet started at j is the corrected wealth times the window product, which is at most D_t^*; sum over j and average over bets and sides. ▫

Theorem S4 (threshold form). For every predictable drift, every $m \geq 1$ and every $K > 0$, $\mathbb{P}(T_A \leq m) \leq Km/A + \mathbb{P}(\max_{t \leq m} D_t^* > K)$.

Proof. On $\{T_A \leq m\}$, $A \leq \bar{M}_{T_A} \leq D_{T_A}^* \tilde{M}_{T_A}$, so either $\max_{t \leq m} D_t^* > K$ or $\tilde{M}_{T_A} \geq A/K$, which has probability at most Km/A by Lemma S3. ▫

A calm spell shrinks the window products and the maximum restarts, so a drift that averages out does not accumulate. No independence is needed: the drift may respond to the data, as under GARCH.

Theorem S5 (a drift that does not respond to the data). Suppose there is a σ-field $\mathcal{D}$ and $\mathcal{D}$-measurable $u_t^{\uparrow}, u_t^{\downarrow}$ as in Lemma S3 with $\mathbb{E}[Y_t \mid \mathcal{F}_{t-1} \vee \mathcal{D}] \leq 1 + u_t^{\uparrow}$ and $\geq 1 - u_t^{\downarrow}$ for every t, and put $\xi_t^{\uparrow} = (u_t^{\uparrow})^+$ and $\xi_t^{\downarrow} = (u_t^{\downarrow})^+$. Then

$$\mathbb{P}(T_A \leq m \mid \mathcal{D}) \leq \min\{1, \frac{1}{2A}[\sum_k w_k^{\uparrow} \sum_{j=1}^{m} \prod_{i=j}^{m} (1 + \lambda_k^{\uparrow} \xi_i^{\uparrow}) + \sum_l w_l^{\downarrow} \sum_{j=1}^{m} \prod_{i=j}^{m} (1 + \lambda_l^{\downarrow} \xi_i^{\downarrow})], (\max_{t \leq m} D_t^*) \frac{m}{A}\},$$

with D^* built from the $\mathcal{D}$-measurable bounds.

Proof. Work with the filtration $\mathcal{F}_t \vee \mathcal{D}$. The first term is Lemma S1 conditionally on $\mathcal{D}$, with the fixed numbers $1 + \lambda \xi_t \geq 1$, followed by the argument of Theorem 9. The second is Theorem S4 conditionally on $\mathcal{D}$ with $K = \max_{t \leq m} D_t^*$, which is $\mathcal{D}$-measurable, so the remainder vanishes. ▫

Examples are a deterministic drift (a known error in the intraday profile), a drift decided by the calibration window (the plug-in), and stochastic volatility independent of the innovations. The maximum over t is needed: with $Y_t = B$ for 7 steps and then 0.95, the default detector alarms at step 7, so $\mathbb{P}(T_A \leq 390) = 1$, while the largest window product ending at step 390 is 17.5 and $17.5 \times 390/7800 = 0.876$; the maximum over $t \leq 390$ is 80,628. For a ±10% drift that flips at every step, Theorem S5 gives 15.3% at one day, where Theorem 9 gives 1 and the measured rate is 0.73%.

What cannot be said. (a) When the drift responds to the data, the formula of Theorem S5 with the realised drift is not a bound: if $Y_1 = B$ with probability $1/B$ and 0 otherwise, the stream then stays at B until the alarm (after $Y_1 = B$) or at 1 (after $Y_1 = 0$), the formula gives 10.2% at one day while the alarm probability is 11.1%. This is the GARCH mechanism, in which a large return raises the next conditional variance, so for GARCH only Theorem S4 is proved. On 64 GARCH(1,1) designs it was below 1 in 22 at one day, all with a small

drift, at 6 to 29 times the measured rate. (b) The mean drift alone gives no bound: a coin decided before monitoring that, with probability p, sets $Y_t = B$ for the 7 steps the detector needs to alarm has $\mathbb{P}(T_A \le m) = p$ while its mean drift is proportional to p/m, so no inequality $\mathbb{P}(T_A \le m) \le C(\bar{\xi})\, m/A$ with C depending only on the mean drift $\bar{\xi}$ holds for all A. A bound in the mean drift exists only with an additive term weighted by A, which exceeds the whole one-day budget once the mean upward drift exceeds 0.026%.

S1.4 Proof of Corollary 11

Fix t, a path $(x_1, \dots, x_t)$ and $\varepsilon \in (0,1)$. Build $P_\varepsilon \in \mathscr{P}^{\downarrow}$ step by step. If the past agrees with $(x_1, \dots, x_{s-1})$ and $x_s^2 \ge m_0$, draw $r_s = x_s$; if $x_s^2 < m_0$, draw $r_s = x_s$ with probability $1-\varepsilon$ and $r_s = z_s$ otherwise, with $z_s^2 = (m_0 - (1-\varepsilon)x_s^2)/\varepsilon$, so that the conditional mean of r_s^2 is exactly m_0. Off the path, and after time t, draw $r_s = \sqrt{m_0}$. Then $P_\varepsilon(r_1 = x_1, \dots, r_t = x_t) \ge (1-\varepsilon)^t$ and, since $W_t \ge 0$, $1 \ge \mathbb{E}_{P_\varepsilon}[W_t] \ge (1-\varepsilon)^t W_t(x_1, \dots, x_t)$. Let $\varepsilon \to 0$. For the last claim, each started process is at most 1, so $M_t \le t$. ▫

S1.5 Proofs of Lemma 1, Theorem 3, Proposition 5 and Proposition 10

Proof of Lemma 1. For each fixed ε, $s \mapsto \min(s^2\varepsilon^2, \tau^2)$ is nondecreasing, with derivative $2s\varepsilon^2$ where $s|\varepsilon| < \tau$ and 0 elsewhere. Since σ_t and τ are $\mathcal{F}_{t-1}$-measurable and ε_t is independent of $\mathcal{F}_{t-1}$, $\mu_t = g_\tau(\sigma_t)$, and monotonicity and the derivative pass through the expectation. ▫

Proof of Theorem 3. If $\mathbb{E}[T_A] = \infty$ there is nothing to show; otherwise $T_A < \infty$ almost surely. Optional stopping of $M_t - t$ (Lemma 2) at the bounded time $T_A \wedge t$ gives $\mathbb{E}[M_{T_A \wedge t}] \le \mathbb{E}[T_A \wedge t] \le \mathbb{E}[T_A]$. As $t \to \infty$, $M_{T_A \wedge t} \to M_{T_A} \ge A$, and Fatou's lemma gives $A \le \mathbb{E}[T_A]$. ▫

Proof of Proposition 5. Both sides satisfy Lemma 2 under the two-sided null, so their average does, and $\max(M^{\uparrow}, M^{\downarrow}) \ge 2A$ implies $\bar{M} \ge A$ because both statistics are nonnegative. ▫

Proof of Proposition 10. For $y \ge 1$ the point mass δ_y lies in $\mathscr{N}_\infty$, so $E(y) \le 1$. For $y < 1$ and $\varepsilon \in (0,1)$, let $z = (1-(1-\varepsilon)y)/\varepsilon$, so that $(1-\varepsilon)\delta_y + \varepsilon\delta_z$ has mean exactly 1. Then $(1-\varepsilon)E(y) + \varepsilon E(z) \le 1$, and since $E \ge 0$, $E(y) \le 1/(1-\varepsilon)$. Let $\varepsilon \to 0$. ▫

S1.6 Proof of Theorem 12 and Corollary 13

The largest probability of one value. Let $y \in [0, B]$. Under the decrease null, a law on $[0, B]$ with mean at least 1 and $p = \mathbb{P}(Y = y)$ satisfies $1 \le py + (1-p)B$, so $p \le (B-1)/(B-y)$ when $y < 1$; the law with mass $(B-1)/(B-y)$ at y and the rest at B has mean exactly 1 and attains it, and for $y \ge 1$ the point mass at y is in the null. Under the increase null, $1 \ge \mathbb{E}Y \ge py$ gives $p \le 1/y$ for $y > 1$, attained by the law on $\{0, y\}$ with mass $1/y$ at y. Under the two-sided null the same two-point laws have mean exactly 1. So π of Section 3.3 is the largest probability of y in each null, attained by a law with mean 1, or by a point mass when $\pi(y) = 1$. For $c = 3$ and Gaussian truncation ($B = 9.045$), $\pi(0) = 0.889$: a null law can put 89% of its mass on a zero return.

Proof of Theorem 12. Fix a path y and n. For $s \le n$ let Q_s be the law just described for y_s, and $Q_s = \delta_1$ for $s > n$; let P_y make $Y_1, Y_2, \dots$ independent with laws $Q_1, Q_2, \dots$. Each conditional mean is that of Q_s, so P_y is in the null, and $P_y(Y_{1:n} = y_{1:n}) = \prod_{s \le n} \pi(y_s) = 1/\Pi_n(y) > 0$. (a) The event $\{T \le n\}$ is a set of values of $Y_{1:n}$; if it

contains $y_{1:n}$, then $n/A \geq P_y(T \leq n) \geq 1/\Pi_n(y)$. (b) If $T = T(Y, U)$ with U uniform and independent of the data, Fubini's theorem gives $n/A \geq P_y(T \leq n) \geq \mathbb{P}_U(T(y, U) \leq n)/\Pi_n(y)$. (c) For any law P in the null and $E_s = \{Y_{1:s} = y_{1:s}\}$, the conditional law of Y_s given E_{s-1}, when $P(E_{s-1}) > 0$, has its mean on the null side, so $P(Y_s = y_s \mid E_{s-1}) \leq \pi(y_s)$ and $P(E_n) \leq 1/\Pi_n(y)$. The rule "stop at n if $Y_{1:n} = y_{1:n}$, otherwise never" therefore has $\mathbb{P}(T \leq k) = 0$ for $k < n$ and $\mathbb{P}(T \leq k) \leq 1/\Pi_n(y) \leq n/A \leq k/A$ for $k \geq n$.

If the null is stated for the returns, the two-point laws can be placed on $\{r_s, \tau\}$ (decrease) or $\{r_s, 0\}$ (increase) with the same masses, so the floor also holds for rules that see r_t, and for rules that use other information whose law the null leaves free. The proof uses independent laws that are not identically distributed. If validity were required only over i.i.d. null laws, the floor would be the largest i.i.d. null probability of the path, which is never larger than $1/\Pi_n(y)$ and equals it on paths with one repeated value, such as the collapse. ▫

Proof of Corollary 13. (a) The decrease null on $[0, \infty)$ contains the decrease null on $[0, B']$ for every $B' > \max_{s \leq n} y_s$, and $(B' - 1)/(B' - y) \to 1$ as $B' \to \infty$, so $\Pi_n(y)$ can be made as close to 1 as we like, and Theorem 12(a) needs $1 \geq A/n$, that is $n \geq A$; by Theorem 12(b) a randomised valid rule alarms by step n with probability at most n/A times a factor that can be made as close to 1 as we like. (b) On the collapse $\Pi_n = (B/(B-1))^n$. ▫

Numbers. With Gaussian truncation the collapse floor for every deterministic valid rule is 23, 33, 45, 73 and 106 steps at $c = 2, 2.5, 3, 4$ and 5 for $A = 7800$ (25, 36, 50, 82 and 120 at $2A$), against 28.7, 41.8, 57.7, 95.9 and 140.8 for a Shiryaev-Roberts e-detector with linear bets and 36, 52, 73, 122 and 180 for the default grid (63.6 and 79 at $2A$ for $c = 3$). On the collapse a single bet just below $\lambda_{\max}$ gives $M_n \leq B(F^n - 1)$ with $F = B/(B-1)$, so no Shiryaev-Roberts e-detector whose bets are valid over the whole null can alarm before $n = \log(1 + A/B)/\log(B/(B-1))$. A deterministic valid rule needs only $\Pi_n = (B/(B-1))^n \geq A/n$ (Theorem 12(a)), while the e-detector needs $(B/(B-1))^n \geq 1 + A/B$. The two thresholds differ by the factor $(1 + A/B)n/A$, which is about n/B because A/B is large, and this factor accounts for the gap between 45 and 57.7 steps at $c = 3$: the e-detector must keep $\mathbb{E}M_n \leq n$ under every null law at once, while a valid rule may spend its whole budget n/A on one path. On a complete explosion, $y_s = B$ for every s, the floor for the increase side is 4 steps at $c = 3$, against 5 for the e-detectors and 7 for the default grid. A rule valid at one horizon only, with $\mathbb{P}(T \leq 390) \leq 0.05$, could alarm on the collapse after 26 steps at $c = 3$, so controlling every n at once costs 19 steps on this path. For randomised rules, the largest probability of an alarm by step n on the collapse is $\min(1, n\Pi_n/A)$: 0.13 at $n = 30$, 0.56 at 40, 0.85 at 43 and 0.98 at 44. Summing $1 - \min(1, n\Pi_n/A)$ over n bounds the mean delay of such a rule on the collapse from below by only 37.8, because each cap spends the whole budget n/A at its own n, while a valid rule must respect every n at once. Let p_k be the probability that the rule stops at step k on the collapse path, with $\sum_k p_k = 1$. Under the i.i.d. law with mass $(B-1)/B$ at 0 and $1/B$ at B, which has mean 1 and lies in the decrease and the two-sided null, the collapse path up to step k has probability $((B-1)/B)^k$, so validity requires $\sum_{k \leq n} p_k((B-1)/B)^k \leq n/A$ for every n; by the argument of Theorem 12(c) these constraints also suffice for a rule that stops only on the collapse path. The smallest mean delay $\sum_k k p_k$ under them, a linear programme, is 43.39 steps. Corollary 13 does not mean that a practical rule detects collapses after 45 steps from any starting point: a rule that tests disjoint blocks of 45 steps for all-zero returns is valid, but finds a collapse that starts inside a block only at the end of the next block.

S1.7 Later days: Proposition 7, Theorem 8, the delay of the restart and the CUSUM statistic

Proof of Proposition 7. For $s \le t \le s+m$ put $S_t = \bar{M}_t + (s+m-t)$. By Lemma 2 (and Proposition 5 for the two-sided rule), S is a nonnegative supermartingale on $\{s, \dots, s+m\}$ given $\mathcal{F}_s$, with $S_s = \bar{M}_s + m$. An alarm implies $\bar{M}_t \ge A$ (for the two-sided rule, $\max(M^{\uparrow}, M^{\downarrow}) \ge 2A$ implies $\bar{M} \ge A$), hence $S_t \ge A$, and Ville's inequality given $\mathcal{F}_s$ gives the first claim. Taking expectations with $s = (d-1)m$ and $\mathbb{E}[\bar{M}_s] \le s$ gives the bound for day d. If $|r_t| = \sqrt{m_0}$, then $X_t = m_0$ because $m_0 \le \tau^2$, so $Y_t = 1$, every factor equals 1, and each recursion gives $M_t = M_{t-1} + 1 = t$. ▫

Proof of Theorem 8. The baseline is known, so Lemma 2 holds at every step. At b_i each side's average is at most ι, so $\bar{M}_{b_i} \le \iota$. Until the first alarm in the period the statistics with and without restarts after alarms coincide, so it suffices to bound the chance that the statistic without such restarts reaches A in the period. For $b_i \le t \le b_i + m$ put $S_t = \bar{M}_t + (b_i + m - t)$; as in the proof of Proposition 7, S is a nonnegative supermartingale given $\mathcal{F}_{b_i}$ with $S_{b_i} \le \iota + m$, and Ville's inequality gives the first claim, since the period ends at $b_{i+1} \le b_i + m$. For the second, let W_t be the sum of $\bar{M}$ at the alarms in $(b_i, t]$, plus the current $\bar{M}_t$, minus $t - b_i$. As in Supplement S1.1, W is a supermartingale with $W_{b_i} = \bar{M}_{b_i} \le \iota$, and each alarm adds at least A, so optional stopping at the bounded time b_{i+1} gives $A\,\mathbb{E}[N_i \mid \mathcal{F}_{b_i}] - m \le \iota$, where N_i is the number of alarms in the period. ▫

A window across a boundary. Let a window $(s, s+m]$ start at s with $b_i \le s \le b_i + m - 1$, an event known at b_i; this includes every window that starts in day i, since $b_{i+1} - b_i \le m$. Let $\tilde{M}$ be the statistic that continues from $\bar{M}_{b_i}$ with no restart after b_i. Every recursion is nondecreasing in its previous value and every statistic is nonnegative, so a restart only lowers the statistic, $\bar{M}_t \le \tilde{M}_t$ for $t \ge b_i$, and an alarm in the window needs $\tilde{M}_t \ge A$ there. Ville's inequality as in the proof of Proposition 7, applied from s, gives $\mathbb{P}(\tilde{M}_t \ge A \text{ for some } t \in (s, s+m] \mid \mathcal{F}_s) \le (\tilde{M}_s + m)/A$, and Lemma 2 from b_i gives $\mathbb{E}[\tilde{M}_s \mid \mathcal{F}_{b_i}] \le \bar{M}_{b_i} + s - b_i \le \iota + s - b_i$. Hence, given $\mathcal{F}_{b_i}$ and so also unconditionally, an alarm in the window has probability at most $(\iota + m + s - b_i)/A \le (2m + \iota - 1)/A$. Given $\mathcal{F}_s$ no such bound holds, since the statistic at s may already be close to A.

Corollary S14 (day restarts with the confidence-sequence baseline). Take one segment of the monitor of Section 4.2: calibration ends at T_0, Ω_G is the event that both bounds hold at T_0 (proof of Theorem 16, Supplement S2), so that $\mathbb{P}(\Omega_G^c) \le \delta$, and both statistics start at 0 at $b_0 = T_0$ and are scaled down at each later boundary, as in Theorem 8, so that each side's average is at most ι; the boundaries $b_0 < b_1 < \dots$ are stopping times with $b_{i+1} - b_i \le m$. If the two inequalities of the tolerance condition hold at every t with $b_i < t \le b_{i+1}$, then

$$\mathbb{P}(\text{an alarm in } (b_i, b_{i+1}] \mid \mathcal{F}_{b_i}) \le 1_{\Omega_G^c} + 1_{\Omega_G}\,\frac{m+\iota}{A}, \qquad \mathbb{P}(\text{an alarm in } (b_i, b_{i+1}]) \le \delta + \frac{m+\iota}{A}.$$

With no assumption on the law of the returns, $\mathbb{P}(\text{an alarm in } (b_i, b_{i+1}]) \le \delta + \mathbb{P}(\mathcal{O}^{(i)}) + (m+\iota)/A$, where $\mathcal{O}^{(i)}$ is the event that one of the two inequalities fails at some $t \in (b_i, b_{i+1}]$.

Proof. $\Omega_G \in \mathcal{F}_{T_0} \subseteq \mathcal{F}_{b_i}$. On Ω_G, Step 2 of the proof of Theorem 16, which uses the tolerance condition only at the step at hand, gives $\mathbb{E}[\bar{M}_t \mid \mathcal{F}_{t-1}] \le \bar{M}_{t-1} + 1$ for $b_i < t \le b_{i+1}$. So $S_t = 1_{\Omega_G}(\bar{M}_t + b_i + m - t)$, $b_i \le t \le b_{i+1}$, is a nonnegative supermartingale given $\mathcal{F}_{b_i}$ with $S_{b_i} \le 1_{\Omega_G}(\iota + m)$, and Ville's inequality gives $\mathbb{P}(\Omega_G \cap \{\text{an alarm in } (b_i, b_{i+1}]\} \mid \mathcal{F}_{b_i}) \le 1_{\Omega_G}(m+\iota)/A$. Add $1_{\Omega_G^c}$ and take expectations. For the last claim, let $T_{\mathcal{O}}$ be the first $t > b_i$ at which an

inequality fails; a failure at t is known at $t-1$, so, as in part 4 of Theorem 16, the statistic that equals $\bar{M}$ before $T_{\mathcal{O}}$ and grows by 1 per step from $T_{\mathcal{O}}$ on gives a supermartingale S on Ω_G with no hypothesis, and on Ω_G an alarm in $(b_i, b_{i+1}]$ either comes before $T_{\mathcal{O}}$ or lies in $\mathcal{O}^{(i)}$. ▫

The bound is marginal. Ω_G is not observed, so given the observed past (for example, given no alarm so far) the probability of an alarm in a day is bounded only by $\mathbb{P}(\Omega_G^c \mid \text{past}) + (m+\iota)/A$, which can exceed $\delta + (m+\iota)/A$. The tolerance must hold throughout the day, not only for the first m steps after T_0 as in part 1 of Theorem 16, but not on earlier days, since the scaling at b_i bounds the statistic on every path; and the statement is per segment: after an alarm the monitor recalibrates, and the next segment has its own Ω_G. With the plug-in baseline nothing per day is proved. The simulations of this section use a known baseline.

The delay of the restart. Let the returns after a change at step v be i.i.d. with some post-change law and independent of the past, let D_0 be the delay, counting the alarm step, of the two-sided detector started at 0 at the change, and let $p = \mathbb{P}(D_0 \le m)$.

Proposition S6 (renewal bound). If the change falls k steps after the start of a day of length m, $0 \le k < m$, and $p > 0$, the delay D_r of the detector restarted at every day boundary satisfies $\mathbb{E}[D_r] \le \mathbb{E}[\min(D_0, m-k)] + \mathbb{P}(D_0 > m-k)\,\mathbb{E}[\min(D_0, m)]/p$, with equality for $k = 0$ in the form $\mathbb{E}[D_r] = \mathbb{E}[\min(D_0, m)]/p$. The plain detector's delay is at most D_0 on every path.

Proof. The recursion $M_t = f_t(M_{t-1} + 1)$ is nondecreasing in M_{t-1} because $f_t \ge 0$, and every statistic is at least 0 at the change, so on the same data the restarted and the plain detector are at least the detector started at 0 at the change, until the next restart. Hence on $\{D_0 \le m-k\}$ the restarted detector alarms by step D_0, and the plain detector always does. Otherwise the restarted detector is set to 0 at the next boundary, $m-k$ steps after the change. From then on, day after day, it is the detector started at 0 on i.i.d. post-change data, independent of the first partial day, and a day contains an alarm with probability p. The number of days without an alarm is geometric, so the expected delay counted from that boundary is $(1-p)m/p + \mathbb{E}[D_0 \mid D_0 \le m] = \mathbb{E}[\min(D_0, m)]/p$. ▫

The restart thus costs little when a change is usually detected within a day from a fresh start and does not fall in the last steps of a day, and it loses changes whose detection from a fresh start needs more than a day. A carry ι keeps part of the evidence across the boundary at the price of $(m+\iota)/A$.

The CUSUM-type statistic. For bet k let $C_{k,t} = f_{k,t}\max(C_{k,t-1}, 1) = \max_{j \le t} \prod_{i=j}^{t} f_{k,i}$, with $C_{k,0} = 0$. Each side averages its bets with weights w (increase) or v (decrease), and $H(w) = -\sum_k w_k \log w_k \le \log K$ for K bets. We call the null exchangeable if, in addition, the law of $(Y_1, \ldots, Y_s)$ is invariant under permutations for every s; i.i.d. returns are the main example.

Theorem S7 (window bound for the CUSUM statistic). Assume a known baseline, an exchangeable null and fixed bets, and let $s \ge 0$ and $m \ge 1$. (a) For one bet at level A, $\mathbb{P}(C_t \ge A \text{ for some } t \in (s, s+m]) \le m/A$. (b) For one side averaging its bets at level A, $\mathbb{P}(\sum_k w_k C_{k,t} \ge A \text{ for some } t \in (s, s+m]) \le (m + \log A + H(w))/A$. (c) For the two-sided rule with each side at $2A$, the probability of an alarm in $(s, s+m]$ is at most $(m + \log 2A + (H(w) + H(v))/2)/A$. (d) The same bounds hold for at least one alarm in the window when the monitor restarts after every alarm, and for the first alarm.

Proof. With fixed bets the factors at time i are a fixed function of Y_i. Reversing the order of $Y_1, \dots, Y_s$ does not change their joint law and maps $C_{k,s}$ to $\max_{j \le s} \prod_{i \le j} f_{k,i}$, so $C_{k,s}$ has the law of the running maximum up to s of the wealth of bet k started at time 1. That wealth is a nonnegative supermartingale started at 1, so Ville's inequality gives $\mathbb{P}(C_{k,s} \ge x) \le 1/x$ for every $x > 0$ and every s, and (a) follows by a union bound over the m steps of the window. For (b), let $\bar{U}_s = \sum_k w_k \max(C_{k,s}, 1)$, put $Z_{k,s+1} = C_{k,s+1}$ and $Z_{k,t} = f_{k,t}(Z_{k,t-1} + 1)$ for $t > s+1$, so that $C_{k,t} \le Z_{k,t}$ by induction. With $S_s = \bar{U}_s + m - 1$ and $S_t = \sum_k w_k Z_{k,t} + (s + m - t)$ for $s < t \le s + m$, S is a nonnegative supermartingale given $\mathcal{F}_s$ (this step needs only the conditional means), so Ville's inequality gives $\mathbb{P}(\text{alarm in } (s, s+m] \mid \mathcal{F}_s) \le \min\{1, (\bar{U}_s + m - 1)/A\}$. Taking expectations, $\mathbb{E} \min\{1, (\bar{U}_s + m - 1)/A\} \le (m-1)/A + \sum_k \mathbb{E} \min(1, w_k U_k / A)$ with $U_k = \max(C_{k,s}, 1)$, and since $\mathbb{P}(U_k \ge x) \le 1/x$ for $x \ge 1$, $\mathbb{E} \min(1, w_k U_k / A) = \int_0^1 \mathbb{P}(U_k \ge vA/w_k)\, dv \le (w_k/A)(1 + \log(A/w_k))$. Summing over k gives (b), and (c) is the union of the two sides, each at level $2A$. For (d), after a restart at time a the restarted statistic $\max_{a < j \le t} \prod_{i=j}^{t} f_{k,i}$ is at most $C_{k,t}$, so every alarm of the restarted monitor in the window, and the first alarm, is a time at which the statistic without restarts is at or above the threshold. ▫

With 6 and 5 equal weights, (c) is 5.146% at $(m, A) = (390, 7800)$ and 5.044% at $(1440, 28, 800)$, and the levels at which it is exactly 5% are $A = 8027.7$ and $29,053.4$. For one bet over all nonnegative factors with mean 1 (an unclipped statistic), (a) is sharp to first order; whether the term $\log A + H(w)$ can be removed for the average over bets is open. Bets that depend on the past, which Lemma 2 allows, break the reversal, so Theorem S7 does not cover them. The bound of Theorem S7 is unconditional: given the state at the start of the window the probability can be far larger (from a state just below the threshold, the next step alarms with probability about $\mathbb{P}(Y > 1)$, 32% for i.i.d. Gaussian returns), and given the past only the conditional bound in the proof of (b), which is Lemma 2 at work, remains. In the terms of Lai (1998; footnote 1 of the main text), Theorem 8 gives, with a known baseline, the window-limited criterion given the past for windows aligned with the days over the whole null, and Theorem S7 an unconditional version for every window over exchangeable nulls.

Proposition S8 (no window bound under martingale differences). Fix the bets and A. For every s there is a null stream with $\mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] = 1$ and $Y_t \in [0, B]$, with $B \ge 2$, at every step under which the two-sided CUSUM-type rule has no alarm up to s and an alarm at $s+1$ with probability at least $\mathbb{P}(T_h \le s)/B$, where T_h is a stopping time, finite almost surely, that does not depend on s. Hence, for every $m \ge 1$ and every A, the largest probability over the null of an alarm in $(s, s+m]$, given none before, is at least $(1 - o(1))/B$ as $s \to \infty$, while the bound of Theorem S7 tends to 0 as A grows.

Proof. Write $\bar{U}_t^{\uparrow}(y) = \sum_k w_k f_k(y) \max(C_{k,t}, 1)$ for the increase side's average at $t+1$ if $Y_{t+1} = y$, and $\bar{U}_t^{\downarrow}(y)$ for the decrease side's, with weights v; the first is nondecreasing and the second nonincreasing in y. Let $Y'_1, Y'_2, \dots$ be i.i.d. on $\{0, B\}$ with $\mathbb{P}(Y'_t = B) = 1/B$, and let T_h be the first $t \ge 0$ at which, on the stream Y', $\bar{U}_t^{\uparrow}(B) \ge 2A$ or $\bar{U}_t^{\downarrow}(0) \ge 2A$: the first time at which one draw could take a side to $2A$. *Hunt:* $Y_t = Y'_t$ for $t \le \min(T_h, s)$. *Park:* $Y_t = 1$ for $T_h < t \le s$. *Strike:* if $T_h \le s$, draw Y_{s+1} from $\{0, B\}$ as before; what follows does not matter. Every draw has conditional mean 1 and lies in $[0, B]$ (returns 0, $\pm\sqrt{m_0}$ or of absolute value at least τ). No alarm comes during the hunt: at each $t \le \min(T_h, s)$ we have $t - 1 < T_h$, so $\bar{U}_{t-1}^{\uparrow}(B)$ and $\bar{U}_{t-1}^{\downarrow}(0)$ are below $2A$, and by monotonicity the actual draw keeps both sides below $2A$. None comes while parked either: $Y = 1$ replaces each C by $\max(C, 1)$ and then leaves it unchanged, and on the increase side $f_k(B) \ge 1$ gives $\max(C_{k,T_h}, 1) \le f_k(B) \max(C_{k,T_h-1}, 1)$, so

the parked average $\sum_k w_k \max(C_{k,T_h}, 1)$ is at most $\bar{U}^{\uparrow}_{T_h-1}(B) < 2A$ (and equals 1 if $T_h = 0$); the decrease side is the same with $f_l(0) \geq 1$. At $s+1$ each statistic is $f(Y_{s+1}) \max(C_{T_h}, 1)$, the value that the same draw would have given at $T_h + 1$. So if $\bar{U}^{\uparrow}_{T_h}(B) \geq 2A$, the draw B, with probability $1/B$, raises an alarm at $s+1$; otherwise $\bar{U}^{\downarrow}_{T_h}(0) \geq 2A$, and the draw 0, with probability $1 - 1/B \geq 1/B$, does. T_h is defined on Y', so it does not depend on s, and it is finite almost surely, because a run of B's whose length depends only on A and the bets takes the increase side to $2A$, and such runs occur almost surely. ▫

The conditional window bound in the proof of Theorem S7(b) still holds under martingale differences, and it predicts the strike: at a parked state $\bar{U}_s$ is close to the level. Corollary 6 also still holds, as for every null stream.

Numbers. All with the true baseline, the default detector and 20 days; 40,000 paths per stream (30,000 for bold play and for the parking stream). For the plain detector and the four i.i.d. laws of Table S4, the probability of a first false alarm on day d, given none before (the hazard), was 0.98% to 1.52% on day 1 and 4.60% to 4.70% on average over days 11 to 20 with 390-step days (2.04% to 2.44% and 4.57% to 4.78% with 1440-step days); for Gaussian returns with 390-step days it averaged 3.0%, 4.1% and 4.7% over days 2 to 5, 6 to 10 and 11 to 20, and no single i.i.d. day was significantly above 5%. For bold play, which watches the detector, it was 5.25% as the average of the daily hazards over days 11 to 20 with 390-step days, and 5.23% pooled over path-days (standard error about 0.06 points; the pooling was chosen after seeing the results), and 5.02% with 1440-step days. With a restart after every alarm the expected number of false alarms per day over days 1 to 20 was 0.037 to 0.048, below the 0.05 of Corollary 6. After $d-1$ days with $Y_t = 1$, a Gaussian day d had a false alarm with probability 11.33%, 28.02% and 65.07% for $d = 5$, 10 and 20 with 390-step days and 15.13%, 34.30% and 70.82% with 1440-step days (a second run of day 10 gave 28.16% and 34.03%); with a two-point day on $\{0, B\}$ instead of a Gaussian one the plain detector reached 31.29% (390) and 35.7% (1440) on day 10 and 70.09% (390) and 73.16% (1440) on day 20. The largest single-day probabilities found, 35.7% on day 10 and 73.2% on day 20, both for a two-point day with 1440-step days, stay well below dm/A (50% and 100%); Proposition 7 shows only that no bound below 1 holds for all days. The restarted detector repeated its first day on every day (0.87% to 1.03% for Gaussian returns with 390-step days); on any day of any stream its false alarm probability was at most 2.66% (390) and 3.52% (1440), and its hazard at most 2.75% and 3.52%, all for bold play. The CUSUM-type rule stayed at or below 0.18% on every day for i.i.d. returns and 0.40% for bold play, far below its bound. The parking stream against the CUSUM-type rule passed the i.i.d. bound only on day 17 with 390-step days (5.43% against 5.146%) and on day 16 with 1440-step days (5.30% against 5.044%), and reached 6.16% and 6.44% on day 20, 7.3 and 9.8 standard errors above the bounds; scaled down to days of 20 steps with $A = 400$, where the i.i.d. bound is 7.1%, it reached 27.0% by day 120. In the delay comparison (20,000 paths per cell, changes at a uniformly random step of day 3, horizon 12 days) the plain detector raised a false alarm before the change on 4.6% to 4.9% of paths with 390-step days and 6.6% to 7.0% with 1440-step days, and the restarted detector on 2.1% to 2.4% and 4.4% to 4.7%, so the delays are not compared at equal false alarm rates. The restarted detector's mean delay was 1.06 to 1.44 times the plain detector's with 390-step days and 1.02 to 1.14 times with 1440-step days. The CUSUM-type rule at its certified level was 1.30 to 1.83 times slower than the plain detector with 390-step days and 1.24 to 1.66 times with 1440-step days, and a carry of m at level $2A$ helped the restarted detector a little for decreases with 390-step days and cost a little elsewhere. The restarted detector's mean delay was below the bound of

Proposition S6 in every cell; that comparison was chosen after the delays had been seen. Part of the plain detector's speed comes from the statistic it builds up under the null, which is also what raises its false alarm probability on later days.

S1.8 The statistic: fourth powers and zero returns

The same detectors applied to $\min(r_t^4, \tau^4)$ are valid too; for pure variance changes they were 1% to 28% faster for increases and 1.8 to 2.6 times slower for decreases. When more than half of a calibration window's returns are exactly zero, $\hat{s} = 0$; the monitor then extends the window, up to twice its length, and if that does not help it uses the scaled median of the nonzero absolute returns. The rule uses past data only, so τ stays predictable and positive.

S1.9 The rank detector, a running truncation level and restarts without a blind period

The results of this section are the proofs behind Section 6.5 and the companion note. They assume more than the conditional-mean null of Section 2: exchangeable or i.i.d. returns.

The rank detector. Let $a_t = |r_t|$ and let $\theta_1, \theta_2, \dots$ be i.i.d. uniform on $[0,1]$ and independent of the returns. The smoothed conformal p-value of step t is

$$p_t = \frac{\#\{i \le t : a_i > a_t\} + \theta_t \, \#\{i \le t : a_i = a_t\}}{t},$$

where the second count includes $i = t$; a small p_t means that $|r_t|$ is large compared with all earlier absolute returns, and ties, such as zero returns, are split at random. Let $\mathcal{G}_t = \sigma(p_1, \dots, p_t)$. A betting function is a measurable $f : [0,1] \to [0,\infty)$ with $\int_0^1 f(u)\,du = 1$. The rank detector chooses betting functions f_t that are predictable in $\mathcal{G}$ (fixed functions are the simplest case), runs $M_t = f_t(p_t)(M_{t-1} + 1)$ from $M_W = 0$ after a warm-up of $W \ge 0$ steps whose returns only enter the reference set, averages several betting functions on each side and alarms when either side reaches $2A$. The default increase bets are the densities of p_t when the new return is a normal draw scaled by 1.25, 1.5 or 2 relative to a normal reference law, and the decrease bets the same for 0.8, 0.67 or 0.5, each clipped at the p-value of a return of 3 standard deviations and renormalised to integrate to 1, with the decrease bets putting weight 0.95 on this density and the rest on the uniform one; normality only tunes the bets and is not used for validity. A band b divides each payoff by $1 + b$.

Proposition S15 (Vovk, Nouretdinov and Gammerman 2003). If $r_1, \dots, r_N$ $(N \le \infty)$ is exchangeable, then $p_1, \dots, p_N$ are independent and uniform on $[0,1]$.

Proof. Fix $n \le N$. For $t \le n$ let $\mathcal{B}_t$ be the bag (multiset) of $a_1, \dots, a_t$ and $\mathcal{H}_t = \sigma(\mathcal{B}_t, a_{t+1}, \dots, a_n, \theta_{t+1}, \dots, \theta_n)$. Then $\mathcal{H}_1 \supseteq \dots \supseteq \mathcal{H}_n$, since $\mathcal{B}_t$ and a_{t+1} give $\mathcal{B}_{t+1}$; each p_s with $s > t$ is a function of $\mathcal{B}_s$, a_s and θ_s, so it is $\mathcal{H}_t$-measurable; and given $\mathcal{H}_t$, by exchangeability a_t is a uniformly random element of the bag $\mathcal{B}_t$, while θ_t is uniform and independent of it. If the bag has distinct values $v_1 > v_2 > \dots$ with multiplicities $k_1, k_2, \dots$ and $a_t = v_j$, which has conditional probability k_j/t, then $p_t = (k_1 + \dots + k_{j-1} + \theta_t k_j)/t$ is uniform on the j-th of consecutive intervals of lengths k_j/t that tile $[0,1]$. So p_t is uniform given $\mathcal{H}_t$, hence independent of $\mathcal{H}_t$ and of $p_{t+1}, \dots, p_n$, and $\mathbb{P}(p_1 \le u_1, \dots, p_n \le u_n) = u_1 \, \mathbb{P}(p_2 \le u_2, \dots, p_n \le u_n)$; induction gives $\prod_{t \le n} u_t$. ▫

The p-values are independent of each other, not of the raw past: given $r_1, \dots, r_{t-1}$, p_t is the rank of a new draw against a fixed sample, which is not uniform. The martingale property therefore holds in $\mathcal{G}$, and the bets may use past p-values only.

Theorem S16 (validity of the rank detector). (a) If the returns are exchangeable, then for every warm-up W and every predictable choice of betting functions, $M_t - (t - W)$, $t \geq W$, is a $\mathcal{G}$-martingale. Hence Theorems 3 and 4 hold for the rank detector, one side at level A or the two-sided rule with each side at $2A$: an alarm within m steps of W has probability at most m/A, and the expected time to the first alarm is at least A, with no baseline, no truncation level and no failure probability. With a band $b > 0$ the statistic minus $t - W$ is a supermartingale and both bounds still hold. (b) Suppose that the detector restarts after every alarm with an empty reference set and bets from the next return, and that, given the returns and tie-breakers up to each alarm, the later returns are exchangeable. Then the bounds of (a) hold for every segment, and the number N_m of alarms within the first m steps satisfies $\mathbb{E}[N_m] \leq m/A$. The condition holds for i.i.d. returns and for every infinite exchangeable sequence.

Proof. (a) By Proposition S15, p_t is uniform and independent of $\mathcal{G}_{t-1}$, so $\mathbb{E}[f_t(p_t) \mid \mathcal{G}_{t-1}] = \int_0^1 f_t = 1$ and $\mathbb{E}[M_t \mid \mathcal{G}_{t-1}] = M_{t-1} + 1$: this is Lemma 2 in the filtration $\mathcal{G}$, and the alarm time is a $\mathcal{G}$-stopping time, so the proofs of Theorems 3 and 4 and of Proposition 5 apply. A band multiplies the conditional mean of each factor by $1/(1+b) < 1$. (b) Let q_t be the p-value that the restarted detector uses at step t, computed within the current segment. It suffices to show that $q_1, q_2, \dots$ are i.i.d. uniform: the detector is then the detector of (a) driven by the q_t, and the argument of Supplement S1.1 in the filtration of the q_t gives $\mathbb{E}[N_m] \leq m/A$. Up to the first alarm T_1, $q_t = p_t$, and whether $T_1 \geq t$ is fixed by $q_1, \dots, q_{t-1}$, so on that event q_t is uniform and independent of $q_1, \dots, q_{t-1}$ by Proposition S15. Given $\mathcal{F}_{T_1}$, which determines $q_1, \dots, q_{T_1}$, the returns after T_1 are exchangeable by assumption and the tie-breakers after T_1 are fresh, so Proposition S15 applied given $\mathcal{F}_{T_1}$ makes the p-values of the second segment i.i.d. uniform and independent of $\mathcal{F}_{T_1}$; induction over segments completes the argument. For i.i.d. returns, the returns after a stopping time are i.i.d. with the same law and independent of the past. For an infinite exchangeable sequence, de Finetti's theorem makes the returns i.i.d. given a random law; given that law, the returns after T_1 are i.i.d. and independent of $\mathcal{F}_{T_1}$, so they are exchangeable given $\mathcal{F}_{T_1}$. ▫

Early p-values after a restart are coarse (the first is just θ), which costs power, not validity.

Lemma S17 (permuted pools at stopping times). Let $Y_1, \dots, Y_N$ be a uniformly random ordering of a finite multiset, and T a stopping time of (Y_t) and of randomisation independent of the ordering. Given T, $Y_1, \dots, Y_T$ and the randomisation used up to T, the remaining values $Y_{T+1}, \dots, Y_N$ are a uniformly random ordering of the values not yet drawn, so they are exchangeable given the past.

Proof. Given $Y_1, \dots, Y_t$, every ordering of the remaining values has the same conditional probability, and the event $\{T = t\}$ is fixed by $Y_1, \dots, Y_t$ and by randomisation independent of the ordering. The same holds given the randomisation up to t, which is independent of the ordering. ▫

On the permuted null streams of Section 7.1 the condition of Theorem S16(b) therefore holds after every alarm until the pool runs out, and the bounds of the restarted rank

detector hold exactly, whatever the pool.

A running truncation level. Let the calibration returns $r_1, \dots, r_{T_0}$ be i.i.d. with an unknown law F, write $m_F(s) = \mathbb{E}_F \min(r^2, s^2)$ for $s > 0$, and fix a band $b \in [0, 1)$. The null asks that, for every $t > T_0$ and every $s > 0$, $\mathbb{E}[\min(r_t^2, s^2) \mid \mathcal{F}_{t-1}] \le (1+b)\, m_F(s)$ (increase side) and $\ge (1-b)\, m_F(s)$ (decrease side); returns that stay i.i.d. with law F satisfy both with $b = 0$. Take for $\mathbb{E}[\min(r_t^2, s^2) \mid \mathcal{F}_{t-1}]$ the version $\int \min(x^2, s^2)\, Q_t(dx)$, where Q_t is a regular conditional distribution of r_t given $\mathcal{F}_{t-1}$, which exists for real-valued returns. Both sides of each inequality are then continuous and nondecreasing in s, so the inequalities hold for all rational s outside one null set and hence for all s at once, and in particular at a level τ_t known at time $t-1$, where $\mathbb{E}[\min(r_t^2, \tau_t^2) \mid \mathcal{F}_{t-1}] = \int \min(x^2, \tau_t^2)\, Q_t(dx)$. Let $\mathscr{S}$ be the set of levels that the truncation level may take. A band at T_0 is a pair of functions $L \le U$ on $\mathscr{S}$, computed from data up to T_0, whose failure event $D = \{m_F(s) \notin [L(s), U(s)] \text{ for some } s \in \mathscr{S}\}$ has $\mathbb{P}(D) \le \delta$. The monitor uses any predictable $\tau_t \in \mathscr{S}$, the statistic $X_t = \min(r_t^2, \tau_t^2)$, the increase baseline $m_t^{\uparrow} = (1+b)\, U(\tau_t)$ and the decrease baseline $m_t^{\downarrow} = (1-b)\, L(\tau_t)$, with decrease bets $\lambda_t = s_l/(B_t - 1)$, $B_t = \tau_t^2/m_t^{\downarrow}$, for the fractions $s_l < 1$ of Section 3.1; at a step with $m_t^{\downarrow} \le 0$ or $B_t \le 1$ the decrease statistic is left unchanged.

Theorem S18 (a running truncation level). Under this null, $\mathbb{P}(\text{an alarm at one of } T_0 + 1, \dots, T_0 + m) \le \delta + m/A$ for every $m \ge 1$, and the expected number of monitored steps before the first alarm is at least $(1-\delta)A$. A band with these properties on a grid $\mathscr{S} = \{s_1, \dots, s_K\}$, fixed by a short pilot before the band's data start, is given by the upper and lower bounds of Lemma S10 for each level, each at error $\delta/(2K)$.

Proof. Let $G = D^c \in \mathcal{F}_{T_0}$. Every factor is positive on every path: $f_t^{\uparrow} \ge 1 - \lambda > 0$, and $f_t^{\downarrow} \ge 1 - s_l > 0$ because $X_t \le \tau_t^2$, whether or not the band holds, which matters because a failure of the band is not observed. On G and for $t > T_0$, $\mu_t = \mathbb{E}[X_t \mid \mathcal{F}_{t-1}] \le (1+b)\, m_F(\tau_t) \le (1+b)\, U(\tau_t) = m_t^{\uparrow}$, so $\mathbb{E}[f_t^{\uparrow} \mid \mathcal{F}_{t-1}] \le 1$, and likewise $\mu_t \ge m_t^{\downarrow}$ when the decrease side is on. This is Step 2 of the proof of Theorem 16 with G in place of Ω_G, and Steps 3 and 4 and part 2 of that proof apply word for word; they need only that, on an event known at T_0, every factor has conditional mean at most 1, and allow predictable bets. For the grid band, the levels are fixed before the bounds start, and under i.i.d. returns $\min(r_i^2, s_k^2)/s_k^2$ has the constant conditional mean $m_F(s_k)/s_k^2$, so Lemma S10 bounds the failure probability of each of the $2K$ one-sided bounds by $\delta/(2K)$; add them up. ▫

Restarts that learn their band. After a stopping time S, for example an alarm, the grid is fixed from returns before S, the bounds of the grid run on $r_{S+1}, r_{S+2}, \dots$, and snapshots $\mathsf{B}(e) = (U, L)$ of the band after e observations are kept for sizes $0 = e_0 < e_1 < \dots$ fixed in advance, with $\mathsf{B}(0)$ the trivial band $(s_k^2, 0)$. The Shiryaev-Roberts chain started at segment step j bets against $\mathsf{B}(e(j))$, where $e(j)$ is the largest snapshot size at most $j-1$, so all of its band's data come before j; the statistic is the sum of the chains' wealths, averaged over bets, and a decrease chain is not started at a step at which its snapshot admits no bet. The truncation level is a predictable $\tau_t \in \mathscr{S}$, as in Theorem S18.

Theorem S19 (restart with a learning band). If the returns after S are i.i.d. with some law given $\mathcal{F}_S$, then an alarm within m steps of S has probability at most $\delta + m/A$, and the expected number of steps to the first alarm is at least $(1-\delta)A$. If the monitor restarts in this way after every alarm and the condition holds in every segment, the number of alarms within the first m monitored steps satisfies $\mathbb{E}[N_m] \le (\delta + m/A)/(1-\delta)$.

Proof. Work given $\mathcal{F}_S$ and number the segment's steps $1, 2, \ldots$. Let N be the first step at which the band fails at some grid level ($N = \infty$ if never). Lemma S10 holds uniformly in time, so, applied given $\mathcal{F}_S$ to each of the $2K$ one-sided bounds at error $\delta/(2K)$ as in the proof of Theorem S18, it gives $\mathbb{P}(N < \infty) \le \delta$, and N is a stopping time. Let $W_t^{(j)}$ be the wealth of the chain started at j and $W_t'^{(j)} = 1\{N > e(j)\} W_t^{(j)}$. The event $\{N > e(j)\}$ is known at step $e(j) \le j - 1$; on it the snapshot covers m_F at every grid level, and $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = m_F(\tau_t)$ because r_t is independent of $\mathcal{F}_{t-1}$ and τ_t is predictable, so every factor of the chain has conditional mean at most 1. Each $W'^{(j)}$ is thus a nonnegative supermartingale from step j on, started at most at 1, and $M'_t = \sum_{j \le t} W_t'^{(j)}$ satisfies $\mathbb{E}[M'_t \mid \mathcal{F}_{t-1}] \le M'_{t-1} + 1$, so Ville's inequality as in Theorem 4 gives $\mathbb{P}(\max_{t \le m} M'_t \ge A) \le m/A$. On $\{N = \infty\}$, $M' = M$, which gives the first claim; the two-sided rule follows as in Proposition 5. For the run length, let ϑ be the first step with $M_\vartheta \ge A$; every chain started at $j \le \vartheta$ has $e(j) \le \vartheta - 1$, so $M'_\vartheta = M_\vartheta$ on $\{N \ge \vartheta\}$, and optional stopping of $M'_t - t$ at $\vartheta \wedge n$ gives $\mathbb{E}[\vartheta \wedge n] \ge A(\mathbb{P}(\vartheta \le n) - \delta)$; let $n \to \infty$. The count follows as in the proof of Corollary 18, with the stopped statistic M' in place of $\bar{M}$, so that every segment contributes a supermartingale, and with an alarm counted as bad when its segment's band failed before it, an event of conditional probability at most δ given the segment's start. ▫

Validity is free in these constructions, power is not: a chain that starts shortly after a restart bets against a wide band, and one that starts after a change bets against a band that already contains post-change returns (companion note, Section 1).

S2. Proofs for Section 4

Throughout, the observations of the confidence sequence are indexed by $i = 1, 2, \ldots$ (global time $n_\tau + i$), $\mathcal{F}_{i-1}$ is the information before the i-th observation, $\chi_i = \min(r_i^2, \tau^2)/\tau^2 \in [0, 1]$ and $\eta_i = \mathbb{E}[\chi_i \mid \mathcal{F}_{i-1}]$. The one-sided ranges are $a_i^\uparrow = 1 - \hat{z}_{i-1}$ and $a_i^\downarrow = \hat{z}_{i-1}$, so that $\hat{z}_{i-1} - \chi_i \ge -a_i^\uparrow$ and $\chi_i - \hat{z}_{i-1} \ge -a_i^\downarrow$ always hold.

The bounds. The first n_τ returns of the calibration window fix $\tau = c\hat{s}$ (the default uses 25% of the window; we did not vary this share). Choose predictably a centre $\hat{z}_{i-1} \in (0, 1)$ and bets $\lambda_i^\uparrow, \lambda_i^\downarrow > 0$ with $\lambda_i^\uparrow(1 - \hat{z}_{i-1}) \le \ell_0$ and $\lambda_i^\downarrow \hat{z}_{i-1} \le \ell_0$ for a fixed $\ell_0 \in (0, 1)$. With $\psi_a(\lambda) = (-\log(1 - \lambda a) - \lambda a)/a^2$, $e_i = (\chi_i - \hat{z}_{i-1})^2$, $\psi_i^\uparrow = \psi_{1-\hat{z}_{i-1}}(\lambda_i^\uparrow)$, $\psi_i^\downarrow = \psi_{\hat{z}_{i-1}}(\lambda_i^\downarrow)$ and error levels $\delta_u, \delta_l \in (0, 1)$, put

$$U_t = \frac{\sum_{i \le t} \lambda_i^\uparrow \chi_i + \log(1/\delta_u) + \sum_{i \le t} \psi_i^\uparrow e_i}{\sum_{i \le t} \lambda_i^\uparrow}, \qquad L_t = \frac{\sum_{i \le t} \lambda_i^\downarrow \chi_i - \log(1/\delta_l) - \sum_{i \le t} \psi_i^\downarrow e_i}{\sum_{i \le t} \lambda_i^\downarrow}.$$

They bound the bet-weighted averages $\bar{\eta}_t^\uparrow = \sum_{i \le t} \lambda_i^\uparrow \eta_i / \sum_{i \le t} \lambda_i^\uparrow$ and $\bar{\eta}_t^\downarrow$ (the same with $\lambda^\downarrow$) at all times at once (Lemma S10). With $1 - \hat{z} = \hat{z} = 1$ these are the predictable plug-in empirical Bernstein bounds of Waudby-Smith and Ramdas (2024); the one-sided ranges let the lower bound use bets up to $\ell_0/\hat{z}$, which matters because $\hat{z} \approx 0.11$ for a clipped squared Gaussian return.

Lemma S9. Let $a > 0$ and $\lambda \ge 0$ with $\lambda a < 1$, and $\psi_a(\lambda) = (-\log(1 - \lambda a) - \lambda a)/a^2 \ge 0$. Then for every $u \ge -a$,

$$\log(1 + \lambda u) \ge \lambda u - \psi_a(\lambda) u^2, \qquad \text{equivalently} \qquad \exp(\lambda u - \psi_a(\lambda) u^2) \le 1 + \lambda u.$$

Proof. For $x > -1$, Taylor's formula with integral remainder gives $\log(1 + x) - x = x^2 \phi(x)$ with $\phi(x) = -\int_0^1 (1 - s)(1 + sx)^{-2}\, ds$. For each $s \in [0, 1]$ the map $x \mapsto (1 + sx)^{-2}$ is nonincreasing on $(-1, \infty)$, so ϕ is nondecreasing. Put $x = \lambda u \ge -\lambda a > -1$. Then $\log(1 + \lambda u) - \lambda u \ge (\lambda u)^2 \phi(-\lambda a)$,

and $\phi(-\lambda a) = (\log(1-\lambda a) + \lambda a)/(\lambda a)^2$, so the right side equals $-\psi_a(\lambda)u^2$. For $\lambda = 0$ both sides are 0. ▫

With $a = 1$ this is the inequality of Fan, Grama and Liu (2015) behind the empirical Bernstein confidence sequences of Howard et al. (2021) and Waudby-Smith and Ramdas (2024).

Lemma S10 (time-uniform coverage). For the bounds above, without any assumption on the dependence between observations or on how η_i varies,

$$\mathbb{P}(\bar{\eta}_t^{\uparrow} \geq U_t \text{ for some } t \geq 1) \leq \delta_u, \qquad \mathbb{P}(\bar{\eta}_t^{\downarrow} \leq L_t \text{ for some } t \geq 1) \leq \delta_l.$$

Proof. Upper bound. Let $K_0 = 1$ and $K_t = \prod_{i \leq t} \exp(\lambda_i^{\uparrow}(\eta_i - \chi_i) - \psi_i^{\uparrow} e_i)$. Write $\eta_i - \chi_i = u_i + (\eta_i - \hat{z}_{i-1})$ with $u_i = \hat{z}_{i-1} - \chi_i \geq -a_i^{\uparrow}$, and note $e_i = u_i^2$. Lemma S9 with $a = a_i^{\uparrow}$ and $\mathbb{E}[u_t \mid \mathcal{F}_{t-1}] = \hat{z}_{t-1} - \eta_t$ give

$$\mathbb{E}[K_t/K_{t-1} \mid \mathcal{F}_{t-1}] \leq \mathbb{E}[1 + \lambda_t^{\uparrow} u_t \mid \mathcal{F}_{t-1}] e^{\lambda_t^{\uparrow}(\eta_t - \hat{z}_{t-1})} = (1-y)e^{y} \leq 1, \qquad y = \lambda_t^{\uparrow}(\eta_t - \hat{z}_{t-1}),$$

since $(1-y)e^{y} \leq 1$ for every real y. So K is a nonnegative supermartingale with $K_0 = 1$, and Ville's inequality gives $\mathbb{P}(K_t \geq 1/\delta_u \text{ for some } t) \leq \delta_u$. Taking logarithms, $K_t \geq 1/\delta_u$ is the event $\sum_{i \leq t} \lambda_i^{\uparrow} \eta_i \geq \sum_{i \leq t} \lambda_i^{\uparrow} \chi_i + \sum_{i \leq t} \psi_i^{\uparrow} e_i + \log(1/\delta_u)$, that is $\bar{\eta}_t^{\uparrow} \geq U_t$. Lower bound: the same argument with $K_t' = \prod_{i \leq t} \exp(\lambda_i^{\downarrow}(\chi_i - \eta_i) - \psi_i^{\downarrow} e_i)$, $u_i = \chi_i - \hat{z}_{i-1} \geq -a_i^{\downarrow}$ and $(1+y)e^{-y} \leq 1$. ▫

Since $0 \leq \bar{\eta} \leq 1$, the clipped bounds $\min(U_t, 1)$ and $\max(L_t, 0)$ fail on the same events. The default predictable choices follow Waudby-Smith and Ramdas (2024): $\hat{z}_t = (1/2 + \sum_{i \leq t} \chi_i)/(t+1)$, a variance estimate $\hat{\sigma}_t^2 = (1/4 + \sum_{i \leq t} (\chi_i - \hat{z}_i)^2)/(t+1)$, and $\lambda_t = \min(\sqrt{2\log(1/\delta)/(n_{\text{plan}}\,\hat{\sigma}_{t-1}^2)},\ \ell_0/a_t)$ with $\ell_0 = 0.75$ and n_{plan} a planned sample size. Validity does not depend on these choices.

Proof of Theorem 16. We write U_{T_0} and L_{T_0} for the bounds computed from the observations up to global time T_0, and similarly $\bar{\eta}_{T_0}^{\uparrow}$, $\bar{\eta}_{T_0}^{\downarrow}$. Let $\Omega^{\uparrow} = \{\bar{\eta}_{T_0}^{\uparrow} < U_{T_0}\}$, $\Omega^{\downarrow} = \{\bar{\eta}_{T_0}^{\downarrow} > L_{T_0}\}$ and $\Omega_G = \Omega^{\uparrow} \cap \Omega^{\downarrow}$; these events are $\mathcal{F}_{T_0}$-measurable.

Step 1 (the bounds hold at the random time T_0). Lemma S10 applies to the observations after n_τ, because τ is fixed before they start. The complement of $\Omega^{\uparrow}$ is contained in the event that the upper bound fails at some time, whatever T_0 is, so $\mathbb{P}((\Omega^{\uparrow})^c) \leq \delta_u$, $\mathbb{P}((\Omega^{\downarrow})^c) \leq \delta_l$ and $\mathbb{P}(\Omega_G^c) \leq \delta$.

Step 2 (on Ω_G both statistics satisfy Lemma 2). Let $T_0 < t \leq T_0 + h$. For $i > n_\tau$, $\tau^2 \eta_i = \mu_i$, and a bet-weighted average of the η_i with $n_\tau < i \leq T_0$ lies between their smallest and largest values. So on Ω_G the tolerance condition gives $\mu_t \leq (1+b)\min_i \mu_i \leq (1+b)\tau^2 \bar{\eta}_{T_0}^{\uparrow} \leq (1+b)\tau^2 \min(U_{T_0}, 1) = m^{\uparrow}$, so $\mathbb{E}[f_t^{\uparrow} \mid \mathcal{F}_{t-1}] = 1 + \lambda(\mu_t/m^{\uparrow} - 1) \leq 1$, and $f_t^{\uparrow} \geq 1 - \lambda > 0$. If the decrease side is on, it gives $\mu_t \geq (1-b)\max_i \mu_i \geq (1-b)\tau^2 \bar{\eta}_{T_0}^{\downarrow} > (1-b)\tau^2 \max(L_{T_0}, 0) = m^{\downarrow}$, so $\mathbb{E}[f_t^{\downarrow} \mid \mathcal{F}_{t-1}] \leq 1$, and $f_t^{\downarrow} \geq 1 - \lambda(B-1) > 0$ because $X_t \leq \tau^2$. The bets are fixed at T_0, hence predictable. As in Lemma 2, $\bar{M}_t = (M_t^{\uparrow} + M_t^{\downarrow})/2$ satisfies $\mathbb{E}[\bar{M}_t \mid \mathcal{F}_{t-1}] \leq \bar{M}_{t-1} + 1$ on $\Omega_G \cap \{T_0 < t \leq T_0 + h\}$; if the decrease side is off, $M^{\downarrow} = 0$ and the same holds. A skipped (missing) return leaves $\bar{M}$ unchanged; when, given the past, whether r_t is missing carries no information about its size, the bound on $\mathbb{E}[f_t \mid \mathcal{F}_{t-1}]$ also holds given that r_t is observed, and the inequality is preserved.

Step 3 (Ville's inequality after a random start). Fix s and let $\mathcal{E}_s = \Omega_G \cap \{T_0 = s\} \in \mathcal{F}_s$. For $s \leq t \leq s + m$ put $S_t = 1_{\mathcal{E}_s}(\bar{M}_t + m - (t-s))$ with $\bar{M}_s = 0$. Since $\mathcal{E}_s \in \mathcal{F}_{t-1}$, Step 2 with $h = m$ gives

$\mathbb{E}[S_t \mid \mathcal{F}_{t-1}] \le S_{t-1}$; moreover $S \ge 0$ and $S_s = m\,1_{\mathcal{E}_s}$. Ville's (1939) inequality gives $\mathbb{P}(\max_{s<t\le s+m} S_t \ge A) \le m\,\mathbb{P}(\mathcal{E}_s)/A$. An alarm requires $\max(M^{\uparrow}, M^{\downarrow}) \ge 2A$, hence $\bar{M} \ge A$ and $S \ge A$. Summing over s, $\mathbb{P}(\Omega_G \cap \{\text{alarm within } m\}) \le m\,\mathbb{P}(\Omega_G)/A$.

Step 4 (part 1). $\mathbb{P}(\text{alarm within } m) \le \mathbb{P}(\Omega_G^c) + \mathbb{P}(\Omega_G)\,m/A \le \delta + m/A$. When $m < A$ the map $p \mapsto p + (1-p)m/A$ is increasing, which gives $\le \delta + (1-\delta)m/A$.

Part 2. Let ϑ be the first $t > T_0$ with $\bar{M}_t \ge A$; the alarm comes no earlier. For $\mathcal{E}_s$ as above and $n \ge 1$, $N = \min(\vartheta, s+n)$ is a bounded stopping time and $1_{\mathcal{E}_s}(\bar{M}_t - (t-s))$, $t \ge s$, is a supermartingale by Step 2 with $h = \infty$. Optional stopping gives $\mathbb{E}[1_{\mathcal{E}_s}(N-s)] \ge \mathbb{E}[1_{\mathcal{E}_s}\bar{M}_N] \ge A\,\mathbb{P}(\mathcal{E}_s \cap \{\vartheta \le s+n\})$, and monotone convergence as $n \to \infty$ gives $\mathbb{E}[1_{\mathcal{E}_s}(\vartheta - s)] \ge A\,\mathbb{P}(\mathcal{E}_s)$ (if $\vartheta = \infty$ with positive probability on $\mathcal{E}_s$ the left side is infinite). Summing over s, $\mathbb{E}[\vartheta - T_0] \ge \mathbb{E}[1_{\Omega_G}(\vartheta - T_0)] \ge A\,\mathbb{P}(\Omega_G) \ge (1-\delta)A$.

Part 3. The same proof with one statistic at level A and only $\Omega^{\uparrow}$ (or $\Omega^{\downarrow}$).

Part 4. Let $T_{\mathcal{O}}$ be the first $t > T_0$ with $\mu_t > m^{\uparrow}$ or, when the decrease side is on, $\mu_t < m^{\downarrow}$. Since μ_t is known at time $t-1$, $\{T_{\mathcal{O}} = t\} \in \mathcal{F}_{t-1}$. Let $\bar{M}'_t = \bar{M}_t$ for $T_0 < t < T_{\mathcal{O}}$ and $\bar{M}'_t = \bar{M}'_{t-1} + 1$ for $t \ge T_{\mathcal{O}}$. On Ω_G the computation of Step 2 holds for every $t < T_{\mathcal{O}}$ with no hypothesis, and trivially for $t \ge T_{\mathcal{O}}$, so Step 3 applies to $\bar{M}'$ and gives $\mathbb{P}(\Omega_G \cap \{\bar{M}' \ge A \text{ within } m\}) \le m\,\mathbb{P}(\Omega_G)/A$. On $\Omega_G \cap \{T_{\mathcal{O}} > T_0 + m\}$, $\bar{M}' = \bar{M}$ at the times $T_0 + 1, \ldots, T_0 + m$. On Ω_G, $T_{\mathcal{O}} \le T_0 + m$ implies $\mathcal{O}_m$, because $\mu_{T_{\mathcal{O}}} > m^{\uparrow} \ge (1+b)\tau^2\bar{\eta}^{\uparrow}_{T_0} \ge (1+b)\min_i \mu_i$, and likewise on the decrease side. Hence $\mathbb{P}(\text{alarm within } m) \le \mathbb{P}(\Omega_G^c) + \mathbb{P}(\mathcal{O}_m) + m/A$. ▫

A weaker condition that depends on the bets. The same proof works under the bet-weighted forms $\mu_t \le (1+b)\tau^2\bar{\eta}^{\uparrow}_{T_0}$ and $\mu_t \ge (1-b)\tau^2\bar{\eta}^{\downarrow}_{T_0}$ for $T_0 < t \le T_0 + h$, but whether these hold depends on the bets. The tolerance condition of Section 4.2 implies them; it involves the law of the returns, the truncation level fixed at n_τ and the end of calibration T_0, but not the bets.

Proof of Corollary 17. Since τ is fixed at time n_τ and the conditional law of r_t is P_0 for $t > n_\tau$ before the change, $\mu_t = \mathbb{E}_{P_0}[\min(r^2, \tau^2)]$ is the same constant μ for all such t, and the tolerance condition holds with $b = 0$. Apply Theorem 16. ▫

Proof of Corollary 18. Let Ω_q be the event that both bounds of segment q hold at its end of calibration, and split $N_m = N_{\text{bad}} + N_{\text{good}}$ according to whether the alarm's segment satisfies Ω_q.

Bad segments. Segment q starts at a stopping time S_q, and its bounds use only returns after S_q, so Lemma S10 applied conditionally on $\mathcal{F}_{S_q}$ gives $\mathbb{P}(\Omega_q^c \mid \mathcal{F}_{S_q}) \le \delta$ on $\{S_q < \infty\}$. Let $C_q \in \mathcal{F}_{S_q}$ be the event that fewer than m monitored steps have passed before S_q. A segment raises at most one alarm, and only segments with C_q can contribute to N_m, so $\mathbb{E}[N_{\text{bad}}] \le \sum_q \mathbb{E}[1_{C_q}\mathbb{P}(\Omega_q^c \mid \mathcal{F}_{S_q})] \le \delta\,\mathbb{E}[\#\{q : C_q\}]$. Segment $q+1$ starts right after alarm q, so C_{q+1} implies that alarm q is among the first m monitored steps; hence $\#\{q : C_q\} \le 1 + N_m$ and $\mathbb{E}[N_{\text{bad}}] \le \delta(1 + \mathbb{E}[N_m])$.

Good segments. Let W_t be the sum of $\bar{M}$ at all past alarms of good segments, plus the current $\bar{M}$ if the current segment is good and monitoring, minus the number of monitored steps so far in good segments. Whether a segment is good is known at the end of its calibration, so by Step 2 above every monitored step of a good segment has $\mathbb{E}[W_t - W_{t-1} \mid \mathcal{F}_{t-1}] \le 0$; at an alarm the statistic moves into the sum unchanged, and in all

other steps W does not move. So W is a supermartingale with $W_0 = 0$. Let ϑ be the time of the m-th monitored step. Each alarm of a good segment adds at least A, so $W_{\vartheta \wedge n} \geq A N_{\text{good}}(\vartheta \wedge n) - m$; optional stopping at $\vartheta \wedge n$ and monotone convergence give $\mathbb{E}[N_{\text{good}}] \leq m/A$.

Adding, $\mathbb{E}[N_m] \leq \delta(1 + \mathbb{E}[N_m]) + m/A$; since $N_m \leq m$ is finite, this rearranges to $\mathbb{E}[N_m] \leq (\delta + m/A)/(1 - \delta)$. ▫

The bound holds unchanged when the statistic is also scaled down at stopping times inside a segment, as in Theorem 8 and Corollary S14: at such a time W only decreases, so it stays a supermartingale.

Why the sample split. If τ is computed from the whole calibration window, each X_i in the window depends through τ on later returns, so X_i is not adapted and K in Lemma S10 need not be a supermartingale. Two other valid routes are a union bound over a grid of truncation levels fixed in advance (costing $\log(K/\delta)$ instead of $\log(1/\delta)$ for K levels) and a band that holds for all truncation levels at once (companion note). Because the coverage of Lemma S10 is uniform in time, the calibration may also stop at a data-dependent time, for example when the relative width $(\min(U_t, 1) - \max(L_t, 0))/\bar{\chi}_t$ first falls below a target; a fixed-n interval would not allow this, since a normal-approximation interval checked at every n excludes the true mean at some n with probability one.

Proof of Theorem 15. Write $m_F(s) = \mathbb{E}_F \min(r^2, s^2)$ and $\hat{m} = n^{-1} \sum_{i \leq n} \min(r_i^2, \hat{\tau}^2)$, and let $d = m_F(\hat{\tau})/\hat{m} - 1$ be the drift of the window: under every stream in $\mathscr{C}(W)$, $\mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] = 1 + d$. Let D_n be the Kolmogorov distance between the empirical law of the $|r_i|$ and the law of $|r|$, and $\hat{B} = \hat{\tau}^2/\hat{m}$.

Step 1 (the drift). Since $\mathbb{E}\min(Z, t) = \int_0^t \mathbb{P}(Z > x)\,dx$ for $Z \geq 0$, applying this to $Z = r^2$ under the empirical law and under F gives $|\hat{m}(s) - m_F(s)| \leq s^2 D_n$ for every s at once, and at $s = \hat{\tau}$, $|d| \leq \hat{B} D_n$. Because at least half of the $|r_i|$ lie on each side of their median, $b_{lo} = 2a^2/(1 + a^2) \leq \hat{B} \leq b_{hi} = 2a^2$ with $a = 1.4826\,c$ (1.904 and 39.57 at $c = 3$).

Step 2 (upper bounds). For $x \geq 0$, $Y_m(x) \leq 1 + x(m + 1)e^{mx}/2$, because $e^u - 1 \leq u e^u$. Inserting this into Corollary 14 gives $\mathbb{P}(T \leq m) \leq (m/A)\,\mathbb{E}_W[\min\{A/m,\ 1 + \frac{m+1}{4}(\bar{\lambda} d^+ + \hat{\varphi} d^-)e^{mx^*}\}]$, where $\bar{\lambda} = 0.2033$ and $\bar{s} = 0.51$ are the mean bets of the two sides (the decrease bets being $s_l/(\hat{B} - 1)$), $\hat{\varphi} = \bar{s}/(\hat{B} - 1)$ and $x^* = \max(0.5\,d^+, 0.95\,d^-/(\hat{B} - 1))$. By Step 1, $\bar{\lambda} d^+ + \hat{\varphi} d^- \leq k_1 D_n$ and $x^* \leq k_2 D_n$, with $k_1 = \max(\bar{\lambda} b_{hi}, \bar{s} b_{lo}/(b_{lo} - 1)) = 8.045$ and $k_2 = \max(0.5\,b_{hi}, 0.95\,b_{lo}/(b_{lo} - 1)) = 19.78$ (0.5 and 0.95 are the largest bets of the two sides). Dropping the cap, the inequality of Dvoretzky, Kiefer and Wolfowitz (1956) with Massart's (1990) constant, $\mathbb{P}(D_n > x) \leq 2e^{-2nx^2}$, gives the non-asymptotic bound $\mathbb{P}(T \leq m) \leq (m/A)(1 + \frac{m+1}{4} k_1\, \mathcal{M}_n(k_2 m))$ with $\mathcal{M}_n(t) = \int_0^1 (1 + tx)e^{tx} \min\{1, 2e^{-2nx^2}\}\,dx$, and, as $m/\sqrt{n} \to 0$, the distribution-free constant $C = k_1 I_0/4 = 1.79$, with $I_0 = \int_0^\infty \min\{1, 2e^{-2u^2}\}\,du = u_0 + \sqrt{2\pi}\,\bar{\Phi}(2u_0) = 0.888$ and $u_0 = \sqrt{\log 2/2}$. For a fixed F with a unique median of $|r|$, $\hat{\tau} \to \tau$ in probability, and $\sqrt{n} d \Rightarrow N(0, \sigma_Y^2)$ with $\sigma_Y^2 = \mathrm{Var}_F(\min(r^2, \tau^2))/m_0^2$; the estimation of τ does not enter at first order, because the null is the truncated mean at the estimated level. Uniform integrability, from the same tail bound, then gives part (a) with $C = (\bar{\lambda} + \bar{s}/(B - 1))\,\sigma_Y/(4\sqrt{2\pi})$: 0.0369 for Gaussian returns ($\sigma_Y = 1.386$, $B = 9.045$) and 0.0471 for Student t_3 returns ($\sigma_Y = 1.583$, $B = 6.362$).

Step 3 (the worst stream). Take the side that d favours, its largest bet β with weight w, and $K = 2A/w$. The stream with $Y_t = 1 + d$ for $t \leq h$, then i.i.d. on $\{0, \hat{B}\}$ with mean $1 + d$ for $h < t \leq m$ (returns $\pm\sqrt{m_F(\hat{\tau})}$, then 0 or $\pm\hat{\tau}$), is in $\mathscr{C}(W)$. After the hold the Shiryaev-Roberts

statistic of β equals $S_h = \sum_{j=1}^{h} (1+\beta|d|)^j$. In the bold phase, optional stopping for the normalised product of the factors gives $\mathbb{P}(T \le m \mid W) \ge (S_h - \sqrt{KS_h}\ \varrho^{m-h})/(F_\beta K)$, where F_β is the bet's largest factor and $\varrho = \mathbb{E}\sqrt{f}/\sqrt{\mathbb{E}f} < 1$ for its factor f. On the event $|d| \ge z_0/\sqrt{n}$, whose probability stays bounded away from 0, a bold phase of length of order $\log A$ and $S_h \ge h(1+\beta|d|(h+1)/2)$ give a ratio to m/A of at least a constant times $1+\beta z_0 m/(4\sqrt{n})$, which tends to infinity under the conditions of (b). If $m/(\sqrt{n}\log A) \to \infty$, the constant stream alone alarms with probability tending to 1.

Step 4 (the margin). On the event $G_z = \{\hat{m} - z\hat{\mathrm{se}} \le m_F(\hat{\tau}) \le \hat{m} + z\hat{\mathrm{se}}\}$ both drifts relative to the moved baselines are at most 0, so Theorem 4 applies conditionally on the window and $\mathbb{P}(T \le m) \le m/A + \mathbb{P}(G_z^c)$. By Step 2 and Slutsky's lemma $(\hat{m} - m_F(\hat{\tau}))/\hat{\mathrm{se}} \Rightarrow N(0,1)$, so $\mathbb{P}(G_z^c) \to 2\bar{\Phi}(z)$, and the constant stream at $m_F(\hat{\tau})$ attains this excess when $m/(\sqrt{n}\log A) \to \infty$. The confidence-sequence baseline of Theorem 16 is the same argument with $\mathbb{P}(G^c) \le \delta$ guaranteed at every n. Measured as an effective z, $(U/m_F(\tau) - 1)\sqrt{n}/\sigma_Y$ with U its upper baseline and $\delta = 0.01$, its width is 3.8 to 5.7 for Gaussian and Student t_3 returns, part of it because it estimates the mean from three quarters of the window. ▫

Simulations of the plug-in. With 390 calibration steps, the one-day false alarm rate of the increase side was 4.58%, 4.68% and 5.10% for Gaussian, Student t_3 and jump returns, against 1.75%, 1.73% and 1.62% with the true baseline (4000 paths, Monte Carlo standard error 0.34 points). The excess grows with the horizon: with i.i.d. returns the five-day rate was 41% to 44%, where a known baseline guarantees at most 25%, and in 30-day streams with recalibration after every alarm the plug-in raised 0.094 (Gaussian) and 0.084 (Student t_3) false alarms per 1440-step day against a nominal 0.05.

Numbers. With $A = 7800$, $c = 3$ and each side at $2A$, the upper bound of Corollary 14 is within 4.1% (Gaussian) and 5.3% (Student t_3) of nominal where $m/\sqrt{n} < 1$, and the first-order formula of Theorem 15(a) is within 1% of it there. The distribution-free constant 1.79 is 48 times the Gaussian one, because it must allow for any law and uses D_n rather than the drift, and its non-asymptotic bound is informative only for m far below $\sqrt{n}$: for $n = 10^6$ it is 1.15 times nominal at $m = 39$ and 2.3 times at $m = 98$, and for $n = 10^8$ 1.021 times at $m = 98$ and 1.15 times at $m = 390$, so m must be below about $\sqrt{n}/20$. A one-day horizon of 390 minutes falls below $\sqrt{n}$ only for windows of more than 152,100 returns (390 trading days of minute data). Where $m/\sqrt{n}$ is large, the proved lower bound for the worst stream is a large multiple of nominal: 6.5 at $n = m = 390$ for Gaussian returns and 8.9 for t_3. Run through the detector after 4000 fresh 390-step Gaussian windows, the hold-then-bold stream alarmed within one day in 36.5% of windows (95% interval 35.0% to 38.0%), between the lower bound 33.8% and the upper bound 42.4% averaged over the same windows; over 20,000 windows these averages of per-window proved bounds are about 33% and 42% (standard errors about 0.3 points). For i.i.d. monitored returns the excess is much smaller: at one day the plug-in was not significantly above nominal in any cell (largest 5.14%, interval 4.72% to 5.59%, Student t_3 at $n = 390$), but at five days after a one-day window it reached 40.8% (Gaussian) and 42.3% (t_3) against 25%, and its ratio to the true baseline at five days fell from 3.4 at $n = 390$ to 1.07 at $n = 23,040$ for Gaussian returns. With $z = 2.576$ the inflated plug-in's bound is at most 1.061 times nominal in every cell. The share of windows whose drift gives i.i.d. returns a positive growth rate, the mechanism behind the heuristic scale $\log A/\sqrt{n}$ (instead of $m/\sqrt{n}$), is 0.83 at $n = 390$ and 0.005 at $n = 92,160$ for Gaussian returns; the scale itself is not proved.

S3. Proofs for Section 5

Notation is that of Section 5.1: $V = v(Y)$ is $Y - 1$ (increase) or $1 - Y$ (decrease), with values in $[v_{\mathrm{lo}}, v_{\mathrm{hi}}]$, $v_{\mathrm{lo}} < 0 < v_{\mathrm{hi}}$, and $\lambda_{\max} = 1/|v_{\mathrm{lo}}|$. Since v is affine and one-to-one, a function of y can be viewed as a function of v.

Proof of Lemma 19. If Q is not absolutely continuous with respect to P, $\mathrm{KL}(Q\|P) = \infty$. Otherwise let $q = dQ/dP$. By Jensen's inequality,
$\mathbb{E}_Q[\log E] - \mathrm{KL}(Q\|P) = \mathbb{E}_Q[\log (E/q)] \le \log \mathbb{E}_Q[E/q] = \log \mathbb{E}_P[E\,1\{q > 0\}] \le \log \mathbb{E}_P[E] \le 0$. ▫

Proof of Lemma 20. (a) Let $E \ge 0$ with $\mathbb{E}_P[E] \le 1$ for every law P on $[0, B]$ with $\mathbb{E}_P[V] \le 0$. First, E is bounded: for $v > 0$, the two-point law with mass $p = |v_{\mathrm{lo}}|/(v + |v_{\mathrm{lo}}|)$ at v and $1 - p$ at v_{lo} has mean 0, so $pE(v) \le 1$ and $E(v) \le 1 + v/|v_{\mathrm{lo}}|$; for $v \le 0$ the point mass gives $E(v) \le 1$. Let φ be the upper concave envelope of E on $[v_{\mathrm{lo}}, v_{\mathrm{hi}}]$. The largest value of $\mathbb{E}_P[E]$ over laws with $\mathbb{E}_P[V] = a$ is $\varphi(a)$ (two-point laws suffice), so $\varphi(a) \le 1$ for every $a \le 0$. As φ is concave and finite and 0 is interior, it has a supergradient s at 0: $\varphi(v) \le \varphi(0) + sv \le 1 + sv$. If $s \ge 0$, take $\lambda = s$; since $0 \le E(v_{\mathrm{lo}}) \le 1 + sv_{\mathrm{lo}}$, $s \le \lambda_{\max}$. If $s < 0$, then $\varphi(v) \le \varphi(0) \le 1$ for $v \ge 0$, and $\varphi \le 1$ for $v \le 0$ by assumption; take $\lambda = 0$.

b. "$\le$": each bet $1 + \lambda v$ is an e-variable for every $P \in \mathscr{M}_B$, so Lemma 19 gives $g(\lambda) \le \mathrm{KL}(Q\|P)$; take the infimum over P and the maximum over λ. "$\ge$": at an interior maximiser, $g'(\lambda^*) = \mathbb{E}_Q[V/(1 + \lambda^* V)] = 0$. The measure $P^*(dy) = Q(dy)/(1 + \lambda^* v(y))$ has total mass $\mathbb{E}_Q[1/(1 + \lambda^* V)] = 1 - \lambda^* \mathbb{E}_Q[V/(1 + \lambda^* V)] = 1$ and mean of V equal to $\mathbb{E}_Q[V/(1 + \lambda^* V)] = 0$, so $P^* \in \mathscr{M}_B$, and $\mathrm{KL}(Q\|P^*) = \mathbb{E}_Q[\log (1 + \lambda^* V)] = g(\lambda^*)$. ▫

Proof of Corollary 21. The first inequality is the data processing inequality, since X is a function of r. The second holds because $P_{0,Y} \in \mathscr{M}_B$, so the infimum defining $\mathrm{KL}_{\inf}(Q)$ is at most $\mathrm{KL}(Q_Y\|P_{0,Y}) = \mathrm{KL}(Q_X\|P_{0,X})$ (Y is a one-to-one function of X). The third holds because each bet is an e-variable for every $P \in \mathscr{M}_B$, so Lemma 19 gives $g(\lambda_k) \le \mathrm{KL}(Q\|P)$ for every such P. ▫

Proof of Corollary 22. Each bet satisfies $1 + \lambda v \le 1 + \lambda_{\max} v_{\mathrm{hi}}$, which is B on the increase side ($\lambda_{\max} = 1$, $v_{\mathrm{hi}} = B - 1$) and $B/(B - 1)$ on the decrease side ($\lambda_{\max} = 1/(B - 1)$, $v_{\mathrm{hi}} = 1$). By Lemma 20(a) every e-variable for $\mathscr{M}_B$ obeys the same bound, and so does $\mathrm{KL}_{\inf} = g(\lambda^*)$ when λ^* is interior (Lemma 20(b)). For the detector, if every factor of bet k equals its maximum F_k, then $M_n^{(k)} = \sum_{j \le n} F_k^{n-j+1}$, and any other data give a smaller value. ▫

Proof of Theorem 23. (a) is Theorem 12(a) applied path by path. (b) For a deterministic rule, $\varphi = 1\{T \le n\}$ is a function of the first n observations with $\mathbb{E}_{P^\circ} \varphi \le n/A$ and, by (a), $\varphi \le 1\{N_A \le n\}$, so $Q(T \le n) = \mathbb{E}_Q \varphi \le \beta_n$; for a randomised rule the same holds with $\varphi = \mathbb{P}_U(T \le n \mid y)$ and the cap $\min\{1, (n/A)\Pi_n\}$ of Theorem 12(b). The supremum β_n is attained by the test that fills the cap in decreasing order of the likelihood ratio dQ/dP° until the budget n/A is spent (the Neyman-Pearson lemma with a cap), and $\mathbb{E}_Q T = \sum_{n \ge 0} Q(T > n)$ gives the delay bound. For i.i.d. Q with an interior maximiser λ^*, $P^*(dy) = Q(dy)/(1 + \lambda^* v(y))$ is a probability law with $\mathbb{E}_{P^*} V = 0$ (Lemma 20), so it lies in all three nulls, and $\log dQ/dP^*$ on the first n steps is $\sum_{s \le n} \log (1 + \lambda^* V_s)$, with mean $n\,\mathrm{KL}_{\inf}(Q)$. ▫

Proof of Theorem 24. (a) For $t \ge v$ all terms of the Shiryaev-Roberts sum are nonnegative, so $M_t \ge w_k L_t^{(v,k)}$. Let $S_n = \sum_{i=0}^{n-1} \Lambda_{k,v+i}$ and $N = \inf\{n \ge 1 : S_n \ge \log (A/w_k)\}$; then $(T_A - v + 1)^+ \le N$. N depends only on post-change data, which are independent of $\mathcal{F}_{v-1}$. S is a random walk with drift $g_k > 0$ and bounded increments, so Wald's identity gives $g_k\,\mathbb{E}[N] = \log (A/w_k) + \mathbb{E}[S_N - \log (A/w_k)]$, and the overshoot is bounded by $\sup \Lambda_k$ and, in mean, by Lorden's (1970) inequality. (b) Each recursion $x \mapsto f(x + 1)$ (or $f \max(x, 1)$) is

nondecreasing in x, so from any state $M_{\nu-1} \ge 0$ the statistic after ν dominates the one started from 0 on the same post-change data. For the two-sided monitor each side alarms at $2A$, and the same argument at that level gives $\log(2A/w_k)$. ▫

Proof of Theorem 25. Write $g^* = g(\lambda^*) = \mathrm{KL}_{\inf}(Q)$. For $0 < b' < \lambda_{\max}$ let $\bar v = \max(|v_{\mathrm{lo}}|, v_{\mathrm{hi}})$, $C = \bar v^2/(1 - b'|v_{\mathrm{lo}}|)^2$, which bounds $|g''|$ and $|\partial_\lambda^2 \log(1+\lambda V)|$ on $[0, b']$, and $o = \log(1 + b'v_{\mathrm{hi}})$, which bounds every log factor of a bet $\lambda \le b'$ and hence every overshoot. By Theorem 24(b) it suffices to take $\nu = 1$.

a. Let $P_r^*(dr) = Q_r(dr)/(1 + \lambda^* v(r))$, where $v(r) = v(Y(r))$. As in Lemma 20(b) it is a probability law with $\mathbb{E}_{P^*}[V] = 0$, so $\mathbb{E}_{P^*}[\min(r^2, \tau^2)] = m^{\uparrow}$ ($m^{\downarrow}$ on the decrease side) and i.i.d. returns from P_r^* satisfy the null. Since $1 + \lambda^* v$ is bounded away from 0 and ∞, Q_r and P_r^* are mutually absolutely continuous, and $\mathrm{KL}(Q_r \| P_r^*) = \mathbb{E}_Q[\log(1 + \lambda^* V)] = g^*$. A rule with average run length at least A under every i.i.d. null law has it under P_r^*, and Lorden's (1971) lower bound for a change from P_r^* to Q_r gives $\mathrm{WADD}_Q(T) \ge (\log A/g^*)(1 + o(1))$.

b. The upper half is Theorem 24(a) with the best bet. For the lower half, let $S_n^k = \sum_{i \le n} \log(1 + \lambda_k V_i)$ and $R_N^k = \max_{0 \le i < n \le N} (S_n^k - S_i^k)$. Writing $S_n^k = g_k n + D_n^k$ with D^k a zero-mean random walk, $R_N^k \le \max(g_k, 0)N + 2\max_{j \le N} |D_j^k| = \max(g_k, 0)N + o(N)$ almost surely by the strong law of large numbers. Since $M_n \le n \max_{k,j} L_n^{(j,k)}$, for all $n \le N$ we have $\log M_n \le \log N + \max_k R_N^k \le g_{\mathrm{grid}} N + o(N)$ almost surely. With $N_A = (1 - \varepsilon)\log A/g_{\mathrm{grid}}$ this gives $\mathbb{P}(T_A \le N_A) \to 0$, so $\mathbb{E}[T_A] \ge N_A(1 - o(1))$; let $\varepsilon \to 0$ and divide by $\log A/\mathrm{KL}_{\inf}$.

c. With K equally spaced bets on $[a, b']$ and spacing $h = (b' - a)/(K - 1)$, some bet satisfies $|\lambda_k - \lambda^*| \le h/2$. Since $g'(\lambda^*) = 0$ and $|g''| \le C$, Taylor's theorem gives $g(\lambda_k) \ge g^* - Ch^2/8$, and Theorem 24(a) with $w_k = 1/K$ gives $\mathrm{WADD}(T_A) \le (\log A + \log K + o)/(g^* - Ch^2/8)$. With $K_A = 1 + \lceil (b' - a)\sqrt{\log A} \rceil$, $h \le 1/\sqrt{\log A}$ and $\log K_A \le \tfrac{1}{2}\log\log A + \log(2 + b' - a)$ for $\log A \ge 1$, and the denominator is $g^*(1 - O(1/\log A))$, which gives the claim, uniformly over laws with $\lambda^* \in [a, b']$ and g^* bounded away from 0. Balancing $\log K$ against $(\log A)C(b' - a)^2/(8g^* K^2)$ shows that $K \asymp \sqrt{\log A}$ minimises the second-order term of this bound.

d. Take $\varepsilon > 0$ with $[\lambda^* - \varepsilon, \lambda^* + \varepsilon] \subset [a, b']$ and write $h_i(\lambda) = \log(1 + \lambda V_i)$. Each h_i is concave, so on the interval it is at least $\Lambda_i = \min(h_i(\lambda^* - \varepsilon), h_i(\lambda^* + \varepsilon))$. Keeping only the start time 1 and the interval, $M_t \ge \int_{\lambda^* - \varepsilon}^{\lambda^* + \varepsilon} \prod_{i \le t} (1 + \lambda V_i)\,\pi(d\lambda) \ge 2\varepsilon p_0 \exp(\sum_{i \le t} \Lambda_i)$. The Λ_i are i.i.d. and bounded, and Taylor's theorem gives $\mathbb{E}[\Lambda_1] \ge g^* - \varepsilon D - C\varepsilon^2/2$ with $D = \mathbb{E}_Q|V/(1 + \lambda^* V)|$. The Wald argument of Theorem 24(a) for the level $\log(A/(2\varepsilon p_0))$ gives $\mathbb{E}[T_A] \le (\log A + \log(1/\varepsilon) + \log(1/(2p_0)) + o)/(g^* - \varepsilon D - C\varepsilon^2/2)$, and $\varepsilon = 1/\log A$ gives the claim. ▫

A Laplace approximation of the mixture suggests that (d) can be sharpened to $(\log A + \tfrac{1}{2}\log\log A + O(1))/\mathrm{KL}_{\inf}$, the order of the refined grid; we have not proved this.

Cost of a gap in the grid. If λ^* is not a bet, $\mathrm{KL}_{\inf} - g(\lambda_k) = \tfrac{1}{2}\sigma_*^2(\lambda_k - \lambda^*)^2 + O(|\lambda_k - \lambda^*|^3)$ with $\sigma_*^2 = \mathbb{E}_Q[V^2/(1 + \lambda^* V)^2]$. For a small change, with $\mathbb{E}_Q V = \mu_V$ and $\mathrm{Var}_Q V = s_V^2$, $\lambda^* \approx \mu_V/s_V^2$, $\mathrm{KL}_{\inf} \approx \mu_V^2/(2s_V^2)$ and $g(\lambda)/\mathrm{KL}_{\inf} \approx 1 - (\lambda/\lambda^* - 1)^2$: a geometric grid whose neighbouring bets differ by a factor ϖ keeps at least $1 - ((\varpi - 1)/(\varpi + 1))^2$ of the growth between its smallest and largest bets (0.89 for $\varpi = 2$). A gap costs $\log A(1/g_{\mathrm{grid}} - 1/\mathrm{KL}_{\inf})$ steps, which grows with A, while an extra bet costs about $\log((K + 1)/K)$ nats, which does not.

Asymptotics of the floor. Write $\gamma_Q = \mathbb{E}_Q \log(1/\pi(Y))$, with π of Section 3.3.

Proposition S11. Let Q be i.i.d. with $\gamma_Q > 0$. (a) $N_A / \log A \to 1/\gamma_Q$ almost surely and in L^1 as $A \to \infty$. (b) $\gamma_Q \geq \mathrm{KL}_{\inf}(Q)$, with equality exactly when $Q(Y > 1) = 0$ on the decrease side or $Q(Y < 1) = 0$ on the increase side. (c) If λ^* is interior, every rule valid at levels n/A over a class that contains P^*, deterministic or randomised, has $\mathbb{E}_Q T \geq (\log A/\mathrm{KL}_{\inf}(Q))(1 - o(1))$.

Proof. (a) $\log \Pi_n$ is a sum of i.i.d. bounded nonnegative increments. The definition of N_A gives $\log \Pi_{N_A} \geq \log A - \log N_A$ and $\log \Pi_{N_A - 1} < \log A - \log(N_A - 1)$, and the strong law of large numbers gives the limit; convergence in L^1 follows from $N_A \leq \min\{n: \log \Pi_n \geq \log A\}$ and the elementary renewal theorem. (b) Every bet factor is an e-variable for the two-point law of Supplement S1.6 at y, so $\pi(y) f_\lambda(y) \leq 1$, that is $\log f_\lambda(y) \leq \log(1/\pi(y))$; taking $\mathbb{E}_Q$ and the maximum over λ gives $\mathrm{KL}_{\inf} \leq \gamma_Q$ by Lemma 20. Equality forces $\log f_{\lambda^*}(Y) = \log(1/\pi(Y))$ almost surely, which on the decrease side fails on $\{Y > 1\}$ unless $Q(Y > 1) = 0$; conversely, if $Q(Y > 1) = 0$ the bet at $\lambda_{\max}$ has factor $(B - y)/(B - 1) = 1/\pi(y)$. The increase side is the same with the bet at 1. (c) For $\varphi = \mathbb{P}(T \leq n \mid y)$, $\Lambda_n = \log dQ/dP^*$ on the first n steps and any a, $Q(T \leq n) \leq e^a \mathbb{E}_{P^*} \varphi + Q(\Lambda_n > a) \leq e^a n/A + Q(\Lambda_n > a)$; take $n = \lfloor (1 - \varepsilon) \log A/\mathrm{KL}_{\inf} \rfloor$ and $a = (1 - \varepsilon/2) \log A$, and apply the weak law of large numbers. ▫

Part (c) is the change-of-measure argument behind the lower bounds of Lorden (1971) and Lai (1998), here for the validity notion of Theorem 4, for randomised rules and with a single null law. By (b) the floor alone gives the slower rate $\log A/\gamma_Q$, which is Lorden's rate exactly when the post-change law has no mass on the far side of 1, as on the collapse. At practical A the floor is close to the root of $n\gamma_Q = \log(A/n)$ rather than to $\log A/\gamma_Q$: at $A = 10^8$ it is 0.72 to 0.86 of $\log A/\gamma_Q$.

Numbers for Theorem 23. With 10,000 paths of 4000 steps per law, the change at the first step, the true baseline, $c = 3$ and $A = 7800$, the bound for deterministic rules was set by the pathwise floor for strong decreases with light tails (Gaussian halving: floor 55.2 steps, Neyman-Pearson bound over every rule 45.6) and by the Neyman-Pearson part for increases and fat tails (Student t_3 at a standard deviation ratio of 0.8: floor 47.8, bound 147.0). The default grid's mean delays for these two laws, 93.2 and 281.3 steps, agree with the simulations of Supplement S4. Randomised rules gain at most 15% on the bound, and less than 1% for every increase. One side at $2A$, its level in the two-sided monitor, took 1.65 to 2.09 times (default grid) and 1.33 to 1.72 times (best single bet) the bound for one-sided rules valid at levels $n/(2A)$; this is not a comparison with the two-sided guarantee at n/A. Whether the remaining gap is the price of controlling every n at once, which the bound does not pay, or the e-detectors' own loss is not settled. Against the bound for randomised rules the default grid's ratios are 1.81 to 2.24 and the best single bet's 1.53 to 1.80. As a check of the code, no alarm on 180,000 paths came before the pathwise floor. As A grows the gap closes slowly: at $A = 10^8$ the best single bet is 1.22 to 1.40 times the bound for deterministic rules.

The decrease bet at its boundary. Write a decrease bet as $\lambda = \lambda_{\max}(1 - \varepsilon)$, so that its factor is $f_\varepsilon(y) = \varepsilon + (1 - \varepsilon) f_0(y)$ with $f_0(y) = (B - y)/(B - 1)$, and let $q_B = Q(Y = B)$, $G_c = \mathbb{E}_Q[\log f_0(Y); Y < B]$ and $g(\varepsilon) = \mathbb{E}_Q \log f_\varepsilon(Y)$.

Proposition S12. Let Q be i.i.d. with $G_c > -\infty$. (a) $(1 - \varepsilon)G_c + q_B \log \varepsilon \leq g(\varepsilon) \leq G_c + q_B \log \varepsilon + \mathbb{E}_Q[\log(1 + \varepsilon/f_0(Y)); Y < B]$. (b) g is concave, and $\lambda^* < \lambda_{\max}$ exactly when $\mathbb{E}_Q[(B - 1)/(B - Y)] > 1$, which holds whenever $q_B > 0$. (c) At an interior

optimum $\varepsilon^* = q_B/h(\varepsilon^*)$, with $h(\varepsilon) = \mathbb{E}_Q[(f_0(Y) - 1)/f_\varepsilon(Y); Y < B]$. (d) Even where $g(\varepsilon) < 0$, the worst-case delay of the single bet stays bounded as $\varepsilon \to 0$.

Proof. (a) On $\{Y = B\}$, $f_\varepsilon = \varepsilon$; on $\{Y < B\}$, f_ε is an average of f_0 and 1, so $\log f_\varepsilon \geq (1 - \varepsilon) \log f_0$, and $f_\varepsilon \leq f_0 + \varepsilon$. (b) $\log f_\varepsilon$ is concave in ε, and $g'(0+) = \mathbb{E}_Q[(B - 1)/(B - Y)] - 1$, which is $+\infty$ when $q_B > 0$. (c) $g'(\varepsilon) = q_B/\varepsilon - h(\varepsilon)$. (d) The statistic at the end of any block of n steps is at least the product of the block's factors, so with $p_n(\varepsilon) = Q(\sum_{i \leq n} \log f_\varepsilon(Y_i) \geq \log A)$, disjoint blocks give $\mathrm{WADD} \leq n/p_n(\varepsilon)$, and $p_n(\varepsilon) \geq (1 - q_B)^n \, Q'(\sum_{i \leq n} \log f_0(Y_i') \geq \log A + n \log(1/(1 - \varepsilon)))$, with Y' drawn from Q given $Y < B$, which does not tend to 0 as $\varepsilon \to 0$. ▫

For Gaussian returns q_B is 2.0×10^{-9} at a halving, 7.5×10^{-6} at a standard deviation ratio of 0.67 and 1.8×10^{-4} at 0.8, so $\lambda^*/\lambda_{\max}$ is $1 - 2.4 \times 10^{-8}$, 0.99987 and 0.9917 for the falls of Table 1, ε^* is 2.4×10^{-8} to 8.3×10^{-3}, and any $\varepsilon \leq 10^{-3}$ gives a delay within 0.4% of the delay at ε^*. For Student t_3 returns q_B is 0.65% to 2.4%, $\lambda^*/\lambda_{\max}$ is 0.93, 0.78 and 0.55 at ratios 0.5, 0.67 and 0.8, and a bet at $\varepsilon = 10^{-6}$ has negative growth at the last two and is 14% and 53% slower than at ε^* (4000 paths of 6000 steps, $A = 7800$). For fat tails the default grid's largest decrease bet, $0.95\,\lambda_{\max}$, is already beyond the optimum.

S4. Second-order behaviour at practical alarm levels

The first-order formula $\log A/g$ is a poor guide at $A = 7800$. For one bet and the change at the first step, $M_n = e^{S_n} \mathscr{V}_n$ with $S_n = \sum_{i \leq n} \log f_i$ and $\mathscr{V}_n = \sum_{j=0}^{n-1} e^{-S_j} \uparrow \mathscr{V}_\infty < \infty$, so $\log M_n$ is a random walk plus a slowly changing term. Nonlinear renewal theory (Woodroofe 1982; Pollak 1987 for the Shiryaev-Roberts statistic) gives, for one bet,

$$\mathbb{E}[T_A] \approx \frac{\log A + \gamma_{\mathrm{ov}} - \zeta}{g}, \qquad \gamma_{\mathrm{ov}} = \frac{\mathbb{E}[J^2]}{2\,\mathbb{E}[J]}, \quad \zeta = \mathbb{E}[\log \mathscr{V}_\infty],$$

with J the first ascending ladder height of S (so γ_{ov} is the limiting mean overshoot); Proposition S20 below states the conditions and checks them. The value of ζ is a further, heuristic step. Approximating S by a Brownian motion with drift g and variance s^2 per step, Dufresne's (1990) identity for exponential functionals gives $\zeta \approx \log(2/s^2) - \psi(2g/s^2)$, with ψ the digamma function. For a growth-optimal bet and a small change $s^2 \approx 2g$, so $\zeta \approx \log(1/g) + \gamma_E$ with Euler's constant $\gamma_E = 0.5772$, and

$$\mathbb{E}[\mathrm{delay}] \approx \frac{\log(Ag) - \gamma_E + \gamma_{\mathrm{ov}}}{g}.$$

The memory term ζ is the reason the first-order formula fails: for a 25% increase in standard deviation $g \approx 0.05$, so $\zeta \approx 3.6$ while $\log A = 8.96$.

Proposition S20 (second-order expansion for one bet). Let the returns be i.i.d. from the first monitored step, take one side with one fixed bet whose factor $f = f(Y)$ has $g = \mathbb{E} \log f > 0$, and suppose that the law of Y has a part with no atoms, as for Gaussian or Student t returns. Let T_A be the first n with $M_n \geq A$. Then ζ and γ_{ov} are finite and $\mathbb{E}[T_A] = (\log A + \gamma_{\mathrm{ov}} - \zeta)/g + o(1)$ as $A \to \infty$.

Proof. Write $\log M_n = S_n + \xi_n$ with $\xi_n = \log \mathscr{V}_n$, so that T_A is the first passage of $S_n + \xi_n$ over $\log A$. The increments $\log f(Y_i)$ are i.i.d. and bounded, because f takes values in a compact subset of $(0, \infty)$, and they are not concentrated on a lattice, because f is continuous and strictly monotone in y and the law of Y has a part with no atoms. The term ξ_n is nonnegative (the term $j = 0$ of $\mathscr{V}_n$ is 1), nondecreasing, and converges almost surely to $\xi_\infty = \log \mathscr{V}_\infty$, since $S_j/j \to g > 0$. Its tail is exponential: for $u \in (0, 1)$, $(\sum_j x_j)^u \leq \sum_j x_j^u$ for $x_j \geq 0$,

so $\mathbb{E}[\mathscr{V}_\infty^u] \le \sum_{j\ge 0} (\mathbb{E}f^{-u})^j$, and $\mathbb{E}f^{-u} < 1$ for small $u > 0$ because f is bounded away from 0 and the derivative of $u \mapsto \mathbb{E}f^{-u}$ at 0 is $-g < 0$; hence $\mathbb{P}(\xi_\infty > x) \le Ce^{-ux}$ for some constant C, and $\zeta = \mathbb{E}\xi_\infty$ is finite. So ξ_n changes slowly (it converges), $\max_{k\le n} \xi_{n+k} \le \xi_\infty$ is uniformly integrable, $\xi_n \ge 0$ excludes large negative values, and an early stop is rare: with $o_{\max}$ the largest log factor and $\varepsilon o_{\max}/g \le 1/2$, $T_A \le \varepsilon \log A/g$ forces $\xi_\infty \ge (\log A)/2$, whose probability is $O(A^{-u/2}) = o(1/\log A)$. These are the conditions under which the nonlinear renewal theorem gives $\mathbb{E}[T_A] = (\log A + \mathbb{E}R_A - \mathbb{E}\xi_\infty)/g + o(1)$, where the mean overshoot $\mathbb{E}R_A$ tends to γ_{ov}, finite because the increments are bounded and not lattice (Woodroofe 1982; Pollak 1987 for the Shiryaev-Roberts statistic). We could not read Woodroofe (1982) in full and use the conditions as secondary accounts state them, so Proposition S20 is subject to the caveat of footnote 1 of the main text. ▫

Table S2 compares the predictions with simulation for all 20 cases (true baseline, $c = 3$, one side at $A = 7800$, change at the first monitored step, 3000 paths per case, mean delays).

Case	g_{grid}	first order $\log A/g_{\text{grid}}$	proved bound (Theorem 24(a))	simulated, default grid	second order, best single bet	simulated, best single bet
Gaussian, sd ratio 1.25	0.0502	179	233	122	109	107
Gaussian, sd ratio 1.5	0.161	56	75	48	42	42
Gaussian, sd ratio 2	0.413	21.7	30	23.1	18.4	18.7
Jumps, sd ratio 1.25	0.0473	190	247	128	115	115
Jumps, sd ratio 1.5	0.154	58	78	51	44	43
Jumps, sd ratio 2	0.401	22.3	31	23.4	18.9	19.2
Student t_5, sd ratio 1.25	0.0320	280	362	178	158	158
Student t_5, sd ratio 1.5	0.110	81	108	66	58	59
Student t_5, sd ratio 2	0.309	29	39	29	24.1	24.0
Gaussian, sd ratio 0.8	0.0340	264	314	194	157	156
Gaussian, sd ratio 0.67	0.0598	150	179	124	99	99
Gaussian, sd ratio 0.5	0.0839	107	127	93	74	74
Jumps, sd ratio 0.8	0.0261	343	410	215	186	182
Jumps, sd ratio 0.67	0.0514	174	208	134	113	113
Jumps, sd ratio 0.5	0.0752	119	142	99	83	83

Case	g_{grid}	first order $\log A/g_{\text{grid}}$	proved bound (Theorem 24(a))	simulated, default grid	second order, best single bet	simulated, best single bet
Student t_5, sd ratio 0.8	0.0222	403	481	240	211	215
Student t_5, sd ratio 0.67	0.0540	166	198	124	105	106
Student t_5, sd ratio 0.5	0.0967	93	111	81	66	66
Gaussian, sd ratio 1.1	0.00929	964	1221	455	405	417
Gaussian, sd ratio 0.9	0.0101	886	1054	434	389	392

Table S2. Simulated data. Predicted and simulated mean delays in steps, one side at level A. Standard errors are below 1.5% of the mean.

In all 20 cases the second-order prediction for a single bet is within 3% of its simulated mean delay, the first-order formula overstates the delay of small changes by up to 112%, and the proved bound is 1.3 to 2.7 times the simulated delay. The default grid costs 9% to 26% against the best single bet. With the change after 500 pre-change steps, as in Table S5, the delays are 17% to 34% shorter, because the Shiryaev-Roberts statistic enters the change with a head start, and they match Table S5 to within about 10%. The asymptotic regime arrives very slowly: at $A = 10^8$ even the best single bet is 5% to 24% faster than $\log A/\mathrm{KL}_{\text{inf}}$ in 15 cases (Gaussian, Student t_3 and jump returns), and the second-order formula explains the gap to within 0.024 in ratio for every $A \geq 10^4$ (not shown).

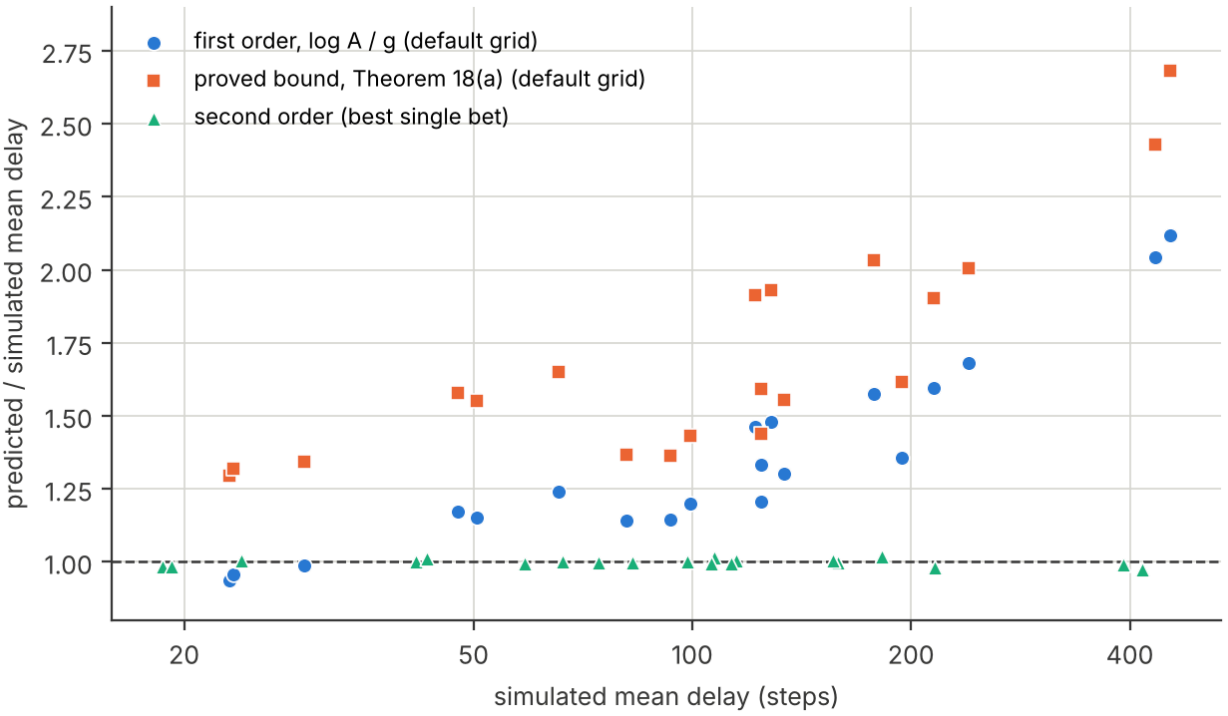


Figure S1. Simulated data. Ratio of predicted to simulated mean delay for the first-order formula, the proved bound of Theorem 24(a) (both for the default grid) and the second-order formula (best single bet), for the 20 cases of Table S2.

S5. Details for Section 6: truncation constant, slack of the bound, delays, competitors and the confidence-sequence baseline

The truncation constant. For increases on clean data the delay barely changes for $c \geq 4$, and $c = 3$ costs 3%, 6% and 12% at sd ratios 1.25, 1.5 and 2 against no truncation. With jumps (probability 0.005 per step, standard deviation 8 times the diffusive one) the growth at sd ratio 1.25 peaks at c between 3.25 and 4.25 depending on the change size, and without truncation it falls to 29% of the $c = 3$ value. A burst of jumps at constant volatility (jump probability raised from 0.005 to 0.02) moves the $c = 3$ detector only 3.5% as much as a real 25% volatility increase, against 39% without truncation. For decreases the speed limit dominates and a smaller c is usually faster: with the true baseline and Gaussian data, $c = 2$ shortens the mean decrease delay by 26% to 49% relative to $c = 3$. The two sides need not share c: with a known baseline, or with the confidence-sequence baseline, which the monitor supports for the split by running the lower bound on $\min(r^2, \tau_\downarrow^2)$ and the upper bound on $\min(r^2, \tau^2)$, each at level $\delta/2$, each side is a valid e-detector for its own truncated statistic and the two-sided rule of Proposition 5 needs only each side's guarantee, so c of about 3 to 4 for increases and 1.5 to 2 for decreases is a legitimate design. The decrease null then concerns the mean of $\min(r^2, \tau_\downarrow^2)$ for the smaller level $\tau_\downarrow$; volatility that has not fallen implies it under a fixed innovation shape (Lemma 1), but a smaller clip reacts more to changes of shape at constant volatility, such as a larger share of zero returns. In the experiments the split was run with a 1440-step plug-in baseline, which no proof covers: over all 18 no-change cells the one-day rate was 5.0% to 6.7%, and 5.0% to 6.05% with $c_\downarrow = c$, so the excess over the 5% bound that holds with the true baseline is there without the split and comes from the baseline. Table S3 gives the median decrease delays. For the Gaussian law the split was also run with the confidence-sequence baseline ($\delta = 0.01$, $A = 36,000$, two confidence sequences at $\delta/2$ when $c_\downarrow \neq c$, on the same paths): its one-day null rate was at most 0.05% for $c_\downarrow = 3$, 2 and 1.5, against 5.65% to 6.10% for the plug-in, and its median decrease delays were 24% to 142% longer (446.5 against 247 steps at a 20% fall with $c_\downarrow = 3$); $c_\downarrow = 2$ was still faster than $c_\downarrow = 3$ in all three cells, and $c_\downarrow = 1.5$ was the fastest for falls to 0.67 and 0.5 times but slower than $c_\downarrow = 3$ for a 20% fall (by 45 steps in mean delay, standard error 9.8), where with the plug-in it was faster. Rounded prices use a tick fixed in units of the pre-change standard deviation, so that more returns are zero after a fall (at a tick of 0.5 standard deviations, 19.7% before the change and 24.5%, 29.1% and 38.3% after falls to 0.8, 0.67 and 0.5 of the standard deviation); a first version of the experiment let the tick shrink with volatility and overstated the gain.

Returns (calibration scale)	**sd × 0.8: $c_\downarrow$ = 3 / 2 / 1.5**	**sd × 0.67**	**sd × 0.5**
Gaussian	247 / 176 / 185	158 / 85 / 73	118 / 58 / 40
Student t_4	334 / 289 / 277	160 / 117 / 106	100 / 62 / 48
Jumps	273 / 196 / 206	161 / 88 / 74	119 / 59 / 40
Ticks of 0.25 sd (scale 1.11)	292 / 185 / 188	191 / 98 / 77	144 / 69 / 45
Ticks of 0.5 sd (scale 0.74)	**188 / 211 / 251**	99 / 80 / 86	70 / 39 / 35
Ticks of 0.75 sd (scale 1.11)	295 / 201 / 207	192 / 100 / 81	144 / 69 / 47

Table S3. Simulated data. Median decrease delay in steps with a separate decrease clip $c_{\downarrow}$ and $c = 3$ for increases, plug-in baseline from 1440 steps, $A = 28,800$, change at the first monitored step; 600 paths per cell for the continuous laws and 2000 for rounded prices, the same paths for the three values of $c_{\downarrow}$; every path detected. "Scale" is the median calibration scale (scaled MAD) in units of the true pre-change standard deviation.

With $c_{\downarrow} = 2$ the median delay fell in all nine cells of the continuous laws, by 13% to 51% (for Gaussian data the mean delay fell by 25%, 44% and 50%, close to the 26%, 43% and 49% that the theory predicts with the true baseline), and in all six cells with ticks of 0.25 and 0.75 standard deviations, by 32% to 52%. At a tick of 0.5 standard deviations it was 12% slower for a 20% fall (median 211 against 188 steps; paired difference of 29 steps, standard error 1.7; slower on 66% of paths) and faster for larger falls, by 19% and 44%. $c_{\downarrow} = 1.5$ was no better than $c_{\downarrow} = 2$ for a 20% fall except under Student t_4 returns, and at the tick of 0.5 standard deviations it was 34% slower than $c = 3$. A reading of the tick case, not a proof: with rounded returns the calibration scale is, on almost every path, a whole number of ticks times 1.4826, so at a tick of 0.5 standard deviations it is 0.74 standard deviations against 1.11 at ticks of 0.25 and 0.75; $c = 3$ then already clips at about 2.2 standard deviations, near the best clip for a 20% fall, and $c_{\downarrow} = 2$ at about 1.5, past it, while the growing share of zero returns after the fall adds to the effect. On the five cryptocurrencies of Section 7, which have few zero returns (1.0% to 4.3% of deseasonalised returns), the choice $c_{\downarrow} = 2$ was neutral to slightly positive for the confirmation rate of decrease alarms, within sampling noise. On data with a coarse tick, $c_{\downarrow} = 2$ should be compared with $c = 3$ at that tick before it is used.

The slack of the false alarm bound. An exact optional stopping identity locates the slack by writing the ratio of the false alarm rate to m/A as a product of a stopping factor, the share F_2 of expected wealth held on paths that crossed A, and an overshoot factor F_3. For the default two-sided monitor (40,000 paths, rate 1.03%) the stopping factor is close to 1, $F_2 = 0.24$ and $F_3 = 0.80$: the slack is overshoot (a factor of 1.25) times wealth stranded on paths that never alarm (a factor of about 4). Of the stranded wealth, 60% is held by the decrease side and 31% by the three smallest increase bets, which build little wealth within a day on Gaussian data. The split of the budget between the two sides costs almost nothing (both sides alarm on the same day in 0.00% of days). The bound itself is nearly sharp: for a single increase bet of 0.5, dynamic programming over the whole nonparametric null finds a law with a one-day false alarm rate of 4.73%, 95% of the bound, against 3.52% for Gaussian data. For the default mixture the best i.i.d. law found reaches 42% of the bound for the increase grid alone, and an adaptive law reaches 52% for the two-sided detector (companion note); a certified sharper bound for mixtures is open. Provable gains are possible by splitting the budget unevenly between the sides and by dropping bets that cannot use it.

Table S4 gives the false alarm rates with the true baseline behind Figure 1 (left).

Horizon m (steps)	**bound m/A**	**Gaussian**	**Student t_3**	**jumps**	**price ticks**
98	1.3%	0.00%	0.18%	0.12%	0.05%
390	5.0%	1.1%	1.4%	1.3%	1.4%
1950	25%	12.8%	14.3%	13.2%	13.7%
3900	50%	29.6%	30.6%	29.7%	31.2%
7800	100%	55.9%	55.9%	56.4%	56.9%

Table S4. Simulated data. False alarm rate within m steps with the true baseline, two-sided rule, $A = 7800$, 4000 paths per law; Student t_3 has an infinite fourth moment. 95% intervals at one day are about ± 0.3 points.

Bet design. The quarticity estimator can enter the detector through the bet. A plug-in bettor with $\lambda_t = (\hat{\mu}_t - 1)/(\hat{v}_t + (\hat{\mu}_t - 1)^2)$, where $\hat{\mu}_t$ and $\hat{v}_t$ are kernel-weighted local mean and variance of Y over the last 60 steps (so that the local second moment is a truncated local quarticity), added with weight one half to the grid, was slower than the grid alone in 14 of 18 cells, by up to 9%. A computation of growth rates explains this negative result: the quadratic approximation behind the bet makes it 1.4 to 4.4 times too small for increases, and it keeps only 49% to 94% of the attainable growth. A grid designed by minimising the worst-case ratio to the best single bet over a range of changes (sd ratio 1.2 to 2 up, 0.5 to 0.85 down) and four tail shapes, $\lambda \in \{0.16, 0.39\}$ up and $\lambda/\lambda_{\max} \in \{0.5, 0.85\}$ down, was at least as fast as the default in all 48 validation cases in its design range, by 0.4% to 10.5% (median 5%), but up to 17% slower for small increases (sd ratio 1.1) under fat tails. A grid tuned on Gaussian data alone was up to 18% faster on Gaussian or jump data and up to 55% slower under Student t_3 data. The gains from bet design are modest, and tuning to one tail shape is risky.

Delays in the designs of Table 2. Table S5 gives the median delays of Section 6.2 after a change 500 steps into monitoring.

Detector	Gaussian	jumps	Student t_5
Increases, sd ratio 1.25, 1.5, 2			
Robust e-detector (default grid)	87, 35, 17 ($\leq$ 4.3%)	102, 36, 18 ($\leq$ 4%)	122, 49, 22 (3%)
Robust e-detector, CUSUM type	154, 51, 23 ($\leq$ 1%)	171, 52, 23 ($\leq$ 0.3%)	215, 72, 32 ($\leq$ 0.3%)
E-detector on raw r_t^2	84, 33, 15 ($\leq$ 4.3%)	246, 60, 23 ($\leq$ 5.3%)	134, 51, 21 ($\leq$ 4%)
Gaussian likelihood-ratio e-detector	78, 26, 10 ($\leq$ 5.3%)	61, 20, 9 (**74 to 78%**)	46, 10, 7 (**96 to 98%**)
Page CUSUM	64, 24, 10 (6 to 10%)	54, 20, 11 (**74 to 79%**)	50, 9, 6 (**97 to 99%**)
Decreases, sd ratio 0.5, 0.67, 0.8			
Robust e-detector (default grid)	71, 92, 147 ($\leq$ 2%)	72, 98, 160 (1%)	68, 96, 168 ($\leq$ 3%)
Gaussian likelihood-ratio e-detector	23, 46, 104 (6 to 9%)	23, 49, 121 (3 to 7%)	29, 81, 282 ($\leq$ 2%)

Table S5. Simulated data. Median delay in steps after a change at step 500 of monitoring; in brackets, the share of paths with a false alarm during the 500 steps before the change, with plug-in baselines from 2000 steps. These shares cover 500 steps, not a day: with a known baseline Theorem 4 would bound them by $500/7800 = 6.4\%$ per side. 300 paths per cell. The decrease detector on raw r_t^2 does not exist (Section 3.3).

BOCPD and the two-window scan. BOCPD of Section 6.2 uses a Gaussian model in each regime. The scan compares the realised variance of the k returns before and after each candidate time, dropping returns above 4 times a local bipower scale; it is our own construction, not the published test of Bibinger, Jirak and Vetter (2017). On clean Gaussian data BOCPD took a median of 13 steps against the e-detector's 20 for a

doubling of the standard deviation. With the confidence-sequence baseline the e-detector's one-day false alarm rate in the online comparison of Section 6.2 was 0.1% with jumps and 0.0% with Student t_5 returns. Offline, on windows of 2000 steps, the scan with $k = 50$ or 125 found a 25% increase in 6% to 30% of windows against 100% for the e-detector; the gap is window length, not betting, since with $k = 500$ the scan found 99%. The e-detector needs no comparison window and no change location, only a change-free calibration stretch, and it runs online, at about 11 microseconds per step against about 300 for BOCPD.

Competitors calibrated for the data at hand: definitions. All rules see the same calibration window (2000 steps) and monitored stream. The *classical rules on the e-detector's statistic* use $Y_t = X_t/m_0$ with the plug-in τ and m_0. Page's two-sided CUSUM is $G_t^{\uparrow} = \max(0, G_{t-1}^{\uparrow} + Y_t - 1 - k_\theta)$ and $G_t^{\downarrow} = \max(0, G_{t-1}^{\downarrow} + 1 - Y_t - k_\theta)$, with the reference value k_θ of the Gaussian likelihood ratio for a change of the variance by a factor θ ($k_\theta = \theta \log \theta/(\theta - 1) - 1$ for $\theta > 1$ and $1 - \theta \log \theta/(\theta - 1)$ for $\theta < 1$): "small" for standard deviation ratios 1.25 and 0.8 ($k_\theta = 0.240$ and 0.207) and "medium" for 1.5 and 0.67 (0.460 and 0.348). The *calibrated betting rule* is the e-detector's own Shiryaev-Roberts statistic with a calibrated threshold in place of $2A$. Thresholds come either from 20,000 clean Gaussian paths, for the e-detector's own first-day rate (1.3%) or for 5%, or from a bootstrap of each path's calibration window (1000 replicates of a resampled window and a resampled 390-step stream, the threshold of each side being the $1 - \alpha/2$ quantile of the replicate maxima for a target α), which assumes i.i.d. returns but no law. The *Student t BOCPD* has a Student t observation model with $\nu = 3$ and an unknown variance on a grid, the prior and short-run window ($w = 250$) of Section 6.2, and a threshold matched to the e-detector's rate on Gaussian paths. The *training-sample rules* use the calibration window as a training sample of size n_{tr} and the detector Σ_j, the sum of Z_t minus its training mean over the first j monitored steps, with $Z_t = r_t^2$ (the setting of the published procedure) or $Z_t = X_t$; they alarm when $|\Sigma_j| \geq c_H \, \hat{\sigma} \sqrt{n_{\mathrm{tr}}} \, (1 + j/n_{\mathrm{tr}}) \, (j/(n_{\mathrm{tr}} + j))^{\gamma}$, where $\hat{\sigma}$ is a Bartlett long-run standard deviation of the training Z (bandwidth 12), $\gamma = 0.25$ and the critical value c_H comes from simulation. This is the boundary of monitoring in linear models of Horváth, Hušková, Kokoszka and Steinebach (2004), applied to squared returns as in Horváth, Kokoszka and Zhang (2006); asymptotic critical values for open-ended monitoring spend almost none of their 5% in the first day for $\gamma \leq 0.25$. A second boundary, that of OLS-CUSUM monitoring in the R package strucchange (Zeileis, Leisch, Hornik and Kleiber 2002), is a check of the type of Chu, Stinchcombe and White (1996). The papers of Horváth, Hušková, Kokoszka and Steinebach (2004), Horváth, Kokoszka and Zhang (2006), Chu, Stinchcombe and White (1996) and Knoblauch, Jewson and Damoulas (2018) could not be read in full (footnote 1 of the main text), so these rules are of the type of the published procedures, not reproductions, and no β-divergence BOCPD was run.

False alarms of the calibrated competitors. On 4000 fresh paths per law (bootstrap rules on 2000), calibrated on Gaussian data, the CUSUMs on X had 5.0% to 8.2% false alarms in the first day under Student t_3 and t_5 returns (target 1.3%) and 14% to 22% (target 5%), and the training-sample rule on r_t^2 10% to 15% under Student t_3 returns and jumps (targets 1.3% and 5%). Per-path bootstrap thresholds gave 1.4% to 2.2% (target 1.3%) and 4.9% to 6.3% (target 5%) under the six i.i.d.-type laws, significantly above target in 5 of these 24 cells (the CUSUM on X at the 1.3% target under Student t_3 returns, jumps and price noise, and both rules at the 5% target under price noise), and the training-sample rule on X held its rate under every i.i.d.-type law (1.1% to 1.6% at a 1.3% target). Under GARCH and slow stochastic volatility every rule alarmed on most

paths, the e-detector included; the training-sample rules, whose long-run variance absorbs part of the clustering, alarmed least (21% to 52% in the first day).

Delays of the calibrated competitors. In the design of Table S5 (1000 paths per law and factor, bootstrap rules on 500), at the e-detector's Gaussian rate the calibrated betting rule detected increases of the standard deviation by 25%, 50% and 100% in 106, 41 and 20 steps (e-detector 100, 39 and 19; here each side of the e-detector alarms at $2A$, against A in Table S5) and a fall to 0.8 times in 144 steps (e-detector 162). The calibrated rule is slower for increases although its false alarm rate before the change is higher (3.2% against 2.6%), because it splits its target first-day rate equally between the two sides, while the e-detector's realised first-day rate falls mostly on the increase side; Page's CUSUM on X was as fast for increases and 15% to 40% faster for decreases. Against thresholds that spend the full 5% in the first day, at 4.5% to 4.9% first-day false alarms, the e-detector took 1.1 to 1.3 times as long for increases and 1.4 to 2.1 times for decreases, and such thresholds hold only if the law is known. With bootstrap thresholds under Student t_3 returns the e-detector detected increases 16% to 24% faster than the CUSUM, and the CUSUM decreases 3% to 10% faster. The training-sample rules were about twice as slow: on r_t^2 at the matched Gaussian rate 199, 82 and 35 steps for increases and 330, 204 and 140 for decreases. This is a property of the boundary: Σ_j accumulates the departure from the training mean since monitoring began, and with $c_H = 1.83$, $\hat{\sigma} = \sqrt{2}$ (Gaussian r_t^2) and $n_{\mathrm{tr}} = 2000$ the boundary is about 104 squared-return units at $j = 600$, so an increase of the variance by 0.5625 per step (standard deviation ratio 1.25) needs about 185 steps after the change, plus the growth of the boundary meanwhile, while Page's CUSUM and the e-detector restart their evidence whenever it turns against a change.

The Student t BOCPD. On the paths of the online comparison of Section 6.2 it had 0.7% false alarms in the first day under jumps and 3.3% under Student t_5 returns, against 73.2% and 53.2% for the Gaussian BOCPD, but 4.9% under Student t_3 and 0.5% under Gaussian data with the same threshold; within 1500 steps it found only 56% of 25% increases and 77% of 20% decreases (e-detector 98%), while it was faster for halvings of the standard deviation (27 against 81 steps).

Width of the confidence-sequence baseline. From 292 observations its median upper bound sits 37% above the true mean for Gaussian returns, against 29% for the best constant bet for the true law, a benchmark that knows the distribution; at 1440 steps and more it is within 3.5 points of that benchmark.

Delays with the confidence-sequence baseline. The false alarm check of Section 6.3 used three laws, calibration of 390, 1440 and 5000 steps, and 10,000 paths in each of the nine settings. Table S6 gives the median delays of Section 6.3.

Returns, sd ratio	390 steps: plug-in / CS	1440 steps: plug-in / CS	5000 steps: plug-in / CS
Gaussian, 1.25	90 / 645 (76%)	88 / 222	88 / 134
Gaussian, 1.5	36 / 103	35 / 59	35 / 46
Gaussian, 2	17 / 31	17 / 24	17 / 20
Gaussian, 0.8	143 / 633 (85%)	137 / 269	139 / 199
Gaussian, 0.67	89 / 207	87 / 135	87 / 114
Gaussian, 0.5	64 / 122	65 / 92	64 / 80
Student t_5, 1.25	128 / 904 (35%)	131 / 391	130 / 219

Table S6. Simulated data. Median delay in steps after a change at step 500, plug-in against confidence-sequence (CS) baseline, by calibration length, 2000 paths per cell. Medians are over the paths that detected the change within 2500 steps; where the share detected (in brackets, shown when below 90%) is low, the CS median is biased downwards. With 390 calibration steps the plug-in detects only 92% to 94% of increases in this sense, because it alarms before the change on 6% to 7.5% of paths.

S6. Clustering, forecasts, the GARCH bound and price noise

Volatility clustering. Under GARCH(1,1) returns with constant unconditional variance, the conditional mean of X_t moves above and below m_0, so the nulls of Section 2 are false. With the true unconditional baseline and $A = 28,800$ for a 1440-step day, the share of days with an alarm is 78% at persistence $\alpha + \beta = 0.95$ (half-life 14 steps) and 100% at 0.99, 0.999 and 0.9999 (Figure 1, right). A band $b = 0.5$ absorbs short-lived clustering (3.6% at persistence 0.95), but no reasonable static band absorbs persistent clustering: even $b = 1$ on the increase side leaves 12% to 20% of days with an alarm at persistence 0.99 or more. For slow log-normal stochastic volatility (log-sd 0.15, half-life 390 steps), whose first-day alarm rates at band 0 were 24.7% (increase) and 26.7% (decrease), a band of 0.25 brought them to 8.3% and 5.0% while 90% of doublings and 94% of halvings of volatility were still detected (68% and 66% at band 0, where false alarms often came first), with median delays of 28 and 111 steps against 19 and 74 at band 0, and a band of 0.5 to 1.7% and 0.0% with 97% and 100% detected; the band is a design choice, not a guarantee. These alarms are not errors of the proof: persistent GARCH clusters are changes in conditional volatility, and the monitor reports them.

Forecast-relative monitoring. Monitoring $r_t/\hat{\sigma}_t$ with a GARCH(1,1) forecast fitted on a separate path and then frozen brings the one-day false alarm rate down to 3.0% to 6.2% at all three persistences (600 paths, plug-in baseline from one day), the level of the oracle with the true conditional volatility and the same plug-in baseline (4.7% to 5.3%). This is not a proved guarantee: the proof needs the true or a confidence-sequence baseline and a correct forecast model, and with the true baseline the one-day rate for i.i.d. returns is 1.1% to 1.4%, so the plug-in roughly quadruples it. An EWMA forecast is only as valid as its match to the clustering (half-life 360 steps: 72% to 100% at band 0). The price is the bound of Proposition S13 below. At persistence 0.99, a doubling of the long-run variance was detected within one day on 34% to 43% of paths and within five days on 93% to 94%, with median delays of 1523 to 1791 steps; without a break the same monitors alarmed within one day on 4% to 6% and within five days on 38% to 44% of paths, the latter far above the 25% that Theorem 4 guarantees with a known baseline. A fourfold increase of the long-run variance was detected within a day on 89% to 90% of paths (median 612 to 712 steps). For the oracle, whose standardised returns are exactly i.i.d., the excess over five days is the plug-in effect of Section 4.1, not a property of the forecast.

Price-level microstructure noise. With i.i.d. noise on log prices, $\mathbb{E}[r_t^2 \mid \mathcal{F}_{t-1}]$ depends on the last noise value, so squared returns are positively autocorrelated and the null fails without any change in volatility. At one-second sampling (23,400 steps per day, A set to give 5% for the first day, plug-in baseline from two days) the one-day false alarm rate of the increase side on raw returns rose from 5.0% without noise to 8.5%, 18.0%, 31.0% and 41.5% at noise-to-signal ratios 0.5, 1, 2 and 4 (200 days per cell); overlapping pre-averaged returns alarmed on every day, because consecutive windows share most of their prices. Pre-averaging over non-overlapping blocks of prices, skipping the one return between blocks, gives each block fresh noise values, so the null holds for the block

statistic whenever the spot variance and the noise law are unchanged, and Theorems 3 and 4 apply to it for every block of $n_b \geq 2$ prices; the block monitors were below the bound or within Monte Carlo error of it in all 58 cells tested, and with $n_b = 10$ were about 8 times slower than raw returns without noise and faster than them from a noise ratio of about 2. One-minute crypto returns were not tested for noise of this kind; the effect is largest at the highest frequencies.

Proposition S13 (a bound on the ratio of means after a break). Let $\alpha, \beta > 0$ with $\alpha + \beta < 1$ and $\omega > 0$. Before a break the returns follow GARCH(ω, α, β), and the forecast is the pre-break filter $s_t = \omega + \alpha r_{t-1}^2 + \beta s_{t-1}$, run without change after the break. From the break on, the returns follow GARCH($\theta\omega, \alpha, \beta$) with $\theta > 0$: $r_t = \sigma_t \varepsilon_t$, $\sigma_t^2 = \theta\omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$, with i.i.d. ε_t of mean 0 and variance 1. Then $\Delta_t = \sigma_t^2 - s_t$ satisfies $\Delta_t = (\theta - 1)\omega + \beta \Delta_{t-1}$ deterministically, so $\Delta_t \to (\theta - 1)\omega/(1 - \beta)$ geometrically at rate β, and in the post-break stationary regime

$$\rho(\theta) := \frac{\mathbb{E}[r_t^2]}{\mathbb{E}[s_t]} = \frac{\theta(1 - \beta)}{1 - \alpha - \beta + \alpha\theta},$$

which does not depend on ω, increases in θ, and satisfies $\rho(\theta) < (1 - \beta)/\alpha$ for every θ, however large the break.

Proof. Subtracting the two recursions, the terms αr_{t-1}^2 cancel, which gives the recursion for Δ_t. In the post-break stationary regime $\mathbb{E}[r_t^2] = \mathbb{E}[\sigma_t^2] = \theta\omega/(1 - \alpha - \beta)$ and $\mathbb{E}[s_t] = \mathbb{E}[\sigma_t^2] - (\theta - 1)\omega/(1 - \beta)$. Dividing and simplifying gives $\rho(\theta)$. Its derivative in θ has the sign of $(1 - \beta)(1 - \alpha - \beta) > 0$, and $\rho(\theta) \to (1 - \beta)/\alpha$ as $\theta \to \infty$. ▫

Proposition S13 is the mirror image of the finding that unmodelled shifts make GARCH fits look nearly integrated (Diebold 1986; Lamoureux and Lastrapes 1990; Mikosch and Starica 2004). Two exact facts sit behind it. A scale break ($r_t \mapsto \sqrt{\theta}\, r_t$) and an intercept break ($\omega \mapsto \theta\omega$) lead to the same post-break model and differ only in the transient right after the break. And the filter absorbs most of the break through its αr^2 term: at persistence $\alpha + \beta = 0.99$ with $\alpha = 0.05$, a doubling of the long-run variance appears after standardisation as an increase of the ratio of means to $\rho(2) = 1.091$, and no break can push it above $(1 - \beta)/\alpha = 1.2$. Here $(1 - \beta)/\alpha$ bounds the post-break ratio of the mean squared return to the mean forecast; it is not a fraction of the shift. The monitor sees the mean of the ratio, $\mathbb{E}[r_t^2/s_t]$, which is a little larger by Jensen's inequality (1.116 for $\theta = 2$ in simulation) and which stayed below 1.28 for θ up to 1000 at this persistence. At persistence 0.95 and $\theta = 1000$ the mean of the ratio, 2.20, exceeds the bound 2.0 on the ratio of means, so the bound does not cover the quantity the monitor sees; we have a proof only for the ratio of means. Figure S2 compares the closed form with simulation: the simulated ratio of means, pooled over paths, agrees with $\rho(\theta)$ within two standard errors at every point. (An average of per-path ratios is biased upward when the path means are noisy; at persistence 0.999 it exceeded the bound, which is why we pool.)

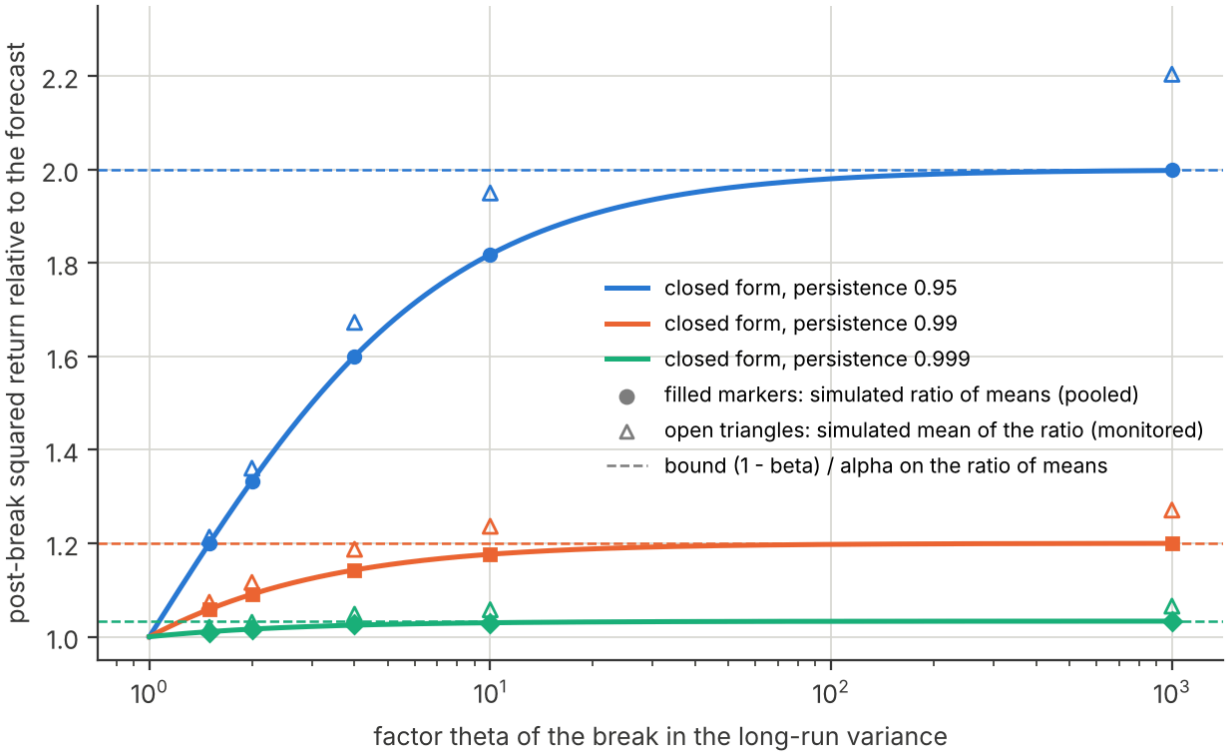


Figure S2. Simulated data. The ratio of means $\rho(\theta)$ after a break of factor θ in the long-run variance (lines, closed form; filled markers, simulation pooled over paths, with bars of two bootstrap standard errors) and the mean of the ratio that the monitor sees (open triangles), with the bounds $(1-\beta)/\alpha$ on the ratio of means (dashed), for three persistences. The mean of the ratio is not covered by the bound and exceeds it at persistence 0.95 for large breaks. 200 paths of 18,000 post-break steps per point; all standard errors are below 0.002.

An information rate. With Gaussian innovations the per-step Kullback-Leibler divergence between the post-break and pre-break conditional laws of r_t is $\frac{1}{2}(q_t - 1 - \ln q_t)$ with $q_t = \sigma_t^2/s_t$. Let I be its average over the post-break path. By Lai (1998), any detector whose average run length to a false alarm is at least A needs, as $A \to \infty$ and under his condition that the post-change log-likelihood ratio grows at rate I uniformly in the change time, a worst-case delay of at least $(\ln A/I)(1+o(1))$. At $A = 28,800$ this is an order-of-magnitude guide, not an exact bound. Table S7 gives the numbers.

$\alpha+\beta$	θ	$\rho(\theta)$	**simulated mean of ratio**	I **(nats per step)**	$\ln A/I$ **(days of 1440 steps)**
0.95	2	1.333	1.361	0.028	0.26
0.99	1.5	1.059	1.074	0.0015	4.7
0.99	2	1.091	1.116	0.0037	1.9
0.99	4	1.143	1.187	0.0093	0.77
0.99	1000	1.200	1.271	0.019	0.38
0.999	2	1.016	1.031	0.00033	21.4

Table S7. Simulated data. Signal left after a frozen GARCH(1,1) forecast and the information rate of a break, $A = 28,800$; I to two significant figures.

The observed median delays of the forecast-relative monitors (1523 to 1791 steps for a doubling, Section 6.4) are about 0.6 times the step at which the cumulative information reaches $\ln A$; the memory term of Section 5.3 can explain a factor of this size on its own (in an i.i.d. version of the same steady state with the true baseline the detector alarms at

0.41 to 0.58 times $\ln A/\mathrm{KL}$), but we did not rerun the GARCH setting to separate it from the effect of the plug-in baseline.

S7. Real-data details

S7.1 The Nifty 50 index and India VIX

Data. Nifty 50 one-minute bars from April 2017 to October 2026, sessions of 375 bars, overnight returns excluded. Stretches of five or more minutes with a frozen index value (exchange outages, the two market-wide halts of March 2020, and the closing auction minutes from 3 August 2026) were set to missing, leaving 2,361 sessions and 882,001 returns. The intraday profile is session-aware, so that the close is not smoothed into the open. Pre-registered configuration: calibration 750 bars, band 0.25; development period 2017 to 2021, held-out period 2022 to October 2026, with nothing changed after the held-out period was seen. A configuration was pre-registered, not a primary test, so the results below are secondary.

Results. The monitor raised 0.26 alarms per session in both periods. In the held-out period, 78% of increase alarms were followed by a session more volatile than the baseline (47% at random times) and 76% of decrease alarms by a calmer one (53%), with 95% intervals of 72% to 84% and 69% to 82% from a block bootstrap over sessions (blocks of 5 sessions; Holm $p = 0.0008$); the Gaussian likelihood-ratio monitor's increase alarms were followed by higher volatility only 61% of the time, while its decrease alarms were as informative. India VIX agreed with the alarms on the same day (in the development years VIX closed higher on 68% of days with an increase alarm, against 47% of all days), which holds after Holm over eight comparisons in the development years and for decrease alarms in the held-out years, but not for held-out increase alarms (Holm $p = 0.25$). VIX did not move in the alarm's direction on the next day more often than on other days (45% to 48% against 46% to 47%): the agreement is concurrent, not a lead. The event analysis below was not part of the pre-registered configuration. Among event days, 9 of 11 Union Budget sessions had an alarm, against 25% of ordinary sessions (Fisher $p = 0.0001$; Holm over the five event types 0.0006; a circular-shift test of the event calendar against the alarm series, which keeps the clustering of alarms, gives $p = 0.0004$ and Holm 0.0021), and so did all 7 budget sessions at which the monitor was already watching at the open, a restriction chosen after the first results. Reserve Bank of India policy days showed no significant excess (32% against 25%, $p = 0.36$), consistent with their volatility being close to the recent norm (median 1.06 times the previous 20 sessions). Monthly derivative expiry days had more alarms (38%, $p = 0.002$; Holm 0.0095, and 0.015 with the circular shift), which may be an artefact of a profile that pools all weekdays. In March 2020 volatility was 2.7 times that of the previous 20 sessions, yet only 6 of 21 sessions had an alarm: after each alarm the monitor recalibrates on the new, already high level. It answers "has volatility changed from the recent baseline?", not "is volatility high?".

S7.2 FOMC statements and CPI releases

Design. Binance BTC, ETH and SOL from 21 January 2024 to 31 December 2025; the 16 FOMC rate decisions (statement at 14:00 US Eastern) and 22 US CPI releases (08:30 US Eastern) of 2024 and 2025 inside the monitored period, with every date checked against a published schedule. Control windows are all days at least two days away from any event day, with the same weekday, at the same Eastern clock time (134 for FOMC, 299 for CPI);

each event is compared with the control windows of its own weekday, and the control rates below are these matched rates averaged over events. The outcome is an increase alarm within six hours of the anchor, and the test is a stratified permutation test, exact when the strata are independent; releases of one weekday and clock time share their control windows, which can raise its level slightly (5.8% at a nominal 5% for seven FOMC statements in the simulation of the test on fresh releases below). The primary family is FOMC and CPI on BTC in the streaming design (the monitor as a user would have run it), with a Holm correction; it is stated as fixed before the first run in the analysis code, which was first saved together with the results. ETH and SOL are secondary replications, and a design in which a fresh monitor is calibrated on the day before every anchor was added after the streaming results were seen and is exploratory.

Results. In the streaming design (the primary family), 9 of 16 FOMC statements on BTC were followed by an increase alarm within six hours, against 14% of control windows (Holm-adjusted $p = 0.00025$); the CPI test was not significant on BTC (5 of 22 against 12%, $p = 0.11$) but was significant on ETH ($p = 0.003$) and SOL ($p = 0.00004$; Holm within the replications 0.0032 and 0.00015). The streaming monitor was recalibrating after its own alarms about 70% of the time and was awake at only 9 of the 16 FOMC statements on BTC; at those it alarmed every time (and at 8 of 8 on ETH and 9 of 9 on SOL), a restriction made after the first run. Exploratory from here on. With a fresh calibration, 14 to 15 of the 16 FOMC statements raised an alarm within six hours, against 29% to 32% of controls, and 59% to 73% of CPI releases, against 23% to 28%. The alarms came after the release (median 23 to 34 minutes, never before 14 minutes on BTC and SOL and 10 minutes on ETH), and there is no sign of more alarms in the hours before a release. The Gaussian likelihood-ratio monitor with a fresh calibration also alarmed at almost every release, but with a median delay of 0 to 2 minutes, on a single return of 8 to 9 robust scale units (median): one large minute is enough to trigger it. It also alarmed in 62% to 76% of control windows. Because it ran at band 0 and the robust monitor at band 0.25, this contrast mixes the band with the truncation; run at band 0 as well, the robust monitor with a fresh calibration alarmed at 14 to 15 of 16 FOMC statements and 16 to 18 of 22 CPI releases, with a median delay of 18 to 22 minutes, and in 39% to 51% of control windows. The truncated detector cannot be moved by one minute (one clipped observation multiplies its statistic by a bounded factor), and its delays and the size of its largest recent return are those of a short volatility regime; the share of release alarms classed as “jump only” did not separate the two monitors, so this reading rests on the delays and the return sizes.

The rerun with proved monitors (pre-registered). The rerun was pre-registered before any of its real-data runs and keeps everything of the design above except the monitor (Table S12). Primary: the confidence-sequence baseline ($\delta = 0.01$, $A = 36,000$, band 0.25) restarted at 00:00 UTC, with a 1440-minute recalibration after each alarm, so that each UTC day is one period of Corollary S14 and the monitor raises at most one alarm a day; its bound, 5% for each day of a segment, is marginal and needs the tolerance during that day. Secondary: the same without the restart (5% for the first day of each segment) and the rank detector restarted without a blind period, with payoffs divided by 1.25 (Theorem S16). In the primary family (BTC, Holm over FOMC and CPI), 8 of 16 FOMC statements were followed by an up alarm within six hours against 10.9% of matched controls ($p = 0.00017$, Holm 0.00033), and 5 of 22 CPI releases against 5.9% (Holm 0.0097); the plug-in had 9 of 16 against 13.9% and 5 of 22 against 11.7%, so for CPI the count of releases is unchanged and the matched control rate halved. Every FOMC cell held after

Holm, on ETH and SOL, without the restart and with the rank detector; CPI did not hold on ETH and SOL under the primary monitor (Holm 0.075) or without the restart on BTC (3 of 22, $p = 0.12$) and SOL. The rank detector's cells repeat an earlier run exactly and add no confirmatory evidence, and the plug-in run through the same code reproduced the results above exactly. The primary monitor watched 80 of the 114 coin and release pairs (without the restart 77, the rank detector 114, the plug-in 55); one BTC FOMC statement (29 January 2025) fell in a segment whose increase side could not alarm. It alarmed on 48.1% to 53.5% of the UTC days on which it monitored at least one minute and raised 0.92 to 1.15 alarms per 1440 monitored minutes, against a long-run bound of 0.0404 per 1440 monitored minutes under the tolerance (Corollary 18 and the remark after its proof), so the tolerance fails on many of these days and the rerun does not test the bound. The restart at 00:00 UTC falls five hours into the six-hour window of FOMC statements made in US Eastern standard time. The data and the plug-in results were known when the rerun was designed, and 16 statements give a wide interval (28% to 72% for the primary rate); the contrast was not confirmed on the releases of 2026 (below).

The test on fresh releases (pre-registered). The pre-registration was committed with the event file before any Binance return of 2026 was downloaded. Releases: the FOMC statements (14:00 US Eastern) and CPI releases (08:30 US Eastern) of 2026 whose six-hour window ends before the data end, 24 October 2026 23:59 UTC, that is 6 FOMC statements (28 January to 16 September; that of 28 October is excluded) and 10 CPI releases (13 January to 14 October), each date checked by web search against the Federal Reserve and BLS announcements. Data: Binance BTC, ETH and SOL, monitored from 1 January 2026 for 297 days with no missing minute. Monitors, control windows (29 for each release, 39 for releases on a Friday) and outcome are those of the rerun. Primary: the confidence-sequence monitor restarted at 00:00 UTC on BTC, with one one-sided stratified permutation test over all 16 releases that passes at $p \le 0.05$; it was chosen over a test of the FOMC statements alone because its power at the rerun's BTC rates was 0.82 against 0.69 for the 17 releases planned (0.78 against 0.62 for the 16 in the data, and 0.37 against 0.28 if the excess were half as large). Result: 2 of 16 releases against 5.9% of matched controls ($p = 0.25$, Fisher 0.29; Wilson interval 3.5% to 36.0%); the same design on the releases of 2024 and 2025 gives 13 of 38 against 8.0%. Secondary families, Holm within each: this monitor pooled on ETH and SOL, 4 and 5 of 16 (Holm 0.079 for both); its six FOMC and CPI cells, none below Holm 0.48; the plug-in, one cell (ETH, FOMC, 4 of 6, Holm 0.040); the rank detector restarted without a blind period, 4 of 6 FOMC statements on every coin (Holm 0.016, 0.033 and 0.055) and no CPI cell; the monitor without the restart, no cell. On BTC the primary monitor was recalibrating at 2 releases and unable to raise an increase alarm ($\tau^2 \le m^\uparrow$) at 3, in two segments of 4.6 and 47.0 days that held 28.0% of its monitored time, and alarmed after 2 of the 11 releases it watched (13 of 26 in 2024 and 2025). It alarmed on 37.8% to 48.0% of the UTC days on which it monitored at least one minute, against 5% under the tolerance. With power 0.78, a failure has probability about 0.22 if the rates of the rerun were true and about 0.63 if the excess were half as large, so the failure does not show that there is no contrast. In the power simulation the stratified test of the individual release types, which treats releases that share control windows as independent strata, had level 5.8% for seven FOMC statements sharing 29 control windows (Monte Carlo standard error 0.17 points) and 4.7% for six; the pooled test stayed below 5%. Post hoc and descriptive: the mean absolute return in the first hour after a BTC release, relative to the six hours before, had medians of 2.11 (FOMC) and 1.73 (CPI) in 2026, against 2.51 and 2.00 in 2024 and 2025.

S7.3 Forecasts and competitors on real data

Table S8 gives the comparison with forecasts of Section 7.2 for the plug-in baseline, with 95% intervals from the block bootstrap over calendar days of Section 7.1. The ratio rules alarm when the robust volatility of the last 60 or 240 minutes, divided by that of the preceding day, crosses thresholds chosen on the same data to give the e-detector's numbers of up and down alarms, with the same dead time; this hindsight favours them. With the hindsight-tuned ratio rule, the one comparison that holds after Holm adjustment (band 0.5, out of sample) has the e-detector's decrease alarms confirmed 10.5 points more often than the 60-minute rule's (Holm $p = 0.0096$). Out of sample the intervals of the differences lie within plus or minus 5 points in 7 of 8 cells at band 0.25, but in July to December 2025 they reach 9.7 points.

Rule	Period	Up alarms: next day above forecast	Down alarms: next day below forecast
Random times (EWMA / HAR)	Jul to Dec 2025	49.4% / 51.3%	50.6% / 48.7%
Robust e-detector (EWMA)	Jul to Dec 2025	78.5% (72.0 to 84.5), $n = 367$	80.5% (74.6 to 86.3), $n = 256$
Robust e-detector (HAR)	Jul to Dec 2025	80.7% (74.7 to 86.4)	78.1% (70.3 to 85.5)
Ratio rule, 60 / 240 min (EWMA)	Jul to Dec 2025	73.9% / 74.2%	78.1% / 82.3%
Random times (EWMA / HAR)	out of sample	50.4% / 48.3%	49.6% / 51.7%
Robust e-detector (EWMA)	out of sample	77.1% (73.5 to 80.5), $n = 951$	70.9% (66.7 to 75.1), $n = 870$
Robust e-detector (HAR)	out of sample	75.9% (72.0 to 79.6), $n = 909$	73.6% (69.9 to 77.3), $n = 834$
Ratio rule, 60 / 240 min (EWMA)	out of sample	77.0% / 75.7%	67.4% / 70.9%

Table S8. Market data, five coins pooled, band 0.25, plug-in baseline. Share of alarms after which the next day's robust volatility was on the flagged side of a forecast made at the alarm, with 95% intervals from the block bootstrap over calendar days of Section 7.1 in brackets. The HAR rows out of sample have fewer alarms because the HAR forecast needs a longer history.

We repeated the comparison of Table S8 for BOCPD and the two-window variance-ratio scan (inspired by Bibinger, Jirak and Vetter 2017) of Section 6.2, run continuously with predictable priors and scales and with the same one-day pause after every alarm, and with their up and down thresholds set per series so that each raises as many up and down alarms as the e-detector at band 0.25 (about 0.68 alarms per day). Table S9 pools the five coins.

Method (window in minutes)	Jul to Dec 2025: up / down	Jan to Jun 2025 (held out): up / down
Robust e-detector (one-day baseline)	78.5% / 80.5% ($n = 367$ / 256)	82.0% / 69.7% ($n = 306$ / 304)
BOCPD ($w = 250$)	67.8% / 61.7%	69.2% / 56.7%
BOCPD ($w = 720$)	70.5% / 66.5%	71.9% / 59.5%
BOCPD ($w = 1440$)	68.4% / 69.9%	74.2% / 63.9%

Method (window in minutes)	Jul to Dec 2025: up / down	Jan to Jun 2025 (held out): up / down
BOCPD on truncated returns (w = 250)	68.0% / 71.7%	71.5% / 67.0%
BOCPD on truncated returns (w = 720)	72.6% / 77.9%	74.8% / 69.3%
BOCPD on truncated returns (w = 1440)	69.6% / 81.7%	73.7% / 71.9%
Variance-ratio scan (k = 120)	65.6% / 58.7%	68.6% / 59.8%
Variance-ratio scan (k = 480)	69.9% / 74.0%	75.7% / 68.3%
Variance-ratio scan (k = 1440)	67.9% / 77.7%	69.3% / 73.7%
Variance-ratio scan, studentised (k = 120)	66.5% / 62.4%	67.8% / 58.3%
Variance-ratio scan, studentised (k = 480)	68.2% / 71.3%	75.5% / 66.8%
Variance-ratio scan, studentised (k = 1440)	66.3% / 77.8%	69.3% / 72.7%

Table S9. Market data. Share of up (down) alarms after which the next day's robust volatility was above (below) an EWMA forecast made at the alarm, at equal numbers of up and down alarms, pooled over five coins; n counts the e-detector's alarms with a complete next day. w is the short-run window of BOCPD and k the window of the two-window variance-ratio scan. At random times the next day is above the forecast 46.5% to 50.1% of the time and below it 49.9% to 53.5% per coin. Rows with windows of 480 to 1440 minutes were added after the first run, with lengths fixed before they were run.

With the first-run windows (2 to 4 hours), the e-detector's alarms were followed by the announced volatility more often than every competitor's in 15 of 16 comparisons (block bootstrap over days of Section 7.1, Holm over the 16 comparisons, adjusted $p \le 0.021$); the exception is the held-out down alarms of BOCPD on truncated returns (67.0% against 69.7%, $p = 0.44$). The outcome, however, asks about the next day, the time scale of the e-detector's one-day baseline. With windows of a day (BOCPD on truncated returns with w = 720 or 1440, and both scans with k = 1440, rows added after the first run), the competitors' down alarms were not significantly less informative (differences -3.9 to $+2.7$ points), but no interval lies within plus or minus 5 points, so equal precision is not shown; plain BOCPD stayed lower (59.5% to 69.9%). For up alarms the e-detector was ahead in 5 of these 8 comparisons after Holm. The best longer-window competitor, chosen after seeing the results, was about 6 points lower (72.6% against 78.5%; 75.7% against 82.0%), with Holm $p = 0.53$ and 0.44 over the 32 longer-window comparisons. Scored on the next four hours instead of the next day (an outcome added after the first run), the short-window competitors did as well as the e-detector or better. So the e-detector is not more precise on crypto in general; its alarms are better aligned with day-long shifts than those of competitors with short windows. Rate matching uses the real data, but not the outcome. The e-detector's settings were frozen in an earlier development run on Bitstamp BTC/USD for July to December 2025, so only January to June 2025 is fully held out for it. On the Nifty 50 (2024 and 2025, 494 sessions) all five methods scored 69% to 83% with the first-run windows and no difference was significant, with intervals more than 20 points wide at 69 to 79 alarms per direction; with session-long windows the scans' decrease alarms were less informative (by 17 to 20 points, Holm $p \le 0.035$).

Classical rules on real data. In the design of Table S9 (five coins, both half-years of 2025, bands 0 and 0.25), the classical rules of Supplement S5, run as the e-detector runs (calibration on 1440 minutes and recalibration after each alarm; the Student t BOCPD with w = 250, 720 and 1440 minutes and a one-day pause), with up and down thresholds matched to the e-detector's alarm counts: the calibrated betting rule and both CUSUMs on X were never significantly less precise than the e-detector in 24 comparisons of the next-day outcome (differences −5.0 to +3.0 points; one significant difference, in favour of the betting rule), the training-sample rules were below it in 13 of 16 comparisons (6 significant) and the Student t BOCPD in 20 of 24 (14 significant), although it improved on the Gaussian BOCPD. These p-values are not adjusted; 21 of the 64 were below 0.05.

S7.4 Re-analysis of the real-data inference

Classes. Of 36 real-data claims in the previous draft of this paper, one rests on the pre-specified primary test of Section 7.3, 19 on secondary tests described in advance (6 of them in a record dated before their results), 12 are exploratory or post hoc, and 4 mix classes. No data set other than the event study has a pre-specified primary test; for the Nifty 50 a primary configuration was fixed, not a primary test. That primary test used the releases of 2024 and 2025; a pre-registered test of the same contrast on the releases of 2026, with a proved monitor, failed (Section 7.3).

Resampling and adjustment. Holm within each family, with no adjustment across families. Shares and their differences are resampled by a circular block bootstrap over calendar days (UTC) or Nifty sessions, with blocks of 7 days or 5 sessions and 9,999 replicates; one set of day weights per replicate is applied to every coin, method, direction and control draw of the period. Intervals are percentile 95% intervals and p inverts them, so the smallest attainable value is 0.0002. Equivalence is shown when the 95% interval of a difference lies within plus or minus 5 points. With blocks of 14 days and 10 sessions, one of the 164 Holm decisions changed. For the Nifty 50 the random-time shares are treated as fixed, which makes those intervals slightly too narrow. The pooled crypto intervals are 1.2 to 1.5 times as wide as intervals that treat alarms as independent.

Robust and Gaussian monitors. At band 0, against the EWMA forecast, the increase alarms of the Gaussian likelihood-ratio monitor were less informative than those of the robust monitor, by 13.5 points in July to December 2025 and 7.2 points out of sample (Holm p = 0.0008), but out of sample its decrease alarms were 5.5 points more informative (95% interval 2.3 to 8.6, Holm p = 0.003).

Forecast regressions. On an hourly grid for the five coins, the outcome $RV(t)$ is the realised variance of the next 1440 minutes, the sum of squared one-minute log returns (360 minutes and bipower variation, Barndorff-Nielsen and Shephard 2004, are secondary). Each forecast $F(t)$ uses past data only and is refitted daily by an expanding-window regression of the log outcome on its regressors and on hour-of-day and day-of-week dummies: EWMA (half-life one day), HAR (Corsi 2009; realised variance over the last 1, 5 and 22 days) and HAR-I (HAR plus the log realised variance of the last hour). With $U(t)$ and $D(t)$ the indicators of an up and a down alarm of the rule in the hour ending at t, the regression, pooled over coins, is

$$\log RV(t) = a_{\text{coin}} + \beta \log F(t) + c_{\uparrow} U(t) + c_{\downarrow} D(t) + e(t),$$

with Driscoll and Kraay (1998) standard errors (scores summed within each UTC day over all coins, Bartlett weights, bandwidth 5 days). The QLIKE loss of a forecast F for an outcome y is $L(y,F) = y/F - \log(y/F) - 1$ (Patton 2011). HAR-I had the lowest mean loss on July to December 2025 (0.537, against 0.562 for HAR and 0.565 for EWMA) and on the held-out period, which starts on 23 February 2024 here because the HAR needs 22 days of history (59,160 grid points). The loss comparison scales F by $\kappa_{\uparrow}$ after an up alarm and by $\kappa_{\downarrow}$ after a down alarm, each the mean of y/F at the July to December 2025 alarms, and compares losses on the held-out period with a test of the Diebold and Mariano (1995) type.

Rule	$c_{\uparrow}$ (SE)	Holm p	$c_{\downarrow}$ (SE)	Holm p	Change in QLIKE	Holm p
Robust e-detector	0.038 (0.027)	0.32	−0.064 (0.025)	0.046	+0.29%	0.11
Ratio rule, 60 min	0.026 (0.024)	0.32	−0.040 (0.022)	0.20	+0.38%	0.025
Ratio rule, 240 min	0.063 (0.026)	0.056	−0.087 (0.022)	0.0007	+0.36%	0.11

Table S10. Market data, five coins, held-out period, band 0.25. Coefficients of the alarm indicators in the regression of the log realised variance of the next day on the log HAR-I forecast (β = 0.952, standard error 0.024), with Driscoll and Kraay standard errors and Holm over the six coefficients; and the relative change in mean QLIKE when the forecast is scaled at alarms by factors fitted on July to December 2025 (positive: the scaled forecast is worse), with Holm over the three rules.

After an e-detector down alarm, next-day realised variance was on average about 6% below the HAR-I forecast; the coefficient is fragile, with Holm $p = 0.060$ under standard errors clustered by day, and no larger than that of the 240-minute ratio rule (Table S10). Against the EWMA and HAR forecasts, in all 32 blocks (two forecasts, two periods, two horizons, two measures, two bands) every alarm coefficient had the sign of its alarm with Holm $p \le 0.002$, from +0.16 to +0.60 for up alarms and from −0.15 to −0.49 for down alarms. The scaling factors did not transfer between periods: at the e-detector's up alarms realised variance was 54% above the HAR-I forecast on average in July to December 2025 but 15% above it in the held-out period. Over all 72 loss comparisons none showed a significant gain.

S7.5 Landmark shocks

Design. The events and reference times T were fixed, from published sources, before any return in the event windows was loaded: T is the announcement if the market was open and otherwise the next open, and for crypto on the two Monday sell-offs it is the Tokyo open (00:00 UTC). Events: Indian general election results, 4 June 2024 (Nifty 50, 09:15 IST); global sell-off, 5 August 2024 and tariff sell-off, 7 April 2025 (five coins at 00:00 UTC, Nifty 50 at 09:15 IST); crypto liquidation cascade, 10 October 2025 (five coins, 20:00 UTC; the triggering post came by 20:57 UTC). The e-detector runs with the settings of Sections 7.2 and 7.4 in half-year streams, each run without interruption, with the plug-in baseline and a one-day recalibration (750 bars on the Nifty 50) after each alarm. The competitors are those of Table S9, with thresholds matched to the e-detector's numbers of up and down alarms outside windows from 8 days before to 2 days after each reference

time, and the same one-day pause after each alarm. With four events and coins that move together within an event, no test was planned or run and the methods are not ranked.

Results. On the Nifty 50 the e-detector was monitoring at the open on 4 June 2024 and flagged the election shock in 11 minutes; on 5 August 2024 and 7 April 2025 it was recalibrating after a down alarm in a quiet session a few days earlier and flagged the shock 104 and 172 minutes after the open. On all fifteen crypto cases it raised an up alarm 4.5 to 6.9 hours before T, as the turbulence began (on Sunday afternoon before the two Monday sell-offs, and after an earlier tariff threat on 10 October 2025), and spent the next day collecting a new baseline. Of the 45 competitor configurations on the coins, 1, 3 and 5 were watching at the three crypto reference times; most of their alarms within 24 hours came in the first minute after a pause ended, and the clean exception is the day-long scan on 10 October 2025, which flagged the cascade within 17 to 73 minutes on all five coins. With a fresh calibration on the day before T, the e-detector flagged all 18 cases: in 18 to 32 minutes on the coins on 5 August 2024, 27 to 207 minutes on 7 April 2025, 60 to 65 minutes on 10 October 2025, and 15, 85 and 26 minutes on the Nifty 50. Plain BOCPD with a 250-minute window and the 120-minute scan were earlier on 10 October 2025, by 4 to 10 minutes on four coins. Streaming from another start date gave identical outcomes for the e-detector. In the seven days before each event the methods together raised 0.76 to 1.10 times the alarms expected at their average rates. These are case studies, not evidence. In an exploratory run, monitors without a blind period (Section 6.5) flagged 15 and 17 of the 18 cases within 24 hours (companion note, Section 1).

S7.6 Null streams from real returns and the confidence-sequence baseline

Design. Pre-registered in a document committed before any real-data run of this study; what was decided later is called post hoc. Data: the deseasonalised returns of the five coins of Section 7.2, July to December 2025, and of the Nifty 50, sessions of January 2022 to October 2026. A day (a session for the Nifty 50) is calm if its robust volatility (1.4826 times the median absolute deseasonalised return) lies within 0.8 to 1.25 times the median of the previous 20; the pool of a data set holds the returns of all its calm days (81 to 100 of 184 days per coin, 639 of 1191 Nifty sessions). Streams are random permutations of a pool (primary), i.i.d. draws from it, or stationary bootstraps of the calm days with mean blocks of 30, 390 and 1440 minutes. Four monitors, all with the default two-sided e-detector, run on each stream: the known baseline (τ and m_0 of the pool, restart at 0 after an alarm), the plug-in (1440 or 750 calibration returns), the confidence-sequence baseline ($\delta = 0.01$, $A = 36{,}000$ or $9{,}375$) and the rank detector of Section 6.5 (Supplement S1.9), with $A = 28{,}800$ for crypto and 7,500 for the Nifty 50 otherwise. The endpoints are the probability of a first alarm within one and within five days, and the alarms per monitored day over 20 monitored days with a restart or recalibration after each alarm.

Exactness. By Lemma S17, after every alarm the rest of a permuted stream is exchangeable given the past, which is what the restarted rank detector needs (Theorem S16(b)), so its bounds hold exactly on these streams. For the known and confidence-sequence baselines the conditional mean of the next X is the mean of the values not yet drawn, which wanders: its largest relative departure from m_0 along a stream had a median of 0.24% to 0.31% for the coins and 0.07% for the Nifty 50, and a maximum of 1.19%. The i.i.d. streams are the exact check for these two baselines.

Results. Five coins pooled, permuted streams, band 0: within five days 15.8% for the known baseline and 16.7% for the rank detector against 25%, and 0.13% for the confidence-sequence baseline against 21%; per monitored day 0.040, 0.037 and 0.0002 false alarms against 0.05, 0.05 and 0.041. The plug-in gave 0.103 false alarms per monitored day against a nominal 0.05, and on the Nifty 50 33% within five days and 0.081 per session. No interval of the three proved monitors lay above its bound in any of the 252 cells (each coin, their pool and the Nifty 50; permuted and i.i.d. streams; three endpoints; bands 0 and 0.25). At band 0.25 almost no monitor alarmed on exchangeable streams (at most 0.1% at one day). On bootstrap streams every monitor alarmed within a day on 86% to 100% of streams at band 0 and raised 2.0 to 7.6 alarms per monitored day; post hoc, these rates match those of the real series (plug-in, band 0: 4.5 to 5.5 per monitored day on the bootstrap streams against 4.9 to 5.2 on the real series). Dividing each calm day's returns by that day's robust volatility (post hoc) lowered the coins' rates by 3% to 42% but left 96% to 99% of streams with an alarm within a day (confidence-sequence baseline 68% to 86%) and 3.1 to 4.9 alarms per monitored day (1.1 to 1.8): most alarms come from volatility changes within calm days.

The confidence-sequence baseline on the five coins. On the series of Section 7.2, out of sample, 80.8% of its up alarms and 90.9% of its down alarms were confirmed, against 77.9% and 84.5% for the plug-in; against the EWMA forecast its shares were 81.4% and 90.7% in July to December 2025 and 82.1% and 76.2% out of sample. On Bitstamp BTC/USD, July to December 2025, it raised 16% to 20% fewer alarms than the plug-in at every band (17% to 22% with the provider's filled minutes treated as missing). On the five coins in July to December 2025 (band 0.25), down alarms that both monitors raised within six hours came a median 40.5 to 142 minutes later with the confidence-sequence baseline, and up alarms 4 to 9 minutes later; it monitored 46% to 49% of the minutes, against 30% to 33% for the plug-in, since it recalibrates less often. A ratio rule matched to its alarm counts was confirmed about as often (83.4% of up and 86.4% of down alarms), so this check shows that its alarms are not noise, not that they add precision.

Blind segments (post hoc). With the confidence-sequence baseline, τ comes from the first quarter of the calibration window and the increase side bets against $m^{\uparrow} = (1+b)\tau^2 \min(U_{T_0}, 1)$. If $\tau^2 \le m^{\uparrow}$, every increase factor is at most 1, each increase statistic grows by at most 1 per step, and no increase alarm can come within the first $2A-1$ monitored steps, whatever the data. This happened in 5 of 7031 segments of the confidence-sequence runs, all on BTC series, lasting 2.6 to 136 days, and in none of 9184 plug-in segments; in one run (BTC, July to December 2025, band 0.5) the monitor was blind to increases for its last 4.5 months. It costs power, never protection against false alarms. A guard that recalibrates when $\tau^2 \le m^{\uparrow}$ selects windows by their confidence bounds, so the kept window's coverage no longer holds as such: with no cap on the retries, on a law built for it, the kept window's bounds fail with probability 1. Splitting the error budget over the windows tried, $\delta_1 + \ldots + \delta_K \le \delta$, restores $\mathbb{P}(\text{the kept window's bounds fail}) \le \delta$ and, under the tolerance relative to each window tried, every bound of Section 4.2. The guard of `avmonitor.guard` (at most 3 windows with budgets $\delta/2$, $\delta/4$, $\delta/4$, a retry when $\tau^2 \le 1.25\, m^{\uparrow}$, and a fallback to the rank detector) detected a later doubling within a day in 86.5% to 100% of simulated paths where the unguarded monitor detected 0% to 0.25%, with no null alarm in 800 paths per design, and on real BTC data it left no monitored minute blind; the real-data results are post hoc (`docs/theory_blind_guard.md`, `docs/report_parts/E24.md`).

The forecast-relative null. With the deseasonalised returns divided by a GARCH(1,1) forecast fitted on January to June 2025 and then frozen, the plug-in monitor at band 0.25 raised 0.033 to 0.054 alarms per day per coin in July to December 2025, 39 alarms in all and 0.89 of the nominal 0.05 per monitored day, and every one was followed by a move in the flagged direction; at band 0 the monitors alarmed 2.5 (confidence-sequence baseline) to 7.4 (plug-in) times the nominal rate. With an EWMA forecast (half-life 360 minutes) they alarmed 17 to 89 times the nominal rate and were confirmed less often (62% to 82%) than on the raw returns. The forecast-relative null changes the question: the monitor flags one or two departures from the forecast per coin per month.

S7.7 An anytime-valid release test

Design (pre-registered). Releases are taken in time order, and release i gets a stratum: its six-hour window and up to eight control windows at the same Eastern weekday and clock time one to four weeks before and after it. A candidate is dropped if its day is less than 2 days from a release day, if it starts within 1800 minutes of the start of the stream or is not inside the data, if an earlier release already uses it, or if its window overlaps the window of any release or of a control kept for an earlier release (the last rule an amendment made before any data of the test were loaded). With Z_i the release window's outcome (an up alarm of the rank detector within six hours), n_i controls, S_i windows with the outcome and $p_i = S_i/(n_i + 1)$, the e-value for an odds ratio ψ is $e_i(\psi) = \psi^{Z_i}/(1 - p_i + \psi p_i)$, and $M_j = \frac{1}{4}\sum_{\psi \in \{2,4,8,16\}} \prod_{i \le j} e_i(\psi)$; the test rejects the first time $M_j \ge 20$.

Guarantee. If, given the outcomes of strata 1 to $i - 1$, the $n_i + 1$ outcomes of stratum i are exchangeable, then $\mathbb{P}(Z_i = 1 \mid \mathcal{G}_{i-1}, S_i) = p_i$, so $\mathbb{E}[e_i(\psi) \mid \mathcal{G}_{i-1}] = 1$, each product is a nonnegative martingale with mean 1, and by Ville's inequality $\mathbb{P}(M_j \ge 20 \text{ for some } j) \le 0.05$ for any number of looks and any stopping rule. No independence between strata is needed, and the disjoint windows are what make the null coherent. The null concerns the monitor's outcomes, not the returns, and no property of the detector implies it; a volatility drift over the eight weeks around a release can break it in either direction, since most strata have unbalanced controls.

Results. Under each of eight null laws, M reached 20 within 200 releases in at most 4.4% of 20,000 simulated sequences. Under the planning model of the fixed-date test (Section 7.3) it reaches 20 by 1 June 2027 with probability 0.43 (the fixed-date test: 0.82), by 1 January 2028 with 0.82 and by 1 January 2029 with 0.98; at the 2026 rates with 0.19, 0.51 and 0.81; under the null with 0.010, 0.016 and 0.022. On releases already seen (post hoc), M reaches 20 in June to August 2024 on all three coins and ends at 3.8 on BTC in 2026. Its first stratum on fresh data completes on 26 November 2026.

S8. Settings of the experiments

All simulations use fixed seeds, in per-step units in which the pre-change diffusive standard deviation is 1 unless stated. "Plug-in" means $\tau = 3$ scaled MADs and m_0 the mean of X over the calibration window; "true baseline" means both computed from the distribution. The settings of the e-detector were fixed before the reported runs; analyses added after first results are named in the text where they appear.

Used in	Returns	Baseline and calibration	Alarm level	Paths	Change design
Tables 2 and S5; Section 6.2	Gaussian; jumps (probability 0.005, sd 8); Student t_5; price noise (sd 0.5 on log prices); slow SV (log-sd 0.15, half-life 390)	plug-in, 2000 steps, same law	$A = 7800$ per side	300 per cell	sd ratio 1.25, 1.5, 2, 0.5, 0.67, 0.8 at monitored step 500
Section 4.1 (one-day plug-in rates); Corollary 6 check	Gaussian, Student t_3, jumps (0.005, sd 8)	true or plug-in (390, 1440 steps)	$A = 7800$; restarts $A = 300$	4000; restarts 400 (true), 40 (streaming)	none
Table S6; Section 6.3	Gaussian, jumps (0.005, sd 8), Student t_5	plug-in, plug-in plus one standard error, or confidence sequence (n_τ = 25% of 390, 1440, 5000 steps; $\delta = 0.01$ per side)	7800 (plug-in), 9750 (CS)	10,000 null, 2000 change	as Table S5, at step 500
Table S4; Figure 1, left	Gaussian, Student t_3, jumps (0.005, sd 10), ticks of 0.5 sd	true (m_0 from 2×10^7 draws; τ floored at three ticks for the tick law)	$A = 7800$, two-sided	4000	none
Figure 1, right; Section 6.4	GARCH(1,1), unit unconditional variance, persistence 0.95 to 0.9999	true unconditional	$A = 28,800$ (1440-step day)	2000	none
Section 6.4; Table S7; Figure S2	GARCH(1,1), $(\alpha, \beta) = (0.05, 0.90), (0.05, 0.94), (0.03, 0.969)$	2000 warm-up steps, plug-in on one 1440-step day; GARCH forecast fitted by QML with variance targeting on a separate 30,000-step path	$A = 28,800$	600 (null), 300 per row (breaks); 200 paths of 18,000 steps (Table S7)	scale or intercept break 200 steps after calibration, 5 days
Table 1; Theorem 24; Section 5.3; Table S2; Figure S1	Gaussian, jumps (0.005, sd 8), Student t	true, $c = 3$ unless varied	$A = 7800$ (and 10^4 to 10^8), one side	3000 per case	change at the first monitored step
Section 5.3 (split clip); Table S3	Gaussian, Student t_4, jumps (0.005, sd 8), ticks	plug-in, 1440 steps	$A = 28,800$	2000 null, 600 per change	sd ratio 0.8, 0.67, 0.5 at step 1

Used in	Returns	Baseline and calibration	Alarm level	Paths	Change design
Section 6.2 (BOCPD and two-window scan)	Gaussian; jumps (0.005, sd 8); Student t_5; price noise (sd 0.5); ticks of 0.5 sd; slow SV; GARCH(1,1) with persistence 0.99	offline: 500 change-free steps of a 2000-step window (plug-in or CS); online: plug-in or CS on 2000 steps	offline: critical values matched to 5% on clean Gaussian windows; online: $A = 7800$, competitors matched to the e-detector's rate in the first 390-step day on clean Gaussian paths	offline 1000 null, 300 per change; online 1000 null, 200 per change	offline at step 1000; online at monitored step 500; sd ratio 1.25, 1.5, 2, 0.8, 0.67, 0.5
Theorem 9; Corollary 14; Supplement S1	default detector; constant drift (two-point, Gaussian, deterministic and adaptive streams); two-state switching drift; GARCH(1,1) designs of Section 6.1; Gaussian plug-in windows	true τ and m_0, or plug-in from 390 to 23,040 steps	$A = 7800$, two-sided (one design row 28,800)	10,000 per i.i.d. law, 20,000 adaptive; 6000 per switching or GARCH design; 100,000 windows (plug-in)	none (validity)
Section 6.1 (adversarial search)	inside the class: i.i.d. two- and three-point laws for Y, scale mixtures, Student t ($\nu \geq 2.1$), rounded returns, shape switching, adaptive policies; outside: GARCH(1,1) on 64 designs, price-level ticks of 0.05 to 6 sd	true τ and m_0 (unconditional for GARCH and ticks); plug-in from 390 or 1440 steps	$A = 7800$, two-sided	search 20,000 (one day) and 6000 to 8000 (five days); fresh seeds 200,000 and 60,000; outside the class 6000 to 30,000 per design	none (validity)
Supplement S1.8 (fourth power); Lemma 1 (tail shape)	Gaussian, jumps, Student t_5 (variance changes); Student t_3, t_5, t_8 and two-point scale mixtures at fixed variance (shape changes)	plug-in, 2000 steps	$A = 7800$ (variance changes); $A = 400,000$ over 20,000 steps (shape changes)	300; 200 per shape design	sd ratio 1.25 to 0.5; shape change at fixed variance

Used in	Returns	Baseline and calibration	Alarm level	Paths	Change design
Section 6.4 (price noise)	one-second returns, 23,400 per day, constant volatility, i.i.d. Gaussian noise on log prices with noise-to-signal ratio 0 to 4	plug-in from two days (46,800 returns)	A = steps per day / 0.05	200 days per cell	none (validity); sd doubles at the start of the day (delay)
Theorems 12, 15 and 23; Supplement S1.6, S2 and S3	collapse and explosion paths; Gaussian, Student t_3 and jumps (0.005, sd 8), sd ratios 0.5, 0.67, 0.8, 1.25, 1.5, 2; plug-in windows of Gaussian and Student t_3 returns, n = 390 to 92,160	true baseline (delay floors); plug-in, plug-in moved by $z = 2.576$ standard errors, or confidence sequence (plug-in regime)	$A = 7800$ (and 10^3 to 10^8 for the asymptotics)	10,000 paths of 4000 steps per law; 20,000 windows per cell (5000 above 23,040); 10,000 paths of 7800 steps per plug-in cell; 4000 for the boundary and the hold-then-bold stream	change at the first step (delay); none (plug-in)
Proposition 7, Theorem 8, Theorem 16 (tolerance); Supplement S1.7	Gaussian, Student t_3, jumps (0.005, sd 10), ticks of 0.5 sd; bold play; $Y_t = 1$ for $d - 1$ days, then one Gaussian or two-point day; parking stream; GARCH(1,1) with persistence 0.95 to 0.9999 and slow log-normal SV (tolerance)	true baseline; default grids; plain, restarted every day, restarted with carry m at $2A$, CUSUM type	$A = 7800$ (390-step days) and $28,800$ (1440); CUSUM type also at 8027.7 and $29,053.4$	40,000 paths of 20 days per stream (30,000 for bold play and the parking stream); 20,000 per delay cell; 2000 per tolerance cell	none (false alarms); sd factor 0.5 to 2 at a uniformly random step of day 3, horizon 12 days (delay)
Section 6.2 (calibrated competitors); Supplement S5	Gaussian, Student t_3 and t_5, jumps (0.005, sd 8), price noise (sd 0.5), ticks of 0.5 sd, slow SV, GARCH(1,1) with $(\alpha, \beta) = (0.05, 0.94)$	plug-in, 2000 steps; thresholds simulated on 20,000 Gaussian paths or bootstrapped per path (1000 replicates); training sample of 2000 steps	e-detector each side at $2A$, $A = 7800$; competitors at 1.3% or 5% in the first 390-step day	4000 null per law (bootstrap 2000); 1000 per change cell (bootstrap 500); the paths of the online comparison for the Student t BOCPD	none (false alarms); sd ratio 1.25, 1.5, 2, 0.8, 0.67, 0.5 at monitored step 500 (delay)

Used in	Returns	Baseline and calibration	Alarm level	Paths	Change design
Supplement S5 (split clip with the confidence-sequence baseline)	Gaussian	plug-in or confidence sequence ($\delta = 0.01$), 1440 steps	$A = 28,800$ (plug-in), $36,000$ (confidence sequence)	2000 null, 600 per change	sd ratio 0.8, 0.67, 0.5 right after calibration

Table S11. Settings of the simulations.

Used in	Data (market)	Settings
Sections 7.1 and 7.2; Table S8	Binance BTC, ETH, XRP, SOL, DOGE (USDT), 1 July to 31 December 2025 and 21 January 2024 to 30 June 2025; Bitstamp BTC/USD, July to December 2025	$A = 28,800$, $c = 3$, calibration 1440 minutes, bands 0, 0.25, 0.5, 20-day past-only intraday profile, recalibration after each alarm; controls: random times with matched delay, EWMA and HAR forecasts, rate-matched ratio rule
Section 7.4; Supplement S7.1	Nifty 50, April 2017 to October 2026, 2,361 sessions	$A = 7500$ per 375-bar session, calibration 375 or 750 bars, bands 0, 0.25, 0.5, session-aware profile; primary 750 bars and band 0.25; development 2017 to 2021, held out 2022 to October 2026; India VIX; event dates verified individually
Section 7.3; Supplement S7.2	Binance BTC, ETH, SOL, 21 January 2024 to 31 December 2025	as Table S8 with band 0.25; 16 FOMC statements, 22 CPI releases; 134 and 299 control windows, matched by weekday; stratified permutation tests, Holm correction over the primary family
Section 7.3; Supplement S7.2 (rerun)	as above	pre-registered; confidence-sequence baseline, $\delta = 0.01$, $A = 36,000$, band 0.25, restart at 00:00 UTC (primary); the same without restart; rank detector restarted without a blind period, $A = 28,800$, payoffs divided by 1.25; plug-in as above
Section 7.3; Supplement S7.2 (fresh releases)	Binance BTC, ETH, SOL, 1 January to 24 October 2026	pre-registered; 6 FOMC statements, 10 CPI releases; monitors as in the rerun; primary: one stratified permutation test over all 16 BTC releases

Used in	Data (market)	Settings
Section 7.2; Table S9	Binance five coins, 1 July to 31 December 2025 and 1 January to 30 June 2025; Nifty 50, 2024 and 2025	e-detector as above; BOCPD with the robust variance of the previous day as prior mean and hazard $1/A$; two-window variance-ratio scan with $k = 120$; in a second run also scans with $k =$ 480 and 1440 and BOCPD with w = 720 and 1440 minutes (375 bars on Nifty); one-day pause after every alarm for every method; competitor thresholds rate-matched per series
Sections 7.1 and 7.2; Supplement S7.4; Table S10	the data of the rows above	circular block bootstrap over calendar days (blocks of 7 days) or Nifty sessions (5 sessions), 9,999 replicates, one set of day weights per replicate for all coins, methods and directions; Holm within families; equivalence margin 5 points; hourly grid, next-day realised variance, EWMA, HAR and HAR-I forecasts refitted daily, Driscoll and Kraay standard errors with bandwidth 5 days; QLIKE with factors fitted on July to December 2025
Section 7.4; Supplement S7.5	Binance five coins, July to December 2024, January to June 2025, July to December 2025; Nifty 50, 2024 and 2025	four events, reference times fixed before any return was loaded; e-detector as in Sections 7.2 and 7.4; competitors as Table S9, thresholds matched outside windows from 8 days before to 2 days after each event; fresh calibration on the day before the reference time; exploratory: the restarted monitors of the companion note with their own levels
Section 7.2; Supplement S7.3 (classical rules)	Binance five coins, 1 July to 31 December 2025 and 1 January to 30 June 2025	as Table S9; rules on X and training-sample rules calibrated on 1440 minutes and recalibrated after each alarm; Student t BOCPD with w = 250, 720 and 1440 minutes and a one-day pause; thresholds matched per series to the e-detector's up and down alarm counts; bands 0 and 0.25

Used in	Data (market)	Settings
Section 7.1; Supplement S7.6 (null streams)	calm days of the five coins, July to December 2025, and of the Nifty 50, January 2022 to October 2026	permuted (primary), i.i.d. and block-bootstrapped streams (mean blocks 30, 390 and 1440 minutes); known baseline, plug-in (1440 or 750 returns), confidence sequence ($\delta = 0.01$, $A = 36{,}000$ or 9,375) and rank detector, otherwise $A = 28{,}800$ or 7,500; bands 0 and 0.25; 20,000 permuted and 10,000 i.i.d. or bootstrap streams for the pooled coins, 4000 permuted per coin; 20 monitored days
Section 7.2; Supplement S7.6 (confidence-sequence baseline, forecast-relative null)	the five-coin series of Sections 7.1 and 7.2	confidence sequence with $\delta = 0.01$ and $A = 36,000$, calibration 1440 minutes, bands 0, 0.25 and 0.5; day-block bootstrap with 2000 resamples; GARCH(1,1) fitted on January to June 2025 and frozen, or EWMA with half-life 360 minutes

Table S12. Settings of the analyses of market data.

S9. Where each table and figure comes from

Every table and figure was produced by a script of the accompanying repository from a committed result file. Figures 1, S1 and S2 are drawn by `paper/make_paper_figures.py` from the result files listed; it runs no simulation. Scripts are in `experiments/` and result files in `results/`.

Table or figure	Data	Script	Result file
Table 1	simulated	`e11_delay_theory.py`	`e11b_information.csv`
Table 2	simulated	`e04_detection.py`	`e04a_null.csv`
Figure 1	simulated	`e09_stress_guarantee.py`; drawn by `paper/make_paper_figures.py`	`e09ab_bound_tightness.csv`, `e09c_garch_band.csv`
Table S1	closed form	`th4_robustness.py`	`th4_sup_inflation.csv`, `th4_design.csv`
Table S2, Figure S1	simulated	`e11_delay_theory.py`; figure drawn by `paper/make_paper_figures.py`	`e11a_delay.csv`
Table S3	simulated	`e11f_split_c.py`	`e11f_split_c_synthetic.csv`
Table S4	simulated	`e09_stress_guarantee.py`	`e09ab_bound_tightness.csv`
Table S5	simulated	`e04_detection.py`	`e04b_change.csv`
Table S6	simulated	`e08_cs_baseline.py`	`e08b_change.csv`
Table S7, Figure S2	simulated	`e10_forecast_relative.py`; figure drawn by `paper/make_paper_figures.py`	`e10d_signal_cap.csv`
Table S8	market	`e06b_five_coins_monitor.py`; intervals from `e20_inference.py`	`e06e_pooled.csv`, `e06e_forecast_hits.csv`, `e20_bootstrap.csv`

Table or figure	Data	Script	Result file
Table S9	market	`e13_benchmarks.py`	`e13d_real_pooled.csv`, `e13d_real_long_windows_pooled.csv`
Table S10	market	`e20_inference.py`	`e20_forecast_regressions.csv`, `e20_forecast_qlike.csv`
Tables S11 and S12	settings	the scripts above	

The limits of Theorems 12, 15 and 23 and Propositions S11 and S12 come from `th5_limits.py` (`th5_collapse.csv`, `th5_floors.csv`, `th5_asymptotic.csv`, `th5_plugin_bound.csv`, `th5_plugin_sim.csv`, `th5_holdbold.csv`, `th5_boundary.csv`). The later-day results of Section 3.1, Supplement S1.7 and the tolerance numbers of Section 4.2 come from `th6_per_period.py` (`th6_hazard_390.csv`, `th6_hazard_1440.csv`, `th6_adversary.csv`, `th6_calm_family.csv`, `th6_park_strike.csv`, `th6_park_scaled.csv`, `th6_delay.csv`, `th6_delay_by_position.csv`, `th6_delay_renewal.csv`, `th6_tolerance.csv`, `th6_laws.csv`). The competitors calibrated for the data at hand (Sections 6.2 and 7.2, Supplement S5 and S7.3) come from `e21_benchmarks.py` (`e21_null.csv`, `e21_change.csv`, `e21_e13_null.csv`, `e21_e13_change.csv`, `e21_thresholds.csv`, `e21_bocpd_thresholds.csv`, `e21_real_pooled.csv`, `e21_real_h1_matched_thresholds.csv`, `e21_real_h2_matched_thresholds.csv`). The null streams, the confidence-sequence configuration and the forecast-relative null of Section 7 and Supplement S7.6, and the split clip with the confidence-sequence baseline in Supplement S5, come from `e19_real_null.py` (`e19_null_pooled.csv`, `e19_null_streams.csv`, `e19_perm_drift.csv`, `e19_null_std.csv`, `e19_cs_pooled.csv`, `e19_cs_primary_bootstrap.csv`, `e19_cs_cost.csv`, `e19_cs_forecast_pooled.csv`, `e19_cs_competitor.csv`, `e19_blind.csv`, `e19_fr_pooled.csv`, `e19_fr_cost.csv`, `e19_split_c_cs.csv`). The adjusted p-values of Section 7 are in `e20_bootstrap.csv` and `e20_holm_existing.csv`, the landmark cases of Supplement S7.5 in `e18_outcomes.csv` and `e18_awake.csv` (`e18_landmark_events.py`), and the rerun of the event test (Section 7.3, Supplement S7.2) in `e22_tests.csv`, `e22_monitor_summary.csv`, `e22_watched.csv`, `e22_windows.csv`, `e22_alarms.csv` and `e22_reproduction.csv` (`e22_guaranteed_events.py`), and the test on fresh releases in `e23_tests.csv`, `e23_monitor_summary.csv`, `e23_watched.csv`, `e23_windows.csv`, `e23_blind_segments.csv`, `e23_power.csv` and `e23_power_actual.csv` (`e23_fresh_releases.py`). Numbers in the text that are not in a table come from the same scripts and from the theory notes and experiment reports of the repository; a tracing script checks each number of the paper against these sources.

---

1. We could not read in full Bibinger, Jirak and Vetter (2017), Chu, Stinchcombe and White (1996), Clerico (2025), Dandapanthula and Ramdas (2025), Horváth, Hušková, Kokoszka and Steinebach (2004), Horváth, Kokoszka and Zhang (2006), Knoblauch, Jewson and Damoulas (2018), Ram and Ramdas (2026) and Shekhar and Ramdas (2024), which we checked from listings, abstracts and search excerpts, nor Lai (1998) and Mei (2008), known to us through secondary sources. What we say about them, and every claim of novelty that rests on it, is subject to this caveat.↩